

SIGNALS AND SYSTEMS

Pingye's Notebook

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CHAPTER 1:

Fundamental Concepts

Chapter 1 Map

Chapter 1 builds the vocabulary for the whole course. The central object is a deterministic signal passing through a system. The course then studies how linear time-invariant systems transmit and process signals in time domain, frequency domain, and complex-frequency domain.

Topic	What you should be able to do
Signals	Distinguish information, message, and signal; classify continuous-time, discrete-time, sampled, analog, and digital signals.
Basic functions	Use step, ramp, impulse, rectangular pulse, and sinusoidal signals.
Signal operations	Perform amplitude scaling, addition, time shift, folding, differentiation, and integration.
Systems	Express systems by input-output equations, differential equations, difference equations, and block diagrams.
System properties	Test causality, memory, linearity, and time invariance.
Examples	Analyze RC circuits, mass-spring-damper systems, moving-average filters, and property-judgment examples.

1.1 Information, Message, and Signal

Definition

- **Information:** meaningful content contained in a message.
- **Message:** a group of symbols or a symbol sequence used to convey information.
- **Signal:** the physical form used to transmit a message; it is the carrier of the message.

Layer	Meaning	Examples
Information	The semantic content to be delivered	temperature rises, a command, speech meaning
Message	Symbolic representation of information	text, binary code, flag semaphore, speech symbols
Signal	Physical carrier of the message	voltage, current, sound pressure, light intensity, pixel value

1.2 Continuous-Time Signals

1.2.1 Definition and classification

Definition

A continuous-time signal is usually written as

$$x(t), \quad t \in \mathbb{R}.$$

It is a real-valued or scalar-valued function of the time variable t . For every fixed t , the value $x(t)$ is a real number.

Classification	Common categories
Time variable	Continuous-time $x(t)$; discrete-time $x[n]$
Periodicity	Periodic; aperiodic
Dimension	1-D, 2-D, N -D
Energy behavior	Energy signal; power signal
Amplitude values	Continuous-valued; discrete-valued

1.2.2 Unit step and unit ramp

Definition

The unit-step function is

$$u(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

The unit-ramp function is

$$r(t) = \begin{cases} t, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

A step with amplitude K is $Ku(t)$; a ramp with slope K is $Kr(t)$.

Rule / Test

The step and ramp functions are related by

$$r(t) = \int_{-\infty}^t u(\lambda) d\lambda, \quad \frac{dr(t)}{dt} = u(t) \quad (t \neq 0).$$

At $t = 0$, ordinary differentiability fails; generalized derivatives are used.

1.2.3 Unit impulse

Definition

The unit impulse $\delta(t)$, also called the Dirac delta distribution, is characterized by

$$\delta(t) = 0 \quad (t \neq 0), \quad \int_{-\varepsilon}^{\varepsilon} \delta(\lambda) d\lambda = 1 \quad (\varepsilon > 0).$$

The important point is its **unit area**, not a finite value at the origin. The value $\delta(0)$ is not defined.

Rule / Test

The sifting property is

$$\int_{-\infty}^{\infty} f(\lambda) \delta(\lambda - t_1) d\lambda = f(t_1),$$

and equivalently

$$f(t) \delta(t - t_1) = f(t_1) \delta(t - t_1).$$

1.2.4 Periodic signals**Definition**

A continuous-time signal $x(t)$ is periodic with period $T > 0$ if

$$x(t + T) = x(t), \quad \text{for all } t.$$

The **fundamental period** is the smallest positive period.

A sinusoidal signal

$$x(t) = A \cos(\omega t + \theta)$$

is periodic with fundamental period

$$T_0 = \frac{2\pi}{|\omega|}, \quad \omega \neq 0.$$

Example 1.1: Sum of continuous-time periodic signals

Let

$$x_1(t) = \cos\left(\frac{\pi t}{2}\right), \quad x_2(t) = \cos\left(\frac{\pi t}{3}\right).$$

Find whether $x(t) = x_1(t) + x_2(t)$ is periodic, and find its fundamental period.

Solution. For $x_1(t)$,

$$\omega_1 = \frac{\pi}{2}, \quad T_1 = \frac{2\pi}{\omega_1} = 4.$$

For $x_2(t)$,

$$\omega_2 = \frac{\pi}{3}, \quad T_2 = \frac{2\pi}{\omega_2} = 6.$$

The ratio is

$$\frac{T_1}{T_2} = \frac{4}{6} = \frac{2}{3},$$

which is rational. Therefore the sum is periodic. A common period is

$$3T_1 = 2T_2 = 12.$$

Hence the fundamental period is

$$T_0 = 12 \text{ s}.$$

1.2.5 Time-shifted signals

Rule / Test

For $t_0 > 0$:

$x(t - t_0)$ is shifted right by t_0 ,

$x(t + t_0)$ is shifted left by t_0 .

Mnemonic: **minus shifts right, plus shifts left.**

Examples:

$u(t - 2)$: unit step delayed to $t = 2$, $u(t + 2)$: unit step advanced to $t = -2$.

1.2.6 Continuity and generalized derivative

Definition

For a continuous-time signal $x(t)$, at $t = t_1$:

$$x(t_1^-) = x(t_1) = x(t_1^+) \Rightarrow \text{continuous at } t_1.$$

If

$$x(t_1^-) \neq x(t_1^+),$$

then $x(t)$ has a jump discontinuity at $t = t_1$.

At a discontinuity, the ordinary derivative is not defined. The generalized derivative includes an impulse term:

$$\left. \frac{dx(t)}{dt} \right|_G = \text{ordinary derivative on continuous intervals} + \sum_k [x(t_k^+) - x(t_k^-)] \delta(t - t_k).$$

Example 1.2: Generalized derivative of a step

Let

$$x(t) = Ku(t).$$

The signal has a jump of height K at $t = 0$. Its ordinary derivative is zero for $t \neq 0$, but its generalized derivative is

$$\boxed{\frac{d}{dt} Ku(t) = K\delta(t)}.$$

For $K = 1$,

$$\boxed{\frac{du(t)}{dt} = \delta(t)}.$$

1.3 Discrete-Time Signals

1.3.1 Definition

Definition

A discrete-time signal is written as

$$x[n], \quad n \in \mathbb{Z}.$$

Here n is only an integer index. The notation $x[n]$ can mean the whole sequence or the value at the time point

$$t = t_n.$$

Example 1.3: Stem-plot sequence

Suppose

$$x[0] = 1, \quad x[1] = 2, \quad x[2] = 1, \quad x[3] = 0, \quad x[4] = -1,$$

and $x[n] = 0$ for all other n . This is a finite-length discrete-time signal. It should be drawn by a **stem plot**, not by connecting samples with a continuous curve.

1.3.2 Sampling

Definition

One common way to obtain discrete-time signals is by sampling continuous-time signals. Under uniform sampling with sampling period T ,

$$x[n] = x(t)|_{t=nT} = x(nT), \quad n \in \mathbb{Z}.$$

Note

A sampled continuous-time signal is discrete in time, but its amplitude may still be continuous. Therefore, a sampled signal is not necessarily a digital signal.

1.3.3 Discrete-time step, ramp, and unit pulse

Definition

The discrete-time unit step and unit ramp are

$$u[n] = \begin{cases} 1, & n = 0, 1, 2, \dots, \\ 0, & n = -1, -2, \dots, \end{cases}$$

$$r[n] = \begin{cases} n, & n = 0, 1, 2, \dots, \\ 0, & n = -1, -2, \dots \end{cases}$$

The discrete-time unit pulse is

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

Rule / Test

For discrete-time signals,

$$u[n] = \sum_{k=-\infty}^n \delta[k], \quad \delta[n] = u[n] - u[n-1].$$

Do not confuse $\delta[n]$ with a sampled version of $\delta(t)$. They are conceptually different objects.

1.3.4 Discrete-time periodicity

Definition

A discrete-time signal $x[n]$ is periodic if there exists a positive integer N such that

$$x[n + N] = x[n], \quad \text{for all } n \in \mathbb{Z}.$$

The smallest such N is the fundamental period.

For a discrete-time sinusoid

$$x[n] = A \cos(\Omega_0 n + \theta),$$

it is periodic if and only if

$$\frac{2\pi}{\Omega_0} \in \mathbb{Q}.$$

Equivalently, there exist integers $N > 0$ and m such that

$$\Omega_0 N = 2\pi m.$$

Example 1.4: Periodicity of a sine sequence

Here

$$\Omega_0 = 0.4 = \frac{2}{5}.$$

Then

$$\frac{2\pi}{\Omega_0} = \frac{2\pi}{0.4} = 5\pi,$$

which is not rational. Hence there is no positive integer N satisfying $\Omega_0 N = 2\pi m$. Therefore

$$x[n] = \sin(0.4n) \text{ is aperiodic.}$$

1.3.5 Digital signals and signal categories

Definition

A digital signal is a discrete-time signal whose values belong to a finite set:

$$x[n] \in \{a_1, a_2, \dots, a_N\}.$$

A binary signal is a digital signal with two allowed amplitude levels.

Signal type	Time variable	Amplitude value
Continuous-time signal	continuous t	arbitrary
Analog signal	continuous t	continuous magnitude
Discrete-time signal	integer n	arbitrary
Sampled signal	discrete n	usually continuous magnitude
Digital signal	discrete n	finite number of values

1.3.6 Discrete-time shift

Rule / Test

For an integer shift $q > 0$:

$x[n - q]$ is shifted right by q ,

$x[n + q]$ is shifted left by q .

The same mnemonic applies: **minus shifts right, plus shifts left**.

1.4 Basic Signal Operations

Operation	Continuous-time form	Meaning
Amplitude scaling	$y(t) = ax(t)$	multiply the signal value
Addition	$y(t) = x_1(t) + x_2(t)$	superpose signals
Time shift	$y(t) = x(t - t_0)$	delay by t_0 when $t_0 > 0$
Folding	$y(t) = x(-t)$	mirror about the vertical axis
Differentiation	$y(t) = \frac{dx(t)}{dt}$	rate of change
Integration	$y(t) = \int_{-\infty}^t x(\lambda) d\lambda$	accumulated area

1.5 Systems

1.5.1 Definition and block representation

Definition

A system is any device or rule that processes signals for analysis, synthesis, enhancement, format conversion, recording, transmission, and other purposes. Mathematically, a system is usually described by the equation relating its input and output signals.

For a single-input single-output continuous-time system,

$$x(t) \longrightarrow \boxed{\mathcal{T}} \longrightarrow y(t) = \mathcal{T}\{x(t)\}.$$

For a single-input single-output discrete-time system,

$$x[n] \longrightarrow \boxed{\mathcal{T}} \longrightarrow y[n] = \mathcal{T}\{x[n]\}.$$

1.5.2 Continuous, discrete, and hybrid systems

System type	Input-output form
Continuous-time system	$x(t) \mapsto y(t)$, where $t \in \mathbb{R}$
Discrete-time system	$x[n] \mapsto y[n]$, where $n \in \mathbb{Z}$
Hybrid system	maps one time type to another, for example ADC: $x(t) \mapsto y[n]$

1.6 Examples of Systems

Example 1.5: RC circuit

For the RC circuit in the lecture, the input-output differential equation is

$$C \frac{dy(t)}{dt} + \frac{1}{R} y(t) = x(t).$$

If the input is the unit step $x(t) = u(t)$ and the initial energy is zero, then for $t \geq 0$,

$$C \frac{dy(t)}{dt} + \frac{1}{R} y(t) = 1.$$

The step response is

$$y(t) = R \left[1 - e^{-t/(RC)} \right], \quad t \geq 0.$$

Interpretation. This is a first-order continuous-time system. The time constant is RC .

Example 1.6: Mass-spring-damper system

The input-output differential equation is

$$M \frac{d^2 y(t)}{dt^2} + D \frac{dy(t)}{dt} + K y(t) = x(t).$$

Here M is mass, D is damping, and K is stiffness. A unit-step input can produce a transient response with overshoot and oscillation, depending on the damping ratio.

Example 1.7: N -point moving-average filter

An N -point moving-average filter is a discrete-time system:

$$y[n] = \frac{1}{N} (x[n] + x[n-1] + \cdots + x[n-N+1]).$$

The next output is

$$y[n+1] = \frac{1}{N} (x[n+1] + x[n] + \cdots + x[n-N+2]).$$

If

$$x[n] = s[n] + e[n],$$

where $s[n]$ is the smooth component and $e[n]$ is noise, then

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} s[n-k] + \frac{1}{N} \sum_{k=0}^{N-1} e[n-k].$$

The averaging operation preserves the slowly varying part and reduces the random noisy part.

1.7 Basic System Properties

1.7.1 Overview

The most important system properties in Chapter 1 are:

causality, memory, linearity, time invariance.

Unless otherwise stated, the output is the zero-state response: no initial energy exists before the input is applied.

1.7.2 Causality

Definition

A system is causal if the output at time t_1 does not depend on input values at times later than t_1 . For continuous-time systems:

$$y(t_1) \text{ must not depend on } x(t), \quad t > t_1.$$

For discrete-time systems:

$$y[n_1] \text{ must not depend on } x[n], \quad n > n_1.$$

Rule / Test

A quick causality test:

- Terms such as $x(t)$, $x(t - 1)$, $x[n]$, $x[n - 1]$ use present or past input: causal.
- Terms such as $x(t + 1)$, $x[n + 1]$ use future input: noncausal.
- Folding, such as $x[-n]$, must be checked carefully. For $n < 0$, $-n > n$, so future samples may be used; usually noncausal.

Example 1.8: Judgment of causality

Classify the following systems.

System	Causality
$y(t) = x^2(t - 1)$	causal; it uses the past value $x(t - 1)$
$y(t) = x(t + 1)$	noncausal; it uses the future value $x(t + 1)$
$y[n] = x[-n]$	noncausal in general; for $n < 0$, $-n > n$
$y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n - 1]$	causal; it uses $x[n - 1]$ and a time-dependent coefficient

Example 1.9: Present plus past vs. present plus future

For

$$y(t) = x(t) + x(t - 2),$$

set $t = 0$:

$$y(0) = x(0) + x(-2).$$

The output uses present and past input values, so the system is causal.

For

$$y(t) = x(t) + x(t + 2),$$

set $t = 0$:

$$y(0) = x(0) + x(2).$$

The output uses a future input value, so the system is noncausal.

1.7.3 Memory and memorylessness

Definition

A system is memoryless if $y(t_1)$ depends only on $x(t_1)$. It has memory if $y(t_1)$ depends on past or future values of the input.

System	Memory property
$y(t) = Kx(t)$	memoryless; output depends only on the current input
$y(t) = x(t - 1)$	has memory; output depends on a past input value
$y(t) = \int_{-\infty}^t x(\lambda) d\lambda$	has memory; output depends on the whole past input
RLC/RC circuits	usually have memory due to stored energy in capacitors and inductors

1.7.4 Linearity

Definition

A system is linear if it satisfies both additivity and homogeneity. Equivalently, if

$$x_1 \rightarrow y_1, \quad x_2 \rightarrow y_2,$$

then for arbitrary scalars a and b ,

$$ax_1 + bx_2 \rightarrow ay_1 + by_2.$$

Rule / Test

Linearity test:

1. First linearly combine the inputs, then pass the result through the system. Call the output y_1 .
2. Pass each input through the system first, then linearly combine the outputs. Call the result y_2 .
3. If $y_1 = y_2$ for all inputs and all scalars, the system is linear.

Example 1.10: A linear time-multiplier

Let

$$x(t) = ax_1(t) + bx_2(t).$$

Then

$$y_1(t) = t[ax_1(t) + bx_2(t)] = atx_1(t) + btx_2(t).$$

But

$$a\mathcal{T}\{x_1(t)\} + b\mathcal{T}\{x_2(t)\} = atx_1(t) + btx_2(t) = y_1(t).$$

Thus

$$y(t) = tx(t) \text{ is linear.}$$

The coefficient t changes with time, but that affects time invariance, not linearity.

Example 1.11: Square-law device is nonlinear

Consider

$$y(t) = x^2(t).$$

For input $ax(t)$, the output is

$$\mathcal{T}\{ax(t)\} = a^2x^2(t),$$

while homogeneity would require

$$a\mathcal{T}\{x(t)\} = ax^2(t).$$

Since $a^2x^2(t) \neq ax^2(t)$ in general, the system is not homogeneous and therefore not linear.

$$y(t) = x^2(t) \text{ is nonlinear.}$$

1.7.5 Time invariance**Definition**

A continuous-time system is time invariant if a time shift of the input causes the same time shift of the output. Mathematically, if

$$x(t) \rightarrow y(t),$$

then for every t_0 ,

$$x(t - t_0) \rightarrow y(t - t_0).$$

Informally, the behavior of the system does not depend on the absolute time origin.

Rule / Test

Time-invariance test:

1. Shift the input first: $x(t) \rightarrow x(t - t_0)$, then pass it through the system. This gives $y_1(t)$.
2. Pass the original input through the system first: $x(t) \rightarrow y(t)$, then shift the output. This gives $y_2(t) = y(t - t_0)$.
3. If $y_1(t) = y_2(t)$ for all inputs and all shifts, the system is time invariant.

Example 1.12: Time invariance of a time-multiplier

Route 1: shift input first.

$$x(t) \rightarrow x(t - t_0), \quad y_1(t) = tx(t - t_0).$$

Route 2: system first, then shift output.

$$y(t) = tx(t), \quad y_2(t) = y(t - t_0) = (t - t_0)x(t - t_0).$$

Since

$$y_1(t) = tx(t - t_0) \neq (t - t_0)x(t - t_0) = y_2(t),$$

the system is time varying.

$$y(t) = tx(t) \text{ is linear but time varying.}$$

Example 1.13: Compare two nonlinear/time-dependent systems

For

$$\text{Sys. I: } y(t) = \cos[x(t)],$$

shift input first:

$$y_1(t) = \cos[x(t - t_0)].$$

System first, then shift output:

$$y_2(t) = y(t - t_0) = \cos[x(t - t_0)].$$

Thus Sys.1 is time invariant.

For

$$\text{Sys.2: } y(t) = x(t) \cos t,$$

shift input first:

$$y_1(t) = x(t - t_0) \cos t.$$

System first, then shift output:

$$y_2(t) = y(t - t_0) = x(t - t_0) \cos(t - t_0).$$

Since $y_1(t) \neq y_2(t)$ in general, Sys.2 is time varying.

Example 1.14: Quick time-invariance judgments

System	Time property
$y(t) = tx(t)$	time varying, because t explicitly changes with time origin
$y(t) = x^2(t + 1)$	time invariant; a fixed advance is not by itself time varying
$y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n - 1]$	time varying; the coefficient depends explicitly on n

1.8 Property Tests for Common Operators

1.8.1 Integrator

Example 1.15: Integrator from $-\infty$ to t

Consider

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

Linearity. For input $ax_1(t) + bx_2(t)$,

$$\int_{-\infty}^t [ax_1(\tau) + bx_2(\tau)] d\tau = a \int_{-\infty}^t x_1(\tau) d\tau + b \int_{-\infty}^t x_2(\tau) d\tau.$$

Thus the system is linear.

Time invariance. Shift the input:

$$y_1(t) = \int_{-\infty}^t x(\tau - t_0) d\tau.$$

Let $\sigma = \tau - t_0$. Then

$$y_1(t) = \int_{-\infty}^{t-t_0} x(\sigma) d\sigma = y(t - t_0).$$

Therefore

$$y(t) = \int_{-\infty}^t x(\tau) d\tau \text{ is linear and time invariant.}$$

Note

If the lower integration limit is a finite constant, such as

$$y(t) = \int_0^t x(\tau) d\tau,$$

then the system is generally time varying. The fixed boundary 0 changes the behavior when the time origin is shifted.

Example 1.16: Sliding-window integrator

Consider

$$y(t) = \int_{t-1}^t x(\lambda) d\lambda.$$

Shift input first:

$$y_1(t) = \int_{t-1}^t x(\lambda - t_0) d\lambda.$$

Let $\sigma = \lambda - t_0$. Then

$$y_1(t) = \int_{t-t_0-1}^{t-t_0} x(\sigma) d\sigma = y(t - t_0).$$

So this system is time invariant. It is also linear because integration is linear.

$$y(t) = \int_{t-1}^t x(\lambda) d\lambda \text{ is linear and time invariant.}$$

1.8.2 Differentiator

Example 1.17: Differentiator

Consider

$$y(t) = \frac{dx(t)}{dt}.$$

Linearity:

$$\frac{d}{dt}[ax_1(t) + bx_2(t)] = a\frac{dx_1(t)}{dt} + b\frac{dx_2(t)}{dt}.$$

Time invariance: if the input is shifted to $x(t - t_0)$,

$$y_1(t) = \frac{d}{dt}x(t - t_0) = x'(t - t_0) = y(t - t_0).$$

Thus

$$\text{the differentiator is linear and time invariant.}$$

1.9 Systems Described by Differential Equations

Example 1.18: Differential-equation system with a time-varying coefficient

A continuous-time system is described by

$$\frac{dy(t)}{dt} + e^{-t}y(t) = x(t).$$

Determine whether it is linear and time invariant.

Linearity. Let $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$. Then

$$y_1'(t) + e^{-t}y_1(t) = x_1(t),$$

$$y_2'(t) + e^{-t}y_2(t) = x_2(t).$$

For input $ax_1(t) + bx_2(t)$, the proposed output is $ay_1(t) + by_2(t)$. Substitution gives

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + e^{-t}[ay_1(t) + by_2(t)] = ax_1(t) + bx_2(t).$$

Thus the system is linear.

Time invariance. If $x(t) \rightarrow y(t)$, then the equation is

$$y'(t) + e^{-t}y(t) = x(t).$$

For a shifted input $x(t - t_0)$, time invariance would require the output $y(t - t_0)$. Substitute it:

$$\frac{d}{dt}y(t - t_0) + e^{-t}y(t - t_0) = x(t - t_0).$$

Let $\tau = t - t_0$, so $t = \tau + t_0$. Then

$$y'(\tau) + e^{-(\tau+t_0)}y(\tau) = x(\tau).$$

This is not the same as

$$y'(\tau) + e^{-\tau}y(\tau) = x(\tau)$$

for general t_0 . Therefore the system is time varying.

The system is linear but time varying.

Example 1.19: Absolute-value system

Determine whether

$$y(t) = |x(t)|$$

is causal, memoryless, linear, and time invariant.

Causality. $y(t_1)$ depends only on $x(t_1)$, not on future input values. Hence it is causal.

Memory. $y(t_1)$ depends only on the current value $x(t_1)$. Hence it is memoryless.

Linearity. Let $x_1(t) > 0$ and $x_2(t) < 0$. Then

$$\mathcal{T}\{x_1(t)\} = x_1(t), \quad \mathcal{T}\{x_2(t)\} = -x_2(t).$$

But

$$\mathcal{T}\{x_1(t) + x_2(t)\} = |x_1(t) + x_2(t)|$$

is not generally equal to

$$\mathcal{T}\{x_1(t)\} + \mathcal{T}\{x_2(t)\} = x_1(t) - x_2(t).$$

Thus it is nonlinear.

Time invariance. Shift input first:

$$y_1(t) = |x(t - t_0)|.$$

System first, then shift output:

$$y_2(t) = y(t - t_0) = |x(t - t_0)|.$$

Thus it is time invariant.

$y(t) = |x(t)|$ is causal, memoryless, nonlinear, and time invariant.

1.10 One-Page Review Tables

1.10.1 Signal formulas

Object	Formula / criterion
CT unit step	$u(t) = 1$ for $t \geq 0$, 0 for $t < 0$
CT unit ramp	$r(t) = t$ for $t \geq 0$, 0 for $t < 0$
Impulse area	$\int_{-\epsilon}^{\epsilon} \delta(t) dt = 1$
Impulse sifting	$\int f(\lambda) \delta(\lambda - t_0) d\lambda = f(t_0)$
CT sinusoid period	$T_0 = 2\pi/ \omega $
DT sinusoid period	periodic iff $2\pi/\Omega_0$ is rational
Time shift	$x(t - t_0)$ shifts right; $x(t + t_0)$ shifts left
DT time shift	$x[n - q]$ shifts right; $x[n + q]$ shifts left

1.10.2 System-property judgment table

Property	Key question	Fast judgment method
Causality	Does $y(t)$ use future inputs?	Future terms $x(t + a)$ or $x[n + k]$ with $a, k > 0$ usually imply noncausal.
Memoryless	Does $y(t)$ depend only on $x(t)$?	Delays, advances, integrals, derivatives, and dynamic circuits usually imply memory.
Linearity	Does superposition hold?	Test $\mathcal{T}\{ax_1 + bx_2\} = a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$.
Time invariance	Does time shift commute with the system?	Compare “shift input then system” with “system then shift output”. Explicit t or n coefficients often imply time variation.

1.10.3 Typical examples

System	Causal	Memory	Linear	Time property
$y(t) = Kx(t)$	yes	no	yes	TI
$y(t) = x(t - 1)$	yes	yes	yes	TI
$y(t) = x(t + 1)$	no	yes	yes	TI
$y(t) = tx(t)$	yes	no	yes	time varying
$y(t) = x^2(t)$	yes	no	no	TI
$y(t) = x(t) $	yes	no	no	TI
$y(t) = \int_{-\infty}^t x(\lambda) d\lambda$	yes	yes	yes	TI
$y(t) = \int_0^t x(\lambda) d\lambda$	yes	yes	yes	time varying
$y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n - 1]$	yes	yes	no	time varying

1.11 Exam-Style Workflow

Rule / Test

When asked to judge a system property, use the following order:

1. **Causality:** locate the time indices inside $x(\cdot)$.
2. **Memory:** check whether the output uses only the current input.
3. **Linearity:** use superposition; ignore whether coefficients depend on time until the time-invariance test.
4. **Time invariance:** perform the two-route shift test.

Note

Most mistakes come from mixing up linearity and time invariance. A system can be linear but time varying, such as $y(t) = tx(t)$. A system can also be nonlinear but time invariant, such as $y(t) = x^2(t)$ or $y(t) = |x(t)|$.

CHAPTER 2:

Time-Domain Models of Systems

Chapter 2 Map

Chapter 2 turns the abstract system notation $y = \mathcal{T}\{x\}$ into computable time-domain models. The discrete-time part uses input-output sums, unit-pulse response, convolution sums, and difference equations. The continuous-time part uses physical laws, differential equations, classical solutions, zero-input/zero-state decomposition, impulse response, and convolution integrals.

Section	Core target
2.1 I/O representation of DT systems	Express causal LTI-DT systems by weighted sums and by their unit-pulse response.
2.2 Convolution of DT signals	Compute $x[n] * v[n]$ and system outputs $y[n] = x[n] * h[n]$.
2.3 Difference-equation models	Use recursion or classical methods to obtain the full response of DT systems.
2.4 Differential-equation models	Derive CT input-output differential equations from circuits and mechanical systems.
2.5 Solution of differential equations	Solve CT models by numerical, classical, and physical response-decomposition methods.
2.6 CT convolution representation	Express zero-state response as $y_{zs}(t) = x(t) * h(t)$.

2.1 Input-Output Representation of Discrete-Time Systems

Definition

A causal LTI discrete-time system can often be represented as a weighted sum of present and past input samples:

$$y[n] = \sum_{i=0}^{N-1} w_i x[n-i].$$

The constants w_i are the **weights**. The system is causal because it uses only $x[n], x[n-1], \dots$

Example 2.1: N -point moving-average filter

The N -point moving-average filter is

$$y[n] = \frac{1}{N} \{x[n] + x[n-1] + \dots + x[n-N+1]\} = \sum_{i=0}^{N-1} \frac{1}{N} x[n-i].$$

Thus

$$w_i = \frac{1}{N}, \quad i = 0, 1, \dots, N - 1.$$

It smooths rapid variations because the output is an average of the most recent N input samples.

Example 2.2: Exponentially weighted moving-average filter

The finite-length exponentially weighted moving-average filter is

$$y[n] = \sum_{i=0}^{N-1} ab^i x[n-i], \quad 0 < b < 1,$$

where the normalizing coefficient is

$$a = \frac{1-b}{1-b^N}.$$

For $N = 3$ and $b = 0.5$,

$$a = \frac{1-0.5}{1-0.5^3} \approx 0.571,$$

so

$$y[n] = 0.571x[n] + 0.286x[n-1] + 0.143x[n-2].$$

Compared with the ordinary three-point MA filter

$$y[n] = 0.333x[n] + 0.333x[n-1] + 0.333x[n-2],$$

this filter gives more weight to recent samples.

Rule / Test

For a causal LTI-DT system with input $x[n] = 0$ for $n < 0$, the output can be written as

$$y[n] = \sum_{i=0}^n h[i]x[n-i], \quad n \geq 0,$$

where $h[n]$ is the **unit-pulse response**. Therefore the system is completely determined by $h[n]$.

2.2 Convolution of Discrete-Time Signals

Definition

For arbitrary discrete-time signals $x[n]$ and $v[n]$, their convolution is

$$(x * v)[n] = \sum_{i=-\infty}^{\infty} x[i]v[n-i].$$

Convolution is commutative:

$$x[n] * v[n] = v[n] * x[n].$$

Rule / Test

If $x[n] = 0$ and $v[n] = 0$ for all $n < 0$, then

$$(x * v)[n] = \begin{cases} 0, & n < 0, \\ \sum_{i=0}^n x[i]v[n-i], & n \geq 0. \end{cases}$$

For a causal LTI-DT system,

$$y[n] = x[n] * h[n] = \sum_{i=0}^n x[i]h[n-i].$$

Note

The graphical or tabular convolution procedure is:

1. Replace the running time index by i : draw $x[i]$ and $v[i]$.
2. Fold $v[i]$ to get $v[-i]$.
3. Shift to get $v[n-i]$.
4. Multiply $x[i]v[n-i]$ pointwise and sum over i .

Example 2.3: Convolution of two exponential sequences

Let

$$x[n] = a^n u[n], \quad v[n] = b^n u[n].$$

For $n \geq 0$,

$$(x * v)[n] = \sum_{i=0}^n a^i b^{n-i} = b^n \sum_{i=0}^n \left(\frac{a}{b}\right)^i.$$

If $a = b$,

$$(x * v)[n] = (n+1)a^n u[n].$$

If $a \neq b$,

$$(x * v)[n] = \frac{b^{n+1} - a^{n+1}}{b - a} u[n].$$

Example 2.4: Rectangular pulse self-convolution

Let

$$p[n] = \begin{cases} 1, & 0 \leq n \leq 9, \\ 0, & \text{otherwise.} \end{cases}$$

Then $p[n] * p[n]$ counts the number of overlapping unit samples between $p[i]$ and $p[n-i]$:

$$(p * p)[n] = \begin{cases} 0, & n < 0, \\ n+1, & 0 \leq n \leq 9, \\ 19-n, & 10 \leq n \leq 18, \\ 0, & n > 18. \end{cases}$$

The result is a discrete triangular sequence.

Example 2.5: Complete response of a causal LTI-DT system

For a causal LTI-DT system,

$$y[n] = y_{zi}[n] + y_{zs}[n].$$

The zero-state part is obtained by convolution:

$$y_{zs}[n] = x[n] * h[n].$$

The zero-input part $y_{zi}[n]$ is produced only by the initial stored state. If all initial conditions are zero, then $y[n] = y_{zs}[n]$.

2.3 Difference-Equation Models**Definition**

A general SISO discrete-time system can be modeled by the constant-coefficient difference equation

$$y[n] + \sum_{i=1}^N a_i y[n-i] = \sum_{i=0}^M b_i x[n-i].$$

Equivalently,

$$y[n] = -\sum_{i=1}^N a_i y[n-i] + \sum_{i=0}^M b_i x[n-i].$$

This form is an N th-order recursion when at least one coefficient a_i is nonzero.

Rule / Test

To compute by recursion, the initial output values must be known. For a start time $n = 0$, specify

$$y[-1], y[-2], \dots, y[-N].$$

For a start time $n = q$, specify

$$y[q-1], y[q-2], \dots, y[q-N].$$

At each step the next output is a linear combination of the previous N output values and the present/past input values.

Example 2.6: Second-order recursion

Given

$$y[n] - 1.5y[n-1] + y[n-2] = 2x[n-2],$$

with $x[n] = u[n]$, $y[-2] = 2$, and $y[-1] = 1$, rewrite as

$$y[n] = 1.5y[n-1] - y[n-2] + 2x[n-2].$$

Then

$$y[0] = 1.5y[-1] - y[-2] + 2x[-2] = 1.5(1) - 2 + 0 = -0.5,$$

$$y[1] = 1.5y[0] - y[-1] + 2x[-1] = 1.5(-0.5) - 1 + 0 = -1.75,$$

$$y[2] = 1.5y[1] - y[0] + 2x[0] = 1.5(-1.75) - (-0.5) + 2 = -0.125,$$

$$y[3] = 1.5y[2] - y[1] + 2x[1] = 1.5(-0.125) - (-1.75) + 2 = 3.5625.$$

Example 2.7: Inventory-level model

If $y[n]$ denotes inventory, $p[n]$ production, and $d[n]$ demand, then

$$y[n] = y[n-1] + p[n] - d[n].$$

Let $x[n] = p[n] - d[n]$. Then

$$y[n] = y[0] + \sum_{i=1}^n x[i] = y[0] + \sum_{i=1}^n (p[i] - d[i]).$$

If $p[n] = d[n-1]$ for $n \geq 2$, then

$$y[n] = y[0] + p[1] - d[n], \quad n \geq 2.$$

Inventory depletion is avoided if

$$y[0] > d[2] - p[1].$$

Method	Convolution summation	Difference-equation recursion
Formula	$y[n] = \sum_{i=0}^n h[i]x[n-i]$	$y[n] = -\sum_{i=1}^N a_i y[n-i] + \sum_{i=0}^M b_i x[n-i]$
Initial values	unnecessary for zero-state output	previous N output values are required
Cost	grows with n	constant for fixed N and M
Response range	zero-state response	full response

2.3.1 Classical solution of difference equations**Rule / Test**

The classical method is

total solution = homogeneous solution + particular solution,

or

$$y[n] = y_h[n] + y_p[n].$$

The homogeneous solution comes from the characteristic equation; the particular solution is chosen according to the input form.

Input form	Trial particular solution
A	constant C
r^n	Cr^n ; if r is a characteristic root, use $C_1 nr^n + C_2 r^n$ as needed
e^{an}	Ce^{an}
$e^{j\omega n}$	$Ce^{j\omega n}$
$\cos \omega n$ or $\sin \omega n$	$A \cos(\omega n + \theta)$ or $A \sin(\omega n + \theta)$
n^k	polynomial $A_k n^k + A_{k-1} n^{k-1} + \dots + A_0$

Example 2.8: First-order difference equation by the classical method

Solve

$$y[n] + 2y[n-1] = 5u[n], \quad y[-1] = 1.$$

For $n \geq 0$, the homogeneous equation gives

$$r + 2 = 0 \Rightarrow r = -2,$$

so

$$y_h[n] = C_1(-2)^n.$$

Because the input is constant for $n \geq 0$, choose $y_p[n] = C$. Substitute:

$$C + 2C = 5 \Rightarrow C = \frac{5}{3}.$$

Find $y[0]$ from the recursion:

$$y[0] + 2y[-1] = 5 \Rightarrow y[0] = 3.$$

Thus

$$3 = C_1 + \frac{5}{3} \Rightarrow C_1 = \frac{4}{3}.$$

Therefore

$$y[n] = \frac{4}{3}(-2)^n + \frac{5}{3}, \quad n \geq 0.$$

Note

For difference equations, the full response can also be decomposed as

$$y[n] = y_{zi}[n] + y_{zs}[n].$$

The zero-input response is caused only by initial conditions. The zero-state response is caused only by the input under zero initial conditions. In many examples, $y_{zs}[n]$ contains both a free component and a forced component.

2.4 Differential-Equation Models**Definition**

Continuous-time physical systems are often modeled by input-output differential equations. The equations are derived from element laws and physical constraints such as Kirchhoff's current/voltage laws or Newton's laws.

Element	Differential law	Integral law
Resistor	$v(t) = Ri(t)$	-
Capacitor	$i(t) = C \frac{dv(t)}{dt}$	$v(t) = \frac{1}{C} \int_{-\infty}^t i(\lambda) d\lambda$
Inductor	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_{-\infty}^t v(\lambda) d\lambda$

Example 2.9: Series RC circuitFor a series RC circuit with input $x(t) = v(t)$ and output $y(t) = v_C(t)$,

$$Ri(t) + y(t) - x(t) = 0, \quad i(t) = C \frac{dy(t)}{dt}.$$

Substitute $i(t)$:

$$RC \frac{dy(t)}{dt} + y(t) = x(t),$$

or

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t).$$

This is a first-order linear constant-coefficient differential equation.

Mechanical element Force law for translation

Mass $x(t) = M \frac{d^2y(t)}{dt^2}$

Damper $x(t) = D \frac{dy(t)}{dt}$

Spring $x(t) = Ky(t)$

Example 2.10: Mass-spring-damper system

For a linear translational mass-spring-damper system,

$$x(t) - Ky(t) - M \frac{d^2y(t)}{dt^2} - D \frac{dy(t)}{dt} = 0.$$

Therefore

$$M \frac{d^2y(t)}{dt^2} + D \frac{dy(t)}{dt} + Ky(t) = x(t).$$

This is a second-order input-output differential equation.

Example 2.11: Motor with load

For a rotational system with torque input $T(t)$, angular position $\theta(t)$, inertia I , and damping k_d ,

$$T(t) - I \frac{d^2\theta(t)}{dt^2} - k_d \frac{d\theta(t)}{dt} = 0,$$

so

$$I \frac{d^2\theta(t)}{dt^2} + k_d \frac{d\theta(t)}{dt} = T(t).$$

2.5 Solution of Differential Equations

Definition

The general N th-order linear constant-coefficient differential equation is

$$y^{(N)}(t) + \sum_{i=0}^{N-1} a_i y^{(i)}(t) = \sum_{i=0}^M b_i x^{(i)}(t).$$

Such a system is linear, time invariant, and finite dimensional when the coefficients are constants and the system is interpreted under proper initial conditions.

Rule / Test

Common solution methods:

- i. classical mathematical method;

2. Laplace-transform method;
3. numerical method by discretizing the differential equation;
4. symbolic computation.

2.5.1 Numerical method: Euler discretization

Example 2.12: First-order Euler model

Consider

$$\frac{dy(t)}{dt} = -ay(t) + bx(t).$$

Set

$$x[n] = x(nT), \quad y[n] = y(nT),$$

and approximate

$$\left. \frac{dy(t)}{dt} \right|_{t=nT} \approx \frac{y[n+1] - y[n]}{T}.$$

Then

$$\frac{y[n+1] - y[n]}{T} = -ay[n] + bx[n],$$

so

$$y[n+1] = (1 - aT)y[n] + bTx[n].$$

Replacing n by $n - 1$ gives

$$y[n] = (1 - aT)y[n-1] + bTx[n-1].$$

If $x[n] = 0$ for all n , then

$$y[n] = (1 - aT)^n y[0].$$

The exact zero-input sample is

$$y[n] = (e^{-aT})^n y[0],$$

so Euler uses $e^{-aT} \approx 1 - aT$.

Example 2.13: Numerical RC model

For the series RC equation

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t),$$

Euler discretization gives

$$y[n] = \left(1 - \frac{T}{RC}\right)y[n-1] + \frac{T}{RC}x[n-1].$$

For a unit-step input and $R = C = 1$, the exact response is

$$y(t) = 1 - e^{-t}, \quad t \geq 0.$$

2.5.2 Classical method

Rule / Test

For the classical method,

$$y(t) = y_h(t) + y_p(t).$$

The homogeneous solution is obtained from

$$y^{(N)}(t) + \sum_{i=0}^{N-1} a_i y^{(i)}(t) = 0.$$

The characteristic equation is

$$\alpha^N + a_{N-1}\alpha^{N-1} + \cdots + a_1\alpha + a_0 = 0.$$

Root condition	Homogeneous-solution form
Distinct real roots α_i	$y_h(t) = \sum_i A_i e^{\alpha_i t}$
K -fold repeated root α	$(A_1 + A_2 t + \cdots + A_K t^{K-1}) e^{\alpha t}$
Complex roots $\alpha \pm j\beta$	$e^{\alpha t} (A_1 \cos \beta t + A_2 \sin \beta t)$
Input or free-term form	Trial particular solution
constant E	K
t^p	$K_1 t^p + K_2 t^{p-1} + \cdots + K_{p+1}$
$e^{\alpha t}$	$K e^{\alpha t}$
$\cos \omega t$ or $\sin \omega t$	$K_1 \cos \omega t + K_2 \sin \omega t$
$t^p e^{\alpha t} \cos \omega t$ or $t^p e^{\alpha t} \sin \omega t$	polynomial times $e^{\alpha t} \cos \omega t$ plus polynomial times $e^{\alpha t} \sin \omega t$

Example 2.14: Homogeneous solution with repeated root

For

$$\frac{d^3 y(t)}{dt^3} + 7 \frac{d^2 y(t)}{dt^2} + 16 \frac{dy(t)}{dt} + 12 y(t) = x(t),$$

set the right-hand side to zero. The characteristic equation is

$$\alpha^3 + 7\alpha^2 + 16\alpha + 12 = 0 = (\alpha + 2)^2(\alpha + 3).$$

Thus

$$\alpha_1 = \alpha_2 = -2, \quad \alpha_3 = -3.$$

The homogeneous solution is

$$y_h(t) = A_1 t e^{-2t} + A_2 e^{-2t} + A_3 e^{-3t}.$$

Example 2.15: Full response by the classical method

Given

$$\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6 v_2(t) = 6 \cos(2t) u(t),$$

with zero initial capacitor voltages, the characteristic equation is

$$\alpha^2 + 7\alpha + 6 = 0, \quad \alpha_1 = -1, \alpha_2 = -6.$$

For the sinusoidal input, choose

$$v_p(t) = B_1 \sin 2t + B_2 \cos 2t.$$

Substitution gives

$$B_1 = \frac{21}{50}, \quad B_2 = \frac{3}{50}.$$

Thus

$$v_2(t) = A_1 e^{-t} + A_2 e^{-6t} + \frac{21}{50} \sin 2t + \frac{3}{50} \cos 2t.$$

Using $v_2(0) = 0$ and $v_2'(0) = 0$ gives

$$A_1 = -\frac{6}{25}, \quad A_2 = \frac{9}{50}.$$

Therefore

$$v_2(t) = -\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t} + \frac{21}{50}\sin 2t + \frac{3}{50}\cos 2t.$$

2.5.3 Physical method: response decomposition

Definition

The complete response can be decomposed in two common ways:

$$y(t) = y_h(t) + y_p(t)$$

or

$$y(t) = y_{zi}(t) + y_{zs}(t).$$

Here $y_{zi}(t)$ is the zero-input response caused by initial stored energy, and $y_{zs}(t)$ is the zero-state response caused by the external input under zero initial state.

Response name	Meaning
Free response	Determined by system natural modes; corresponds to homogeneous terms.
Forced response	Determined by the input form; corresponds to the particular solution.
Transient response	Components that vanish as $t \rightarrow \infty$ in stable systems.
Steady-state response	Remaining part after transient terms vanish.
Zero-input response	Response with $x(t) = 0$, caused by initial stored energy.
Zero-state response	Response with zero initial state, caused by the input.

Example 2.16: Zero-input response

For

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = x(t),$$

with

$$y_{zi}(0^-) = 1, \quad \dot{y}_{zi}(0^-) = 2,$$

set $x(t) = 0$. The characteristic equation is

$$\alpha^2 + 3\alpha + 2 = 0, \quad \alpha_1 = -1, \alpha_2 = -2.$$

Hence

$$y_{zi}(t) = c_1 e^{-t} + c_2 e^{-2t}.$$

Use

$$c_1 + c_2 = 1, \quad -c_1 - 2c_2 = 2,$$

so

$$c_1 = 4, \quad c_2 = -3.$$

Therefore

$$y_{zi}(t) = (4e^{-t} - 3e^{-2t})u(t).$$

Example 2.17: Jump initial value by impulse balance

Given

$$\frac{dy(t)}{dt} + 3y(t) = 2\frac{dx(t)}{dt}, \quad x(t) = u(t), \quad y(0^-) = 1,$$

we have $\dot{x}(t) = \delta(t)$. Balance the impulse terms:

$$\dot{y}(t) \text{ must contain } 2\delta(t).$$

Therefore

$$y(0^+) - y(0^-) = 2.$$

Since $y(0^-) = 1$,

$$y(0^+) = 3.$$

2.6 Convolution Representation of Continuous-Time Signals

Definition

For arbitrary continuous-time signals $x(t)$ and $v(t)$, convolution is defined by

$$(x * v)(t) = \int_{-\infty}^{\infty} x(\lambda)v(t - \lambda) d\lambda.$$

If both signals are zero for $t < 0$, then

$$(x * v)(t) = \begin{cases} 0, & t < 0, \\ \int_0^t x(\lambda)v(t - \lambda) d\lambda, & t \geq 0. \end{cases}$$

Rule / Test

For a causal LTI continuous-time system with impulse response $h(t)$,

$$y_{zs}(t) = x(t) * h(t) = \int_0^t x(\tau)h(t - \tau) d\tau.$$

The complete response is

$$y(t) = y_{zi}(t) + y_{zs}(t).$$

Note

The continuous-time graphical convolution procedure is the same idea as the discrete-time procedure:

1. graph $x(\lambda)$ and $v(\lambda)$;
2. fold $v(\lambda)$ into $v(-\lambda)$;
3. shift to get $v(t - \lambda)$;
4. integrate the product $x(\lambda)v(t - \lambda)$ over the overlap interval.

Example 2.18: RC circuit impulse response and output

For the RC equation

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t),$$

with impulse input $x(t) = \delta(t)$, the impulse response is

$$h(t) = \frac{1}{RC}e^{-t/(RC)}u(t).$$

For $R = C = 1$ and a unit rectangular input of duration 1, i.e.

$$x(t) = u(t) - u(t - 1),$$

we get

$$y(t) = \int_0^t x(\lambda)e^{-(t-\lambda)} d\lambda.$$

Thus

$$y(t) = \begin{cases} 1 - e^{-t}, & 0 \leq t \leq 1, \\ e^{-t}(e - 1), & t \geq 1. \end{cases}$$

Example 2.19: Convolution of two pulses

Let

$$x(t) = u(t) - 2u(t - 1) + u(t - 2),$$

and

$$v(t) = u(t) - u(t - 1).$$

The overlap calculation gives

$$(x * v)(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \leq t \leq 1, \\ -2t + 3, & 1 \leq t \leq 2, \\ t - 3, & 2 \leq t \leq 3, \\ 0, & t \geq 3. \end{cases}$$

Equivalently,

$$\begin{aligned} (x * v)(t) &= t[u(t) - u(t - 1)] + (-2t + 3)[u(t - 1) - u(t - 2)] \\ &\quad + (t - 3)[u(t - 2) - u(t - 3)]. \end{aligned}$$

2.7 Review Tables

2.7.1 Discrete-time formulas

Object	Formula
Finite weighted-sum system	$y[n] = \sum_{i=0}^{N-1} w_i x[n-i]$
MA filter	$y[n] = \frac{1}{N} \sum_{i=0}^{N-1} x[n-i]$
EWMA filter	$y[n] = \sum_{i=0}^{N-1} a b^i x[n-i], a = (1-b)/(1-b^N)$
DT convolution	$(x * v)[n] = \sum_{i=-\infty}^{\infty} x[i] v[n-i]$
Causal DT convolution	$(x * v)[n] = \sum_{i=0}^n x[i] v[n-i]$
System output	$y[n] = y_{zi}[n] + x[n] * h[n]$
Difference equation	$y[n] + \sum_{i=1}^N a_i y[n-i] = \sum_{i=0}^M b_i x[n-i]$
Classical solution	$y[n] = y_h[n] + y_p[n]$

2.7.2 Continuous-time formulas

Object	Formula
General differential equation	$y^{(N)}(t) + \sum_{i=0}^{N-1} a_i y^{(i)}(t) = \sum_{i=0}^M b_i x^{(i)}(t)$
Classical solution	$y(t) = y_h(t) + y_p(t)$
Characteristic equation	$\alpha^N + a_{N-1} \alpha^{N-1} + \dots + a_0 = 0$
Complete response	$y(t) = y_{zi}(t) + y_{zs}(t)$
CT convolution	$(x * v)(t) = \int_{-\infty}^{\infty} x(\lambda) v(t-\lambda) d\lambda$
Causal CT convolution	$(x * v)(t) = \int_0^t x(\lambda) v(t-\lambda) d\lambda$
System zero-state response	$y_{zs}(t) = x(t) * h(t)$
Euler approximation	$\dot{y}(nT) \approx [y[n+1] - y[n]]/T$

2.7.3 Exam workflow for Chapter 2

Rule / Test

Use the following decision path:

1. If the system is DT-LTI with known $h[n]$, use convolution for zero-state response.
2. If initial conditions are nonzero, add the zero-input response or use the difference equation recursion.
3. If the system is CT and physical, first derive a differential equation from element laws.
4. For CT differential equations, choose among Euler discretization, classical solution, Laplace transform, or convolution depending on the required response type.
5. Always distinguish 0^- initial stored state from 0^+ initial condition when impulses, steps, or derivatives of impulses appear.
6. Memorize the separation: convolution gives zero-state response; recursion/classical solution gives full response when initial conditions are included.

CHAPTER 3:

The Fourier Series and Fourier Transform

Chapter 3 Map

Chapter 3 moves the course from time-domain descriptions to frequency-domain descriptions. The key idea is that signals can be represented by sinusoidal or complex-exponential components. Periodic signals use Fourier series; aperiodic signals use the Fourier transform. Sampling and modulation are direct engineering applications of these representations.

Topic	What to remember
Frequency components	A sum of sinusoids is characterized by frequencies, amplitudes, and phases.
Trigonometric Fourier series	Periodic CT signals are expanded by sine and cosine harmonics.
Complex Fourier series	Periodic CT signals are expanded by complex exponentials at integer multiples of ω_0 .
Fourier transform	Aperiodic CT signals are represented by a continuous spectrum $X(\omega)$.
Common spectra	Rectangular pulse, exponential signal, impulse, step, and sinusoids have standard transform pairs.
Properties	Linearity, shifting, scaling, modulation, differentiation, integration, convolution, multiplication, Parseval, and duality.
Sampling theorem	Bandlimited signals can be reconstructed from samples if $\omega_s \geq 2B$.
Modulation	Multiplication in time shifts or replicates spectra in frequency.

3.1 Frequency-Component Representation

Definition

A finite CT signal made of sinusoidal components can be written as

$$x(t) = \sum_{k=1}^N A_k \cos(\omega_k t + \theta_k), \quad t \in \mathbb{R}.$$

It is completely characterized by three sets:

$\{\omega_k\}$ frequencies, $\{A_k\}$ nonnegative amplitudes, $\{\theta_k\}$ phases.

Note

Frequency content tells us *which sinusoids are present*. Amplitude tells us *how strong each component is*. Phase tells us *where each component is shifted in time*. A time waveform is therefore not just a curve; it is a structured

combination of spectral components.

Example 3.1: Sum of sinusoids

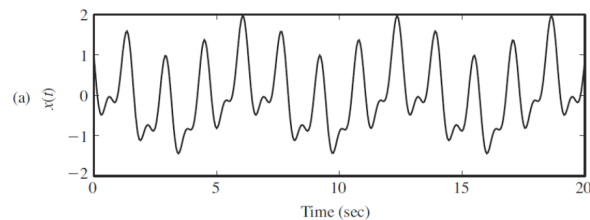
For

$$x(t) = A_1 \cos t + A_2 \cos(4t + \pi/3) + A_3 \cos(8t + \pi/2),$$

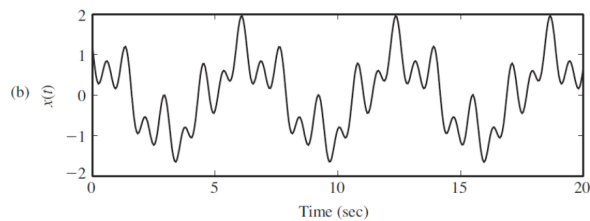
the frequency components are at 1, 4, and 8 rad/s. The amplitudes are A_1, A_2, A_3 and the phases are $0, \pi/3, \pi/2$. Changing the relative amplitudes changes the waveform shape while the component frequencies remain fixed.

Example: Sum of Sinusoids –Cont'd

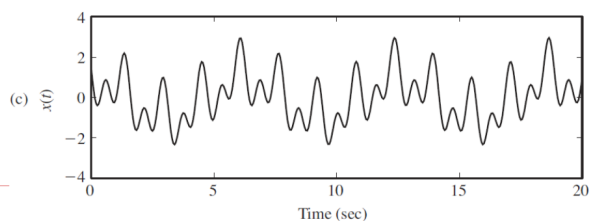
$$\begin{cases} A_1 = 0.5 \\ A_2 = 1 \\ A_3 = 0.5 \end{cases}$$



$$\begin{cases} A_1 = 1 \\ A_2 = 0.5 \\ A_3 = 0.5 \end{cases}$$



$$\begin{cases} A_1 = 1 \\ A_2 = 1 \\ A_3 = 1 \end{cases}$$



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Example figure: different amplitude choices for the same three sinusoidal frequencies produce different time-domain waveforms.

3.1.1 Amplitude and phase spectra

Definition

For a finite sum of sinusoids, an **amplitude spectrum** plots A_k versus ω_k , and a **phase spectrum** plots θ_k versus ω_k .

Amplitude Spectrum(幅值谱/幅度谱)

Plot of the amplitudes A_k of the sinusoids making up $x(t)$ vs. ω_k

Example:

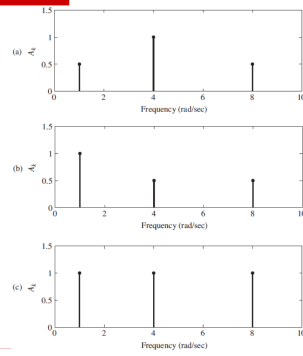


FIGURE 3.3 Amplitude spectra of the versions of $x(t)$ plotted in Figure 3.2.

Amplitude spectra.

Phase Spectrum(相位谱)

□ Plot of the phases θ_k of the sinusoids making up $x(t)$ vs. ω

□ Example:

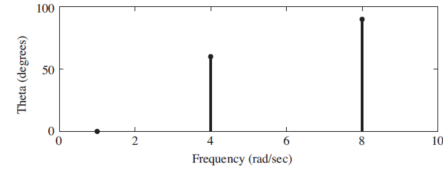


FIGURE 3.4 Phase spectrum of signal $x(t)$ defined by (3.2).

Phase spectrum.

3.2 Trigonometric Fourier Series

Definition

A CT signal $x(t)$ is periodic with period $T > 0$ if

$$x(t + T) = x(t), \quad \forall t.$$

The fundamental period is the smallest positive T satisfying the equality. The fundamental angular frequency is

$$\omega_0 = \frac{2\pi}{T}.$$

Rule / Test

If a periodic signal satisfies the Dirichlet conditions, it can be represented by the trigonometric Fourier series

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)].$$

The coefficients over any complete period are

$$a_0 = \frac{1}{T} \int_T x(t) dt,$$

$$a_k = \frac{2}{T} \int_T x(t) \cos(k\omega_0 t) dt, \quad b_k = \frac{2}{T} \int_T x(t) \sin(k\omega_0 t) dt.$$

3.2 Trigonometric Fourier Series

$$\omega_0 = 2\pi/T \quad \text{fundamental frequency (rad/sec) 基频}$$

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)], \quad -\infty < t < \infty$$

$$a_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$a_k = \frac{2}{T} \int_0^T x(t) \cos(k\omega_0 t) dt, \quad k = 1, 2, \dots$$

$$b_k = \frac{2}{T} \int_0^T x(t) \sin(k\omega_0 t) dt, \quad k = 1, 2, \dots$$

$$a_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(k\omega_0 t) dt, \quad k = 1, 2, \dots$$

a_k and b_k can be computed by integration over any full period.

The representation is called the **trigonometric Fourier series**

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Coefficient formulas for the trigonometric Fourier series.

3.2.1 Amplitude-phase form

The same series can be written as

$$x(t) = a_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k),$$

where

$$A_k = \sqrt{a_k^2 + b_k^2}, \quad a_k = A_k \cos \theta_k, \quad b_k = -A_k \sin \theta_k.$$

A common quadrant-aware definition is

$$\theta_k = \begin{cases} \tan^{-1} \left(-\frac{b_k}{a_k} \right), & a_k \geq 0, \\ \pi + \tan^{-1} \left(-\frac{b_k}{a_k} \right), & a_k < 0. \end{cases}$$

3.2.2 Dirichlet conditions

Note

A periodic signal has a Fourier series representation if, over any period, it is absolutely integrable, has only finitely many maxima/minima, and has only finitely many discontinuities. These conditions are sufficient conditions used in engineering analysis.

3.2.3 Even/odd simplifications

Rule / Test

For an even periodic signal, only cosine terms remain:

$$b_k = 0, \quad a_k = \frac{4}{T} \int_0^{T/2} x(t) \cos(k\omega_0 t) dt.$$

For an odd periodic signal, only sine terms remain:

$$a_k = 0, \quad b_k = \frac{4}{T} \int_0^{T/2} x(t) \sin(k\omega_0 t) dt.$$

3.2 Trigonometric Fourier Series

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)], -\infty < t < \infty$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad a_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(k\omega_0 t) dt, k = 1, 2, \dots$$

$$b_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(k\omega_0 t) dt, k = 1, 2, \dots$$

$x(t)$ is an **even function**



$$a_k = \frac{4}{T} \int_0^{T/2} x(t) \cos(k\omega_0 t) dt, k = 1, 2, \dots$$

$$b_k = 0, k = 1, 2, \dots$$

$x(t)$ is an **odd function**



$$a_k = 0, k = 1, 2, \dots$$

$$b_k = \frac{4}{T} \int_0^{T/2} x(t) \sin(k\omega_0 t) dt, k = 1, 2, \dots$$

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Symmetry shortcuts for Fourier-series coefficients.

Example 3.2: Rectangular pulse train

For the periodic rectangular pulse train with $T = 2$ and $\omega_0 = \pi$, the computation gives

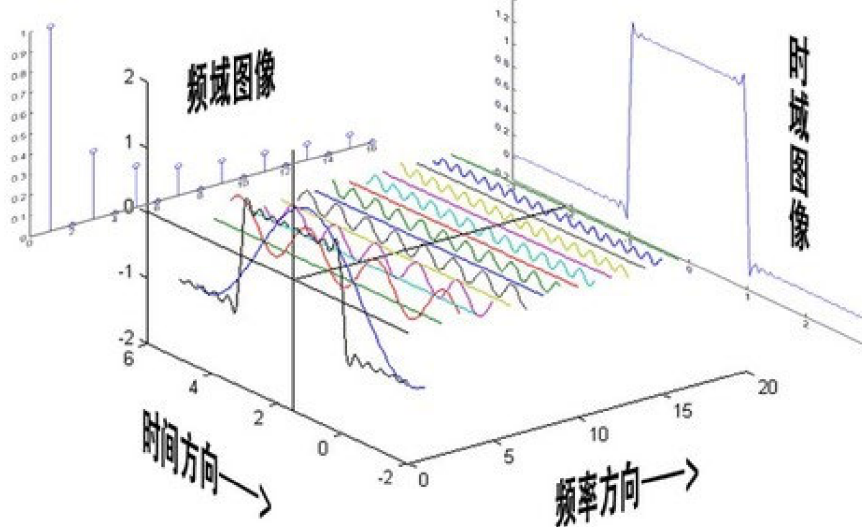
$$a_0 = \frac{1}{2},$$

$$a_k = \frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right), \quad b_k = 0.$$

Therefore

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{\substack{k=1 \\ k \text{ odd}}}^{\infty} \frac{1}{k} \sin\left(\frac{k\pi}{2}\right) \cos(k\pi t).$$

3.2 Trigonometric Fourier Series



$$x(t)_N = \frac{1}{2} + \frac{2}{\pi} \sum_{\substack{k=1 \\ k=\text{odd}}}^N \frac{1}{k} \sin\left(\frac{k\pi}{2}\right) \cos(k\pi t), -\infty < t < \infty$$

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Fourier synthesis of a rectangular pulse train: adding more harmonics improves the approximation but does not remove the jump overshoot.

3.2.4 Gibbs phenomenon and Parseval theorem

Note

At jump discontinuities, Fourier partial sums exhibit an overshoot of about 9% of the jump magnitude. This overshoot does not vanish as the number of harmonics $N \rightarrow \infty$; it only becomes narrower around the discontinuity.

For a periodic signal represented by

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t},$$

Parseval's theorem gives average power:

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2.$$

3.3 Complex Exponential Fourier Series

Definition

The complex exponential Fourier series of a periodic CT signal is

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T}.$$

The coefficients are

$$c_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt, \quad k = 0, \pm 1, \pm 2, \dots$$

Note

The complex exponential representation contains positive and negative frequencies. Negative frequencies arise algebraically from Euler's formula; for real signals they form conjugate-symmetric pairs and should not be interpreted as separate physical sinusoidal oscillations.

Relations between trigonometric coefficients and complex coefficients are

$$c_k = \frac{1}{2}(a_k - jb_k), \quad c_{-k} = \frac{1}{2}(a_k + jb_k), \quad c_0 = a_0.$$

For the amplitude-phase form,

$$|c_k| = \frac{A_k}{2}, \quad \angle c_k = \theta_k.$$

3.4 Fourier Transform

Definition

The Fourier transform of an aperiodic CT signal $x(t)$ is

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt.$$

The inverse Fourier transform is

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

We write the transform pair as

$$x(t) \longleftrightarrow X(\omega).$$

Note

Periodic signals have line spectra at a discrete set of harmonic frequencies. Aperiodic signals generally have spectra defined over a continuum of real frequency values.

3.4.1 From Fourier series to Fourier transform

The rectangular pulse example shows how a periodic pulse train tends to a single pulse as $T \rightarrow \infty$. Its line spectrum becomes denser and approaches a continuous spectrum.

Frequency Content of the Rectangular Pulse

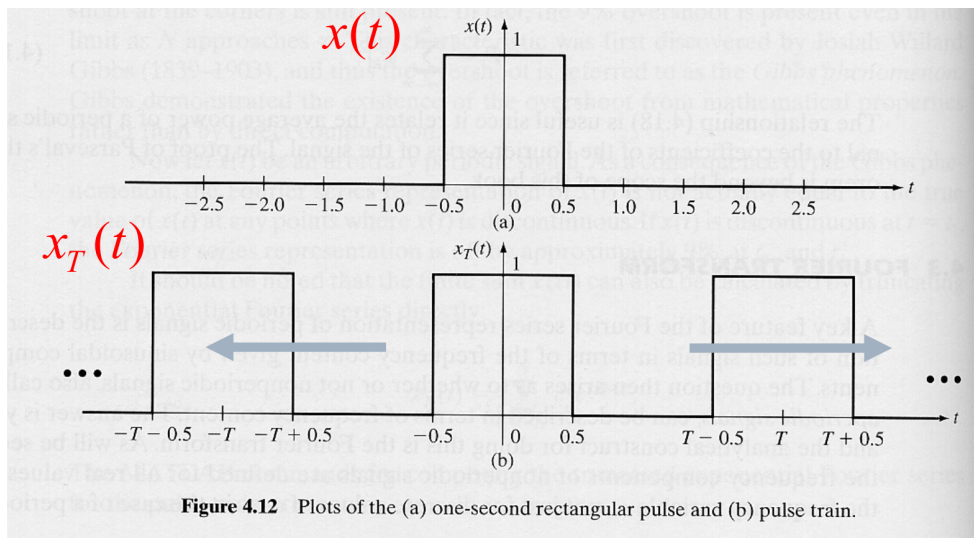


Figure 4.12 Plots of the (a) one-second rectangular pulse and (b) pulse train.

$$x(t) = \lim_{T \rightarrow \infty} x_T(t)$$

3

A single rectangular pulse as the limit of a periodic rectangular pulse train.

Frequency Content of the Rectangular Pulse – Cont'd

plots of $T |c_k|$
vs. $\omega = k\omega_0$
for $T = 2, 5, 10$

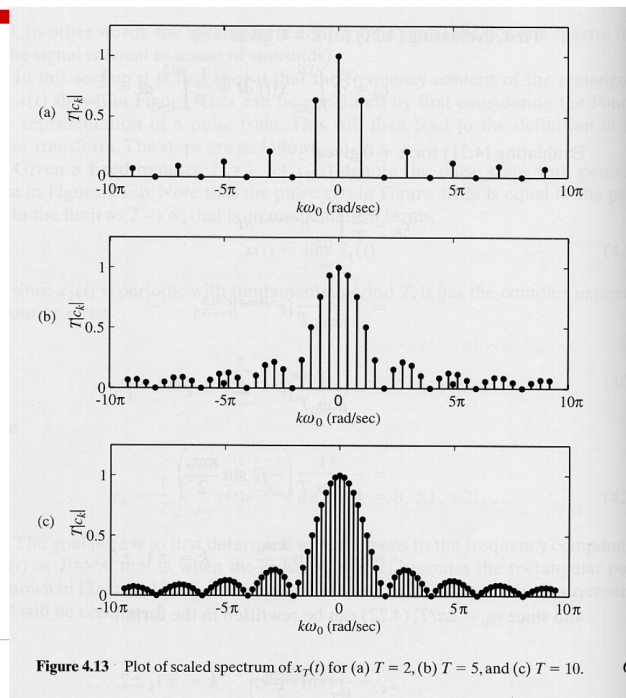


Figure 4.13 Plot of scaled spectrum of $x_T(t)$ for (a) $T = 2$, (b) $T = 5$, and (c) $T = 10$.

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As the period grows, frequency lines become denser and approximate a continuous sinc-shaped spectrum.

Example 3.3: Fourier transform of a rectangular pulse

For the even rectangular pulse

$$p_\tau(t) = \begin{cases} 1, & |t| \leq \tau/2, \\ 0, & |t| > \tau/2, \end{cases}$$

we use even symmetry:

$$X(\omega) = 2 \int_0^{\tau/2} \cos(\omega t) dt = \frac{2}{\omega} \sin\left(\frac{\omega\tau}{2}\right).$$

Thus

$$p_\tau(t) \longleftrightarrow \tau \operatorname{sinc}\left(\frac{\omega\tau}{2\pi}\right),$$

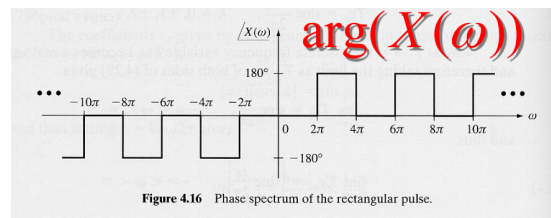
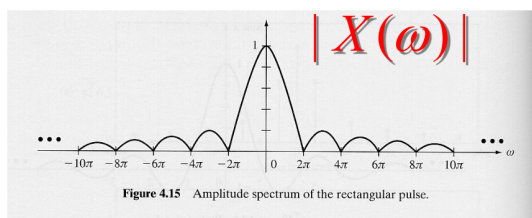
where

$$\operatorname{sinc}(\lambda) = \frac{\sin(\pi\lambda)}{\pi\lambda}.$$

Fourier Transform of the Rectangular Pulse

The Fourier transform of **the rectangular pulse** $x(t)$ is defined to be the limit of Tc_k as $T \rightarrow \infty$, i.e.,

$$X(\omega) = \lim_{T \rightarrow \infty} Tc_k = \operatorname{sinc}\left(\frac{\omega}{2\pi}\right), \quad \omega \in \mathbb{R}$$



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Fourier transform of the rectangular pulse: the amplitude is sinc-shaped and the phase alternates by sign changes.

3.4.2 Ordinary existence and bandlimited signals

Rule / Test

A signal has a Fourier transform in the ordinary sense if the defining integral converges. A standard sufficient condition is absolute integrability:

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty.$$

A signal is bandlimited if

$$X(\omega) = 0 \quad \text{for } |\omega| > B.$$

A time-limited signal cannot also be exactly bandlimited, so practical bandwidth often means that $|X(\omega)|$ is

negligible beyond a convenient B .

3.4.3 Rectangular and polar forms

Since $X(\omega)$ is generally complex,

$$X(\omega) = R(\omega) + jI(\omega),$$

where

$$R(\omega) = \int_{-\infty}^{\infty} x(t) \cos(\omega t) dt,$$

$$I(\omega) = - \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt.$$

Its polar form is

$$X(\omega) = |X(\omega)|e^{j\angle X(\omega)},$$

with

$$|X(\omega)| = \sqrt{R^2(\omega) + I^2(\omega)}, \quad \angle X(\omega) = \tan^{-1} \frac{I(\omega)}{R(\omega)}.$$

Example 3.4: Exponential signal

For

$$x(t) = e^{-bt}u(t), \quad b > 0,$$

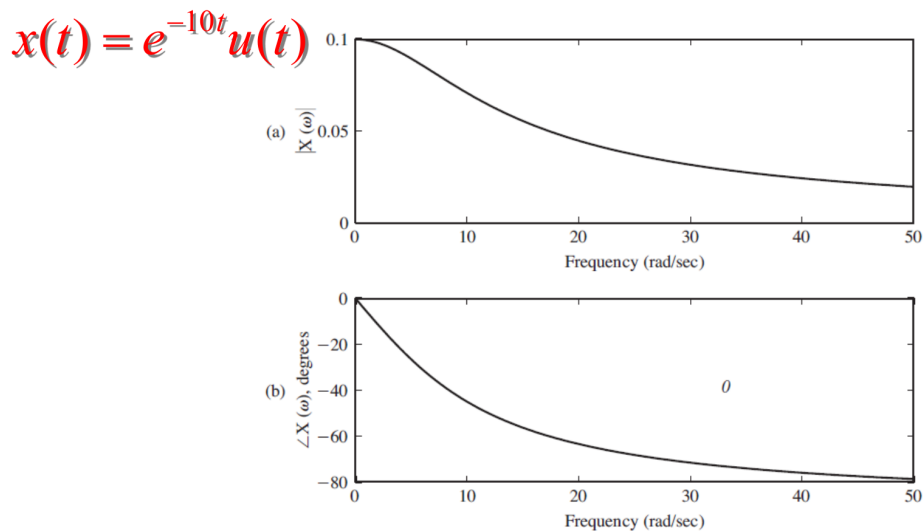
the Fourier transform is

$$X(\omega) = \int_0^{\infty} e^{-(b+j\omega)t} dt = \frac{1}{b + j\omega}.$$

Therefore

$$|X(\omega)| = \frac{1}{\sqrt{b^2 + \omega^2}}, \quad \angle X(\omega) = -\tan^{-1} \left(\frac{\omega}{b} \right).$$

Example: Amplitude and Phase Spectra of the Exponential Signal



Plots of the (a) amplitude and (b) phase spectra of $x(t) = \exp(-10t)u(t)$

3.5 Common Fourier Transform Pairs

Time-domain signal	Fourier transform
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$\delta(t - t_0)$	$e^{-j\omega t_0}$
$e^{-bt}u(t), b > 0$	$\frac{1}{b + j\omega}$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$p_\tau(t)$	$\tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

Common Fourier Transform Pairs

TABLE 4.2 COMMON FOURIER TRANSFORM PAIRS

$1, \quad -\infty < t < \infty \leftrightarrow 2\pi\delta(\omega)$
$-0.5 + u(t) \leftrightarrow \frac{1}{j\omega}$
$u(t) \leftrightarrow \pi\delta(\omega) + \frac{1}{j\omega}$
$\delta(t) \leftrightarrow 1$
$\delta(t - c) \leftrightarrow e^{-j\omega c}, \quad c \text{ any real number}$
$e^{-bt}u(t) \leftrightarrow \frac{1}{j\omega + b}, \quad b > 0$
$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0), \omega_0 \text{ any real number}$
$p_\tau(t) \leftrightarrow \tau \operatorname{sinc} \frac{\tau\omega}{2\pi}$
$\tau \operatorname{sinc} \frac{\tau t}{2\pi} \leftrightarrow 2\pi p_\tau(\omega)$
$\left(1 - \frac{2 t }{\tau}\right)p_\tau(t) \leftrightarrow \frac{\tau}{2} \operatorname{sinc}^2\left(\frac{\tau\omega}{4\pi}\right)$
$\frac{\tau}{2} \operatorname{sinc}^2\left(\frac{\tau t}{4\pi}\right) \leftrightarrow 2\pi \left(1 - \frac{2 \omega }{\tau}\right)p_\tau(\omega)$
$\cos \omega_0 t \leftrightarrow \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$\cos(\omega_0 t + \theta) \leftrightarrow \pi[e^{-j\theta}\delta(\omega + \omega_0) + e^{j\theta}\delta(\omega - \omega_0)]$
$\sin \omega_0 t \leftrightarrow j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$\sin(\omega_0 t + \theta) \leftrightarrow j\pi[e^{-j\theta}\delta(\omega + \omega_0) - e^{j\theta}\delta(\omega - \omega_0)]$

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Reference table of common CT Fourier transform pairs.

3.6 Properties of the Fourier Transform

Property	Transform relation
Linearity	$ax(t) + by(t) \leftrightarrow aX(\omega) + bY(\omega)$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$
Time scaling	$x(at) \leftrightarrow \frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time reversal	$x(-t) \leftrightarrow X(-\omega)$
Frequency shift	$x(t)e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$
Multiplication by cosine	$x(t) \cos \omega_0 t \leftrightarrow \frac{1}{2}[X(\omega - \omega_0) + X(\omega + \omega_0)]$
Time differentiation	$\frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$
Time integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Convolution in time	$x(t) * y(t) \leftrightarrow X(\omega)Y(\omega)$
Multiplication in time	$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(\omega) * Y(\omega)$
Parseval	$\int x(t) ^2 dt = \frac{1}{2\pi} \int X(\omega) ^2 d\omega$

Chapter Summary

FT Properties

Linearity ^o	$ax(t) + bv(t) \leftrightarrow aX(\omega) + bV(\omega)$
Time Shifting ^o	$x(t - c) \leftrightarrow e^{-j\omega c} X(\omega)$
Time Scaling ^o	$x(at) \leftrightarrow X(\omega/a)/ a \quad a > 0$
Time Reversal ^o	$x(-t) \leftrightarrow X(-\omega) = \overline{X(\omega)}$
Multiplication by a power of t/n ^o	$t^n x(t) \leftrightarrow j^n (d^n X(\omega)/d\omega^n) \quad n = 1, 2$
Multiplication by a complex exponential ^o	$e^{j\omega_0 t} x(t) \leftrightarrow X(\omega - \omega_0) \quad \omega_0 \text{ real}$
Multiplication by $\cos(\omega_0 t)$ ^o	$x(t) \cos(\omega_0 t) \leftrightarrow (1/2)[X(\omega + \omega_0) + X(\omega - \omega_0)]$
Multiplication by $\sin(\omega_0 t)$ ^o	$x(t) \sin(\omega_0 t) \leftrightarrow (j/2)[X(\omega + \omega_0) - X(\omega - \omega_0)]$
Differentiation in the time domain ^o	$d^n/dt^n x(t) \leftrightarrow (j\omega)^n X(\omega)$
Integration in the time domain/Summation ^o	$\int_{-\infty}^t x(\lambda) d\lambda, x(-\infty) = 0 \leftrightarrow \frac{X(j\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Convolution in the time domain ^o	$x(t) * v(t) \leftrightarrow X(\omega)V(\omega)$
Multiplication in the time domain ^o	$x(t)v(t) \leftrightarrow \frac{1}{2\pi} X(\omega) * V(\omega)$
Parseval's theorem ^o	$\int_{-\infty}^{\infty} x(t)v(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\overline{V(\omega)}d\omega$ $\int_{-\infty}^{\infty} x^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$
Duality ^o	$X(t) \leftrightarrow 2\pi x(-\omega)$

Summary table of Fourier-transform properties.

3.7 Generalized Fourier Transform

Some important signals are not absolutely integrable, but are still used in systems analysis through generalized Fourier transforms.

Rule / Test

Key generalized transform pairs:

$$\delta(t) \longleftrightarrow 1, \quad 1 \longleftrightarrow 2\pi\delta(\omega).$$

For sinusoids,

$$\cos(\omega_0 t) \longleftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)],$$

$$\sin(\omega_0 t) \longleftrightarrow j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)].$$

Generalized Fourier Transform of Sinusoidal Signals

$$\cos(\omega_0 t) \longleftrightarrow \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

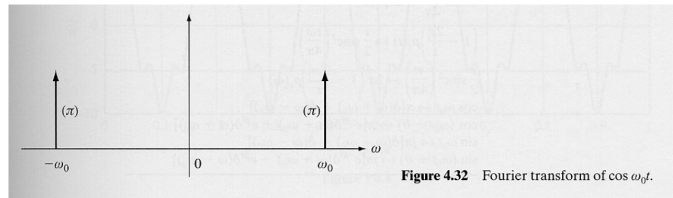


Figure 4.32 Fourier transform of $\cos \omega_0 t$.

$$\sin(\omega_0 t) \longleftrightarrow j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

3

Generalized Fourier transforms of sinusoidal signals: sinusoidal components appear as impulses at positive and negative frequencies.

For a periodic signal

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t},$$

its Fourier transform is a line spectrum:

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0).$$

3.8 Sampling and Sampling Theorem

3.8.1 Impulse-train sampling

Let the sampling period be T and the sampling angular frequency be

$$\omega_s = \frac{2\pi}{T}.$$

The impulse train is

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$

The sampled signal is

$$x_s(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT).$$

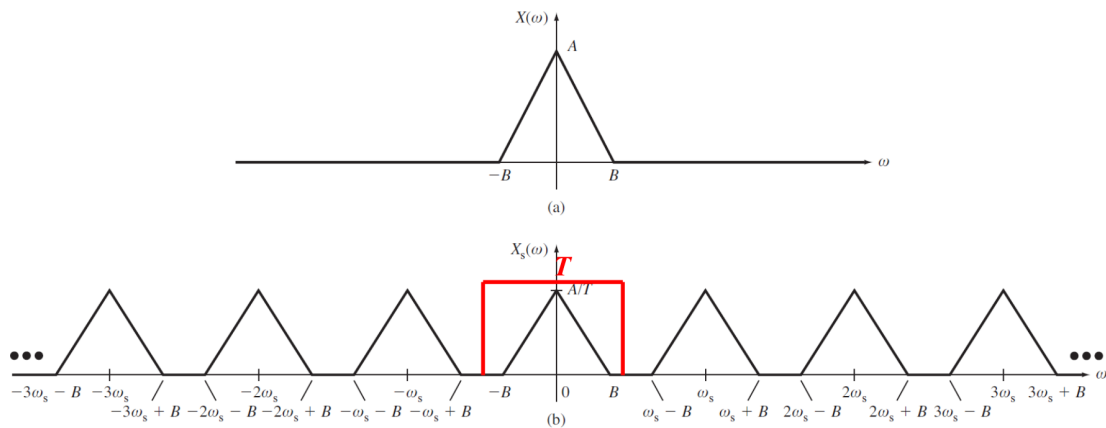
In the frequency domain,

$$X_s(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

Sampling therefore creates shifted spectral replicas spaced by ω_s .

3.8 Sampling and Sampling Theorem

$$x(t)p(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T} x(nT) e^{jk\omega_s t} \leftrightarrow X_s(\omega) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(\omega - k\omega_s)$$



Sampling creates repeated copies of the original spectrum at integer multiples of the sampling frequency.

3.8.2 Sampling theorem

Rule / Test

Suppose $x(t)$ is bandlimited with bandwidth B :

$$X(\omega) = 0 \quad \text{for } |\omega| > B.$$

If

$$\omega_s \geq 2B,$$

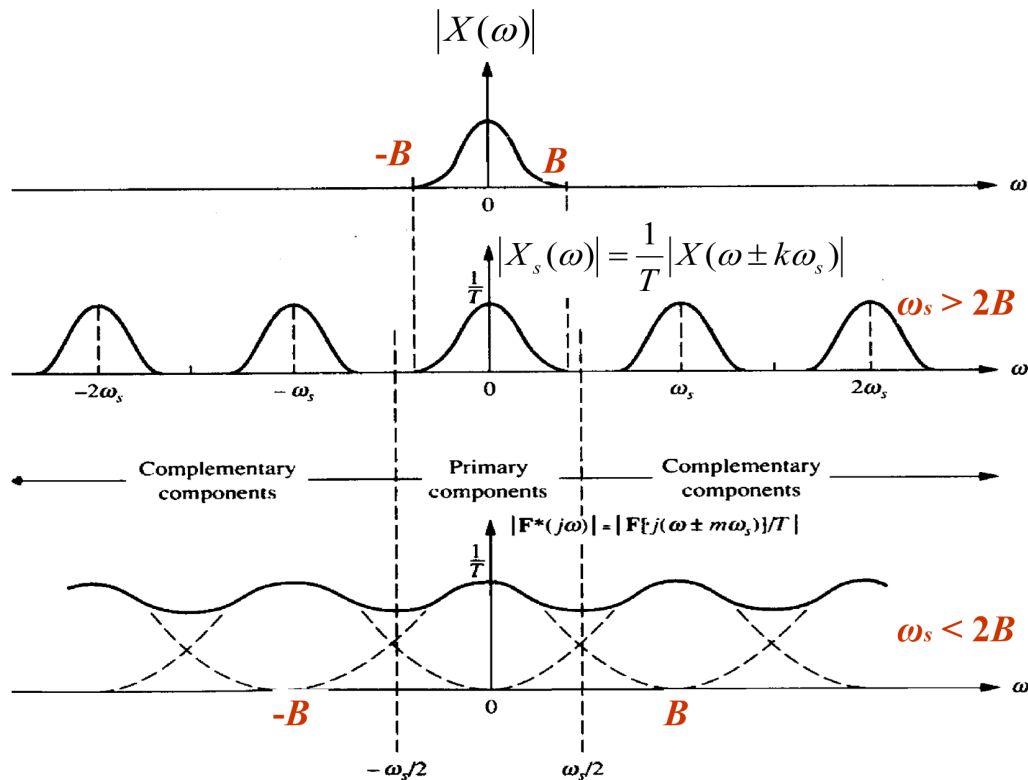
then the shifted spectra do not overlap. An ideal low-pass filter can pass the central copy and reject the others, so $x(t)$ can be completely reconstructed from its samples.

Note

The minimum sampling angular frequency is the Nyquist rate:

$$\omega_s = 2B.$$

If $\omega_s < 2B$, spectral replicas overlap. This is **aliasing**; after reconstruction, high-frequency components can appear as false low-frequency components.



Sampling above Nyquist keeps spectral copies separated; sampling below Nyquist causes aliasing by spectral overlap.

3.8.3 Interpolation formula

For ideal reconstruction, the low-pass interpolation filter has

$$H(\omega) = \begin{cases} T, & |\omega| \leq B, \\ 0, & \text{otherwise.} \end{cases}$$

The corresponding impulse response is sinc-shaped, and the reconstructed signal can be written as a sinc interpolation sum:

$$x(t) = \frac{BT}{\pi} \sum_{n=-\infty}^{\infty} x(nT) \operatorname{sinc}\left(\frac{B}{\pi}(t - nT)\right).$$

3.8 Sampling and Sampling Theorem

Interpolation Formula

$$H(\omega) = \begin{cases} T & -B \leq \omega \leq B \\ 0 & \text{all other } \omega \end{cases} \quad \Rightarrow \quad h(t) = \frac{BT}{\pi} \text{sinc}\left(\frac{B}{\pi}t\right)$$

The output $y(t)$ of the interpolating filter is given by

$$y(t) = h(t) * x_s(t) = \int_{-\infty}^{\infty} x_s(\tau) h(t - \tau) d\tau$$

$$\text{where } x_s(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(t)\delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$$

$$y(t) = h(t) * x_s(t) = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x(nT)\delta(\tau - nT)h(t - \tau) d\tau$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(nT)\delta(\tau - nT)h(t - \tau) d\tau = \sum_{n=-\infty}^{\infty} x(nT)h(t - nT)$$

$$= \frac{BT}{\pi} \sum_{n=-\infty}^{\infty} x(nT) \text{sinc}\left(\frac{B}{\pi}(t - nT)\right)$$

But $y(t) = x(t)$



$$x(t) = \frac{BT}{\pi} \sum_{n=-\infty}^{\infty} x(nT) \text{sinc}\left(\frac{B}{\pi}(t - nT)\right)$$

Sinc interpolation reconstructs a bandlimited signal from ideal samples.

3.9 Application: Modulation and Demodulation

3.9.1 Frequency shifting by modulation

Multiplication by a sinusoidal carrier shifts spectra. If

$$s(t) = Ax(t) \cos(\omega_c t),$$

then

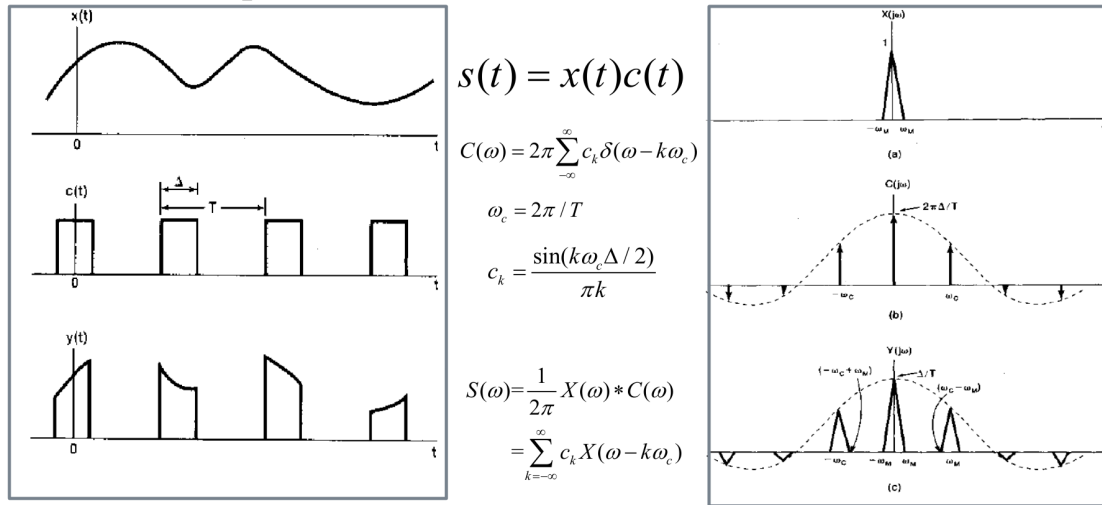
$$S(\omega) = \frac{A}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)].$$

This is the basic spectral mechanism behind amplitude modulation.

Modulation type	Formula
AM	$s(t) = Ax(t) \cos(\omega_c t)$
Alternative AM	$s(t) = A[1 + kx(t)] \cos(\omega_c t)$
PM	$s(t) = A \cos[\omega_c t + k_p x(t)]$
FM	$s(t) = A \cos\left[\omega_c t + 2\pi k_f \int_0^t x(\tau) d\tau\right]$
PAM	$s(t) = x(t)c(t)$, where $c(t)$ is a periodic pulse carrier

3.9 Application to Signal Modulation Demodulation

Pulse-Amplitude Modulation-PAM脉冲幅度调制



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Pulse-amplitude modulation: multiplication by a pulse train produces shifted and weighted spectral copies.

Chapter Summary

Signal Modulation Demodulation

Amplitude Modulation-AM $c(t) = A \cos(\omega_c t)$ $s(t) = Ax(t) \cos(\omega_c t)$

$$S(\omega) = \frac{A}{2} [X(\omega + \omega_c) + X(\omega - \omega_c)]$$

Alternative Form of AM $s(t) = A[1 + kx(t)] \cos \omega_c t$

$$S(\omega) = \pi A [\delta(\omega + \omega_c) + \delta(\omega - \omega_c)] + \frac{Ak}{2} [X(\omega + \omega_c) + X(\omega - \omega_c)]$$

Phase Modulation-PM $s(t) = A \cos[\omega_c t + k_p x(t)]$

Frequency Modulation-FM $s(t) = A \cos[\omega_c t + 2\pi k_f \int_0^t x(\tau) d\tau]$

Pulse-Amplitude Modulation-PAM

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Chapter summary figure for AM, PM, FM, and PAM modulation formulas.

3.10 Review Table

Concept	Core formula or idea
Frequency components	$x(t) = \sum A_k \cos(\omega_k t + \theta_k)$
Periodic condition	$x(t + T) = x(t), \omega_0 = 2\pi/T$
Trigonometric FS	$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)]$
Complex FS	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$
Fourier transform	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
Inverse FT	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$
Rectangular pulse	$p_\tau(t) \leftrightarrow \tau \text{sinc}(\tau\omega/2\pi)$
Impulse	$\delta(t) \leftrightarrow 1$
Constant	$1 \leftrightarrow 2\pi\delta(\omega)$
Sinusoid	$\cos \omega_0 t \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
Sampling	$X_s(\omega) = \frac{1}{T} \sum X(\omega - k\omega_s)$
Sampling theorem	$\omega_s \geq 2B$
Convolution property	$x(t) * y(t) \leftrightarrow X(\omega)Y(\omega)$
Multiplication property	$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(\omega) * Y(\omega)$

CHAPTER 4:

Fourier Analysis of Discrete-Time Signals

Chapter 4 Map

Chapter 4 moves the frequency-domain tools from continuous-time signals to discrete-time signals. The main sequence is

$$\text{CTFT} \rightarrow \text{DTFT} \rightarrow \text{DFT} \rightarrow \text{FFT}.$$

The practical idea is simple: the DTFT is exact but continuous in frequency; the DFT samples frequency and is directly computable; the FFT is a fast algorithm for computing the DFT.

Topic	What to remember
DTFT	Frequency representation of an aperiodic discrete-time signal; periodic in frequency with period 2π .
DFT	Finite set of uniformly spaced samples of a DTFT, normally taken over one period.
Truncation	Multiplication by a finite window in time causes convolution in frequency; this produces spectral spreading.
DFT properties	Circular time shift, circular frequency shift, time reversal, circular convolution, and circular multiplication.
FFT	A fast computational method for the DFT based on symmetry and periodicity of twiddle factors.

4.1 Discrete-Time Fourier Transform

4.1.1 From CTFT to DTFT

A sampled continuous-time signal is obtained from

$$x[n] = x_c(nT_s),$$

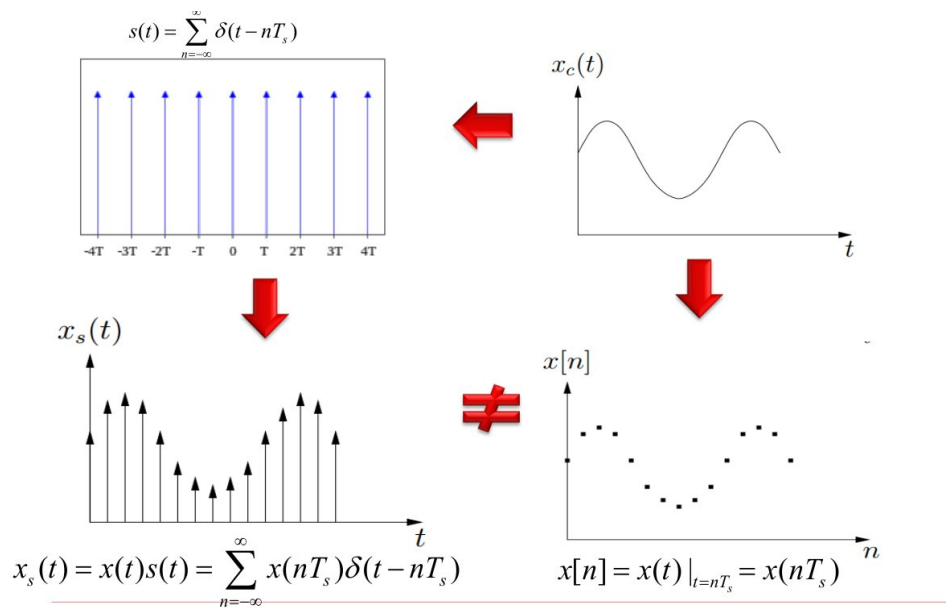
where T_s is the sampling interval. The normalized angular frequency of the discrete-time signal is

$$\Omega = \omega T_s.$$

Therefore, Ω is dimensionless and measured in radians/sample, while ω is measured in radians/second.

Note

Do not confuse the sampled impulse train $x_s(t)$ with the discrete-time sequence $x[n]$. The former is still a continuous-time signal made of impulses; the latter is a sequence indexed by integers.



Impulse-train sampling produces a continuous-time sampled signal; conversion then gives the discrete-time sequence.

4.1.2 Definition and inverse transform

Definition

For a discrete-time signal $x[n]$, the DTFT is

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}.$$

The inverse DTFT is

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega)e^{j\Omega n} d\Omega.$$

The DTFT is a continuous function of the frequency variable Ω , but its input signal is discrete in time. This is the first major distinction from the DFT.

4.1.3 Periodicity

Rule / Test

The DTFT is always periodic in frequency:

$$X(\Omega + 2\pi) = X(\Omega).$$

Thus one complete frequency interval, such as $-\pi \leq \Omega < \pi$ or $0 \leq \Omega < 2\pi$, determines the whole DTFT.

The highest distinct discrete-time angular frequency is $\Omega = \pi$. Frequencies outside one 2π interval are repetitions, not new spectral content.

DTFT of $x[n]$ is a periodic function of Ω with period 2π

$$X(\Omega + 2\pi) = X(\Omega)$$

An important consequence of periodicity of $X(\Omega)$ is that $X(\Omega)$ is completely determined by computing $X(\Omega)$ over any 2π interval such as $0 < \Omega < 2\pi$ or $-\pi < \Omega < \pi$

The DTFT repeats every 2π in the normalized frequency variable Ω .

4.1.4 Rectangular and polar forms

The DTFT is generally complex:

$$X(\Omega) = R(\Omega) + jI(\Omega).$$

For real-valued sequences,

$$|X(-\Omega)| = |X(\Omega)|, \quad \angle X(-\Omega) = -\angle X(\Omega).$$

Thus the magnitude spectrum is even and the phase spectrum is odd.

Sequence symmetry	DTFT structure
$x[n]$ real and even	$X(\Omega)$ is real and even.
$x[n]$ real and odd	$X(\Omega)$ is imaginary and odd.
General real $x[n]$	$X(-\Omega) = X^*(\Omega)$, so magnitude is even and phase is odd.

4.1.5 Example: finite exponential sequence

For

$$x[n] = \begin{cases} a^n, & 0 \leq n \leq q, \\ 0, & \text{otherwise,} \end{cases}$$

its DTFT is a finite geometric sum:

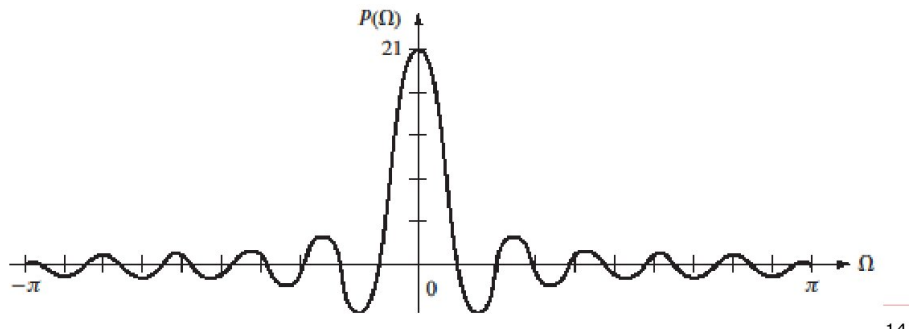
$$X(\Omega) = \sum_{n=0}^q a^n e^{-j\Omega n} = \frac{1 - (ae^{-j\Omega})^{q+1}}{1 - ae^{-j\Omega}}.$$

This example is important because time-limited sequences always have an ordinary DTFT.

Example 4.3 DTFT of Rectangular Pulse

Given a positive integer q , let $p[n]$ denote the discrete-time rectangular pulse function defined by

$$p[n] = \begin{cases} 1, & n = -q, -q+1, \dots, -1, 0, \dots, q \\ 0, & \text{all other } n \end{cases}$$



DTFT of a discrete-time rectangular pulse. The sinc-like shape appears because a finite block in time spreads in frequency.

4.1.6 Low-frequency and high-frequency examples

For

$$x[n] = (0.5)^n u[n],$$

the spectrum is concentrated near $\Omega = 0$, so it is a low-frequency sequence. For

$$x[n] = (-0.5)^n u[n],$$

the alternating sign shifts the spectral concentration toward high normalized frequencies near $\Omega = \pi$.

2. Spectrum of a discrete-time signal

Example 4.4 Signal with Low-Frequency Components

Consider the discrete-time signal $x[n] = (0.5)^n u[n]$

which is plotted in Figure 4.1.

$$X(\Omega) = \frac{1}{1 - 0.5e^{-j\Omega}}$$

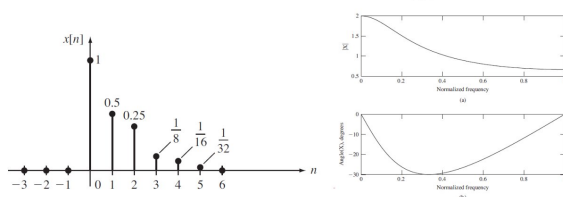


Figure 1: Low-frequency discrete-time signal.

2. Spectrum of a discrete-time signal

Example 4.5 Signal with High-Frequency Components

Consider the discrete-time signal $x[n] = (-0.5)^n u[n]$

Its Fourier Transform is

$$X(\Omega) = \frac{1}{1 + 0.5e^{-j\Omega}}$$

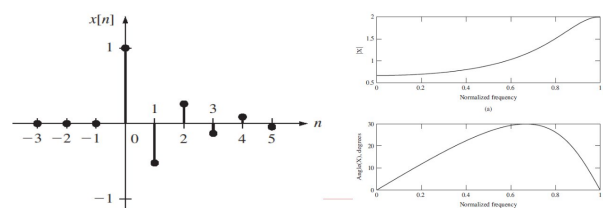


Figure 2: High-frequency discrete-time signal.

4.1.7 CTFT and DTFT property comparison

Property	CTFT	DTFT
Transform	$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$	$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$
Inverse	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega)e^{j\Omega n} d\Omega$
Frequency domain	Continuous and generally aperiodic	Continuous and always 2π -periodic
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$	$x[n - n_0] \leftrightarrow e^{-j\Omega n_0} X(\Omega)$
Convolution	$x(t) * h(t) \leftrightarrow X(\omega)H(\omega)$	$x[n] * h[n] \leftrightarrow X(\Omega)H(\Omega)$

4.2 Discrete Fourier Transform

4.2.1 Why the DFT is needed

The DTFT is a function of a continuous frequency variable Ω , so it cannot be stored directly in a computer using a finite number of values. The DFT solves this by evaluating the frequency representation at a finite number of equally spaced frequencies.

Definition

For a length- N sequence $x[n]$, the N -point DFT is

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

The inverse DFT is

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N-1.$$

This leads to the concept of the discrete Fourier transform. The N -point discrete Fourier transform (DFT) X_k of $x[n]$ is defined by

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \quad k = 0, 1, \dots, N-1$$

X_k is a function of the discrete integer variable k .

X_k , in general, is complex number

Definition of the N -point DFT. The index k is discrete.

4.2.2 DFT as sampled DTFT

For a finite-duration discrete-time signal, the DFT samples its DTFT at

$$\Omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1.$$

Thus

$$X_k = X(\Omega) \Big|_{\Omega=2\pi k/N}.$$

The DFT can therefore be viewed as frequency-domain discretization of the DTFT.

Sampling By Impulse Train

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \leftrightarrow S(\omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T_s})$$

$$\begin{aligned} x_s(t) &= x(t)s(t) = \sum_{n=0}^{N-1} x(nT_s)\delta(t - nT_s) \\ X_s(\omega) &= \sum_{n=0}^{N-1} x(nT_s) \int_{-\infty}^{\infty} \delta(t - nT_s) e^{-j\omega t} dt \\ &= \sum_{n=0}^{N-1} x(nT_s) e^{-j\omega nT_s} = \sum_{n=0}^{N-1} x(nT/N) e^{-j\omega nT/N} \end{aligned}$$

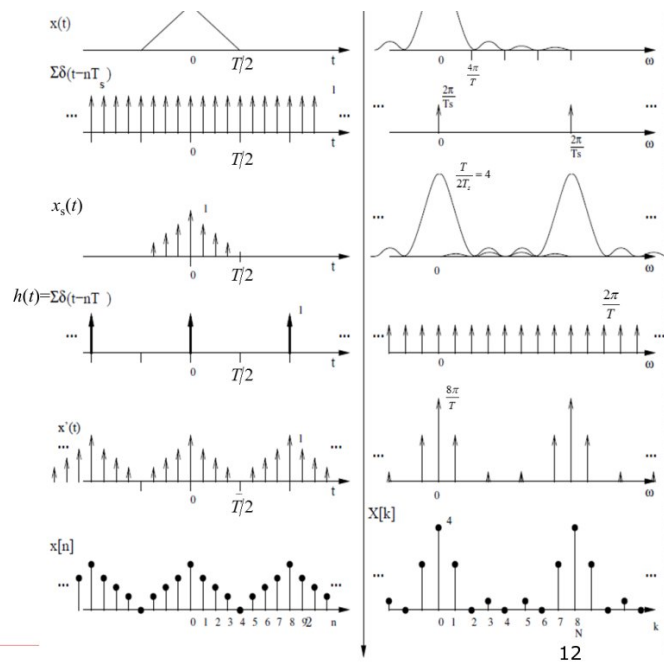
Periodic Extension(周期延拓)

$$x'(t) = x_s(t) * h(t) = x_s(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$\begin{aligned} X'(\omega) &= X_s(\omega)H(\omega) \\ &= \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T}) \sum_{n=0}^{N-1} x(nT/N) e^{-j\omega nT/N} \\ &= \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T}) \left(\sum_{n=0}^{N-1} x(nT_s) e^{-j\frac{2\pi kn}{N}} \right) \end{aligned}$$

$$X_k = \frac{T}{2\pi} \text{area}[X'(\frac{2\pi k}{T})] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}}$$

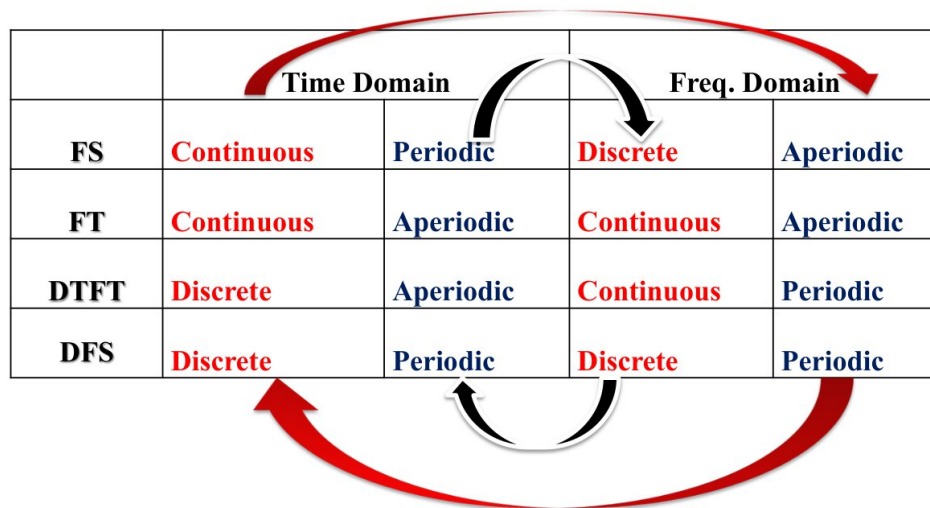
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi kn}{N}}$$



DFT construction: finite sampling in one domain corresponds to periodic extension in the other domain.

4.2.3 Transform-domain classification

Transform	Time domain		Frequency domain	
FS	Continuous	Periodic	Discrete	Aperiodic
FT	Continuous	Aperiodic	Continuous	Aperiodic
DTFT	Discrete	Aperiodic	Continuous	Periodic
DFS	Discrete	Periodic	Discrete	Periodic



4.2.4 Polar and rectangular forms

The DFT value X_k is generally complex. It can be written as

$$X_k = R_k + jI_k$$

or

$$X_k = |X_k|e^{j\angle X_k}.$$

For real-valued $x[n]$, the DFT has conjugate symmetry:

$$X_{N-k} = X_k^*.$$

Therefore, $|X_k|$ is symmetric about $k = N/2$, and the phase has odd symmetry.

4.2.5 Example: four-point DFT

For

$$x[0] = 1, x[1] = 2, x[2] = 2, x[3] = 1,$$

the four-point DFT is computed by

$$X_k = \sum_{n=0}^3 x[n]e^{-j\pi kn/2}, \quad k = 0, 1, 2, 3.$$

The result is

$$X_0 = 6, \quad X_1 = -1 - j, \quad X_2 = 0, \quad X_3 = -1 + j.$$

Hence

$$R_k = \begin{cases} 6, & k = 0 \\ -1, & k = 1 \\ 0, & k = 2 \\ 1, & k = 3 \end{cases}$$

$$I_k = \begin{cases} 0, & k = 0 \\ -1, & k = 1 \\ 0, & k = 2 \\ 1, & k = 3 \end{cases}$$

$$X_k = \begin{cases} 6, & k = 0 \\ -1 - j, & k = 1 \\ 0, & k = 2 \\ -1 + j, & k = 3 \end{cases}$$

the
rectangular
form



$$X_k = \begin{cases} 6e^{j0}, & k = 0 \\ \sqrt{2}e^{j5\pi/4}, & k = 1 \\ 0e^{j0}, & k = 2 \\ \sqrt{2}e^{j3\pi/4}, & k = 3 \end{cases}$$

the polar
form

Example of writing DFT results in rectangular and polar form.

4.3 DFT of Truncated Signals

4.3.1 Finite observation window

In practice, only a finite segment of a discrete-time signal is usually known:

$$x[0], x[1], \dots, x[N-1].$$

The truncated sequence is

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N-1, \\ 0, & \text{otherwise.} \end{cases}$$

The N -point DFT is computed from this finite sequence.

In most applications, a discrete-time signal $x[n]$ would be known over only a **finite interval of time**:

That is, only the value $x[n]$, $x[1]$, ..., $x[N-1]$ would be known.

The N -point DFT is computed based on the values $x[0]$, $x[1]$, ..., $x[N-1]$, the **N -point DFT** of the truncated signal.

$$\tilde{x}[n] = \begin{cases} x[n] & n = 0, 1, \dots, N-1 \\ 0 & n < 0 \text{ or } n \geq N \end{cases}$$

A truncated signal keeps only N known samples and sets the rest to zero.

4.3.2 Time truncation means frequency convolution

A finite observation window multiplies the original signal in time:

$$\tilde{x}[n] = x[n]p[n].$$

By the multiplication property, this corresponds to convolution in frequency. Hence a sharp line spectrum becomes a broadened spectrum shaped by the DTFT of the rectangular window.

Note

The main practical consequence is **spectral leakage**: energy that ideally belongs to one frequency spreads into nearby DFT bins when the sinusoid frequency is not exactly aligned with the DFT grid.

Example DFT of Truncated Sinusoid

$$\begin{aligned}
 x[n] &= \cos \Omega_0 n, -\infty < n < \infty \\
 &\downarrow \text{DTFT} \\
 \sum_{i=-\infty}^{\infty} \pi [\delta(\Omega + \Omega_0 - 2\pi i) + \delta(\Omega - \Omega_0 - 2\pi i)] \\
 \tilde{x}[n] &= (\cos \Omega_0 n) p\left[n - \frac{N-1}{2}\right], 0 \leq \Omega_0 \leq \pi \\
 \tilde{x}[n] &= \begin{cases} \cos \Omega_0 n, & n = 0, 1, 2, \dots, N-1 \\ 0, & \text{all other } n \end{cases} \\
 &\downarrow \text{DTFT} \\
 P(\Omega) &= \frac{\sin[N\Omega/2]}{\sin(\Omega/2)} e^{-j(N-1)\Omega/2} \\
 \tilde{X}(\Omega) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} P(\Omega - \lambda) \pi [\delta(\lambda + \Omega_0) + \delta(\lambda - \Omega_0)] d\lambda
 \end{aligned}$$

shifted rectangular pulse

Truncating a sinusoid multiplies it by a rectangular window, so its spectrum becomes the shifted window spectrum.

4.3.3 Coherent sampling

If

$$\Omega_0 = \frac{2\pi r}{N}$$

for an integer r , the sinusoid frequency lands exactly on a DFT bin. Then the DFT has nonzero samples only at the corresponding positive and negative frequency bins:

$$k = r \quad \text{and} \quad k = N - r.$$

If this condition is not met, the DFT samples the sidelobes of the window spectrum and leakage appears.

Finally, the DFT $\tilde{X}_k$ is given by

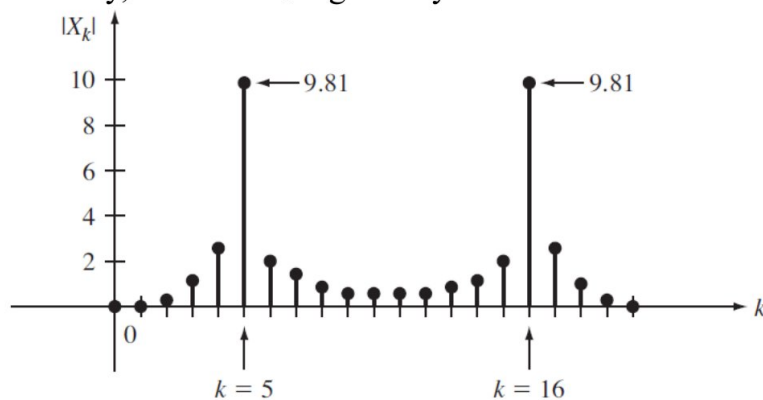


FIGURE 4.20

DFT $\tilde{X}_k$ of the truncated signal $\tilde{x}[n] = (\cos \Omega_0 n) p\left[n - \frac{N-1}{2}\right]$ when $N = 21$ and $\Omega_0 = 9.5\pi/21$.



When the sinusoid frequency is exactly on the DFT grid, the DFT energy is concentrated at two bins.

4.4 DFT Properties and Circular Convolution

4.4.1 Linearity and circular time shift

Rule / Test

Linearity:

$$\text{DFT}\{a_1x_1[n] + a_2x_2[n]\} = a_1X_1[k] + a_2X_2[k].$$

For signals of different lengths, the effective DFT length is chosen after zero-padding to a common length.

Circular time shift is defined modulo N :

$$x[n - q, \text{mod } N].$$

Its DFT is

$$x[n - q, \text{mod } N] \leftrightarrow W_N^{qk} X[k], \quad W_N = e^{-j2\pi/N}.$$

2. Circular Time - shift

Define a time shift (mod N) as : $x[n - q, \text{mod } N]$:

$$x[n - q, \text{mod } N] = \begin{cases} x[N - q], & n = 0 \\ x[N - q + 1], & n = 1 \\ \vdots & \\ x[N - 1], & n = q - 1 \\ x[0], & n = q \\ x[1], & n = q + 1 \\ \vdots & \\ x[N - q - 1], & n = N - 1 \end{cases}$$

$$x[n - N, \text{mod } N] = x[n];$$

$$x[n - q, \text{mod } N] = x[n - q \pm Nr, \text{mod } N] \quad r \text{ is any integer}$$

Circular time shift wraps samples around the length- N interval.

4.4.2 Time reversal and circular frequency shift

Circular time reversal is

$$x[-n, \text{mod } N] \leftrightarrow X[-k, \text{mod } N].$$

Multiplication by a complex exponential shifts the DFT circularly in frequency:

$$x[n]e^{j2\pi qn/N} \leftrightarrow X[k - q, \text{mod } N].$$

This is the DFT version of modulation and frequency translation.

4.4.3 Linear convolution

For ordinary linear convolution,

$$y[n] = x[n] * h[n] = \sum_m x[m]h[n - m].$$

If $x[n]$ has length L and $h[n]$ has length M , then the linear convolution result has length

$$L + M - 1.$$

5. Circular Convolution

$$x[n] \circledast_N v[n] = \sum_{i=0}^{N-1} x[i]v[n-i, \text{mod } N]$$

区别: Linear Convolution $x[n] * v[n] = \sum_{i=-\infty}^{\infty} x[i]v[n-i]$

or $x[n] * v[n] = \sum_{i=0}^{N-1} x[i]v[n-i]$

(If both $x[n]$ and $v[n]$ are zero for $n < 0$ and $n > M$)
Linear convolution slides one sequence across the other without modulo wrap-around.

4.4.4 Circular convolution

Length- N circular convolution is

$$y[n] = x[n] \circledast_N h[n] = \sum_{m=0}^{N-1} x[m]h[n-m, \text{mod } N].$$

Its DFT property is

$$x[n] \circledast_N h[n] \leftrightarrow X[k]H[k].$$

Conversely,

$$x[n]h[n] \leftrightarrow \frac{1}{N}X[k] \circledast_N H[k].$$

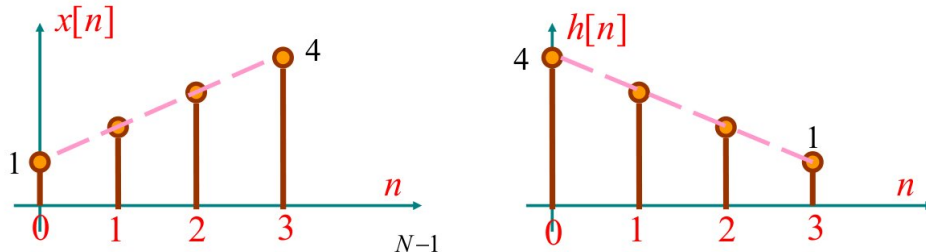
Note

Circular convolution equals linear convolution only if enough zero-padding is used. To compute linear convolution by DFT, choose

$$N \geq L + M - 1.$$

Example: Circular Convolution

$$x[n] = (n+1)p_4[n]; \quad h[n] = (4-n)p_4[n]$$



$$y[n] = x[n] \circledast_N h[n] = \sum_{m=0}^{N-1} x[m]h[n-m, \text{mod } N]p_N[n]$$

$$= \sum_{m=-\infty}^{N-1} x[n-m, \text{mod } N]h[m]p_N[n]$$

Circular convolution reuses samples by modulo indexing, so the shifted sequence wraps around.

TABLE 7.3 PROPERTIES OF THE DFT

Property	Transform Pair
Linearity	$ax[n] + bv[n] \leftrightarrow aX_k + bV_k$
Circular time shift	$x[n - q, \text{mod } N] \leftrightarrow X_k e^{-j2\pi kq/N}$
Time reversal	$x[-n, \text{mod } N] \leftrightarrow X_{-k, \text{mod } N}$
Multiplication by a complex exponential	$x[n]e^{j2\pi qn/N} \leftrightarrow X_{k-q, \text{mod } N}$
Circular convolution	$x[n] \circledast v[n] \leftrightarrow X_k V_k$
Multiplication in the time domain	$x[n]v[n] \leftrightarrow (1/N)X_k \circledast V_k$
Parseval's theorem	$\sum_{n=0}^{N-1} x[n]v[n] \leftrightarrow (1/N) \sum_{i=0}^{N-1} X_i \bar{Y}_i,$ where $\bar{Y}_i$ = complex conjugate of Y_i

Summary table of DFT properties.

4.5 FFT Algorithm

4.5.1 Computational cost of direct DFT

A direct N -point DFT requires approximately

$$N^2$$

complex multiplications and

$$N(N-1)$$

complex additions. For $N = 1024$, this means over one million complex multiplications. This is why the FFT is needed.

Rule / Test

The FFT is not a different transform. It is a fast algorithm for computing the same DFT values.

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-jn \frac{2\pi k}{N}} \quad k=0,1,\dots,N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j \frac{2\pi k}{N} n} \quad n=0,1,\dots,N-1$$

***N* points DFT**

Complex multiplications: N^2 (1024²=1048576)

Complex additions: $N(N-1)$

One complex multiplication=4*real
multiplications+2* additions

FFT

It requires only $(N \log_2 N)/2$ multiplications by
using a Fast Fourier Transform algorithm.

Direct DFT cost versus FFT cost. FFT reduces computation from order N^2 to order $N \log_2 N$.

4.5.2 Twiddle factor identities

Define

$$W_N = e^{-j2\pi/N}.$$

Important identities are

$$W_N^0 = 1, \quad W_N^{N/2} = -1, \quad W_N^{N+r} = W_N^r, \quad W_N^{N/2+r} = -W_N^r.$$

These identities allow repeated decomposition of one DFT into smaller DFTs.

Approach based on decimation in time

Let $W_N = e^{-j \frac{2\pi}{N}}$

$$\begin{aligned} \text{We have:} \quad W_N^0 &= 1 & W_N^{N/2} &= -1 & W_N^{2r} &= W_{N/2}^r \\ W_N^{N+r} &= W_N^r & W_N^{N/2+r} &= -W_N^r \end{aligned}$$

Only N independent values

Twiddle-factor periodicity and symmetry are the algebraic basis of the FFT.

4.5.3 Decimation-in-time decomposition

For even N , split the sequence into even-indexed and odd-indexed samples:

$$A(k) = \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{rk},$$

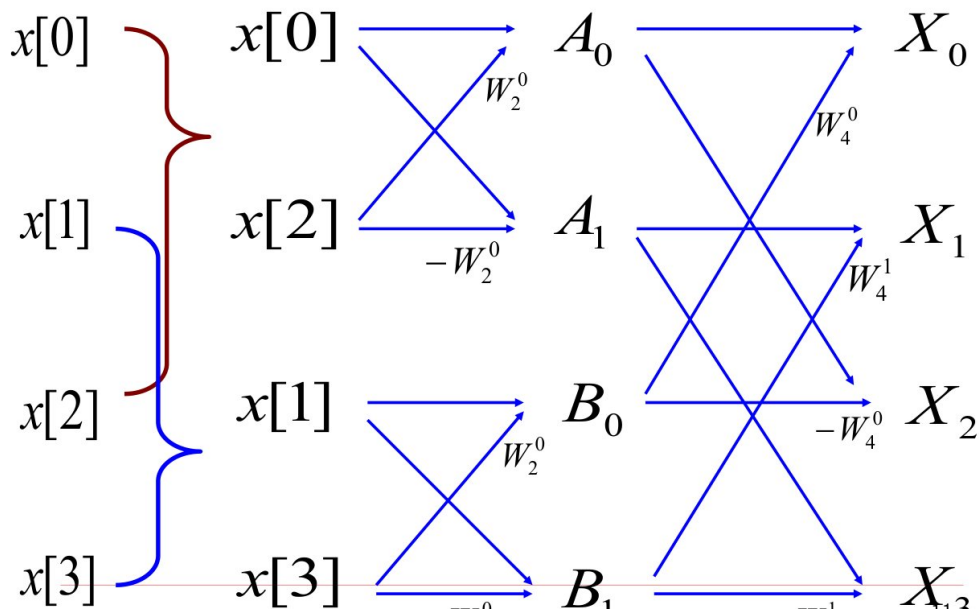
$$B(k) = \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{rk}.$$

Then

$$X(k) = A(k) + W_N^k B(k), \quad 0 \leq k \leq N/2 - 1,$$

$$X(k + N/2) = A(k) - W_N^k B(k), \quad 0 \leq k \leq N/2 - 1.$$

This pair of equations is the butterfly operation.



Basic FFT butterfly relation for combining even-index and odd-index DFTs.

4.5.4 Flow graph and bit reversal

When $N = 2^m$, the decomposition can continue until two-point DFTs are reached. The flow graph contains $m = \log_2 N$ stages. In a radix-2 decimation-in-time FFT, the input order is commonly bit-reversed, while the output order is natural.

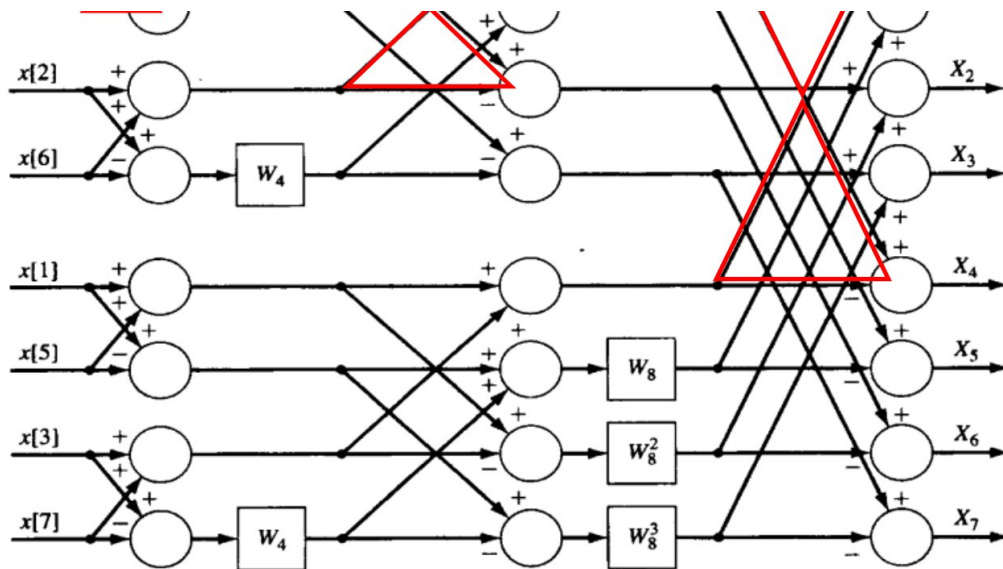


Figure 7.33 Block diagram of FFT algorithm when $N = 8$.

Example FFT flow graph for $N = 8$.

4.6 Chapter 4 Review Table

Concept	Core formula or rule
DTFT	$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$
Inverse DTFT	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega)e^{j\Omega n} d\Omega$
DTFT periodicity	$X(\Omega + 2\pi) = X(\Omega)$
DFT	$X_k = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$
Inverse DFT	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}$
DFT grid	$\Omega_k = 2\pi k/N$
Time truncation	Multiplication by a finite window in time causes convolution in frequency.
Circular shift	$x[n - q, \text{mod } N] \leftrightarrow W_N^{qk} X[k]$
Circular convolution	$x[n] \otimes_N h[n] \leftrightarrow X[k]H[k]$
Linear convolution by DFT	Zero-pad to $N \geq L + M - 1$.
FFT	Same DFT result, lower computational cost: order $N \log_2 N$.

Definition of DTFT

$$X(\Omega) = \sum_{n=0}^{N-1} x[n]e^{-jn\Omega} \qquad x[n] = \frac{1}{2\pi} \int_{2\pi} X(\Omega)e^{jn\Omega} d\Omega$$

Definition of DFT

$$X_k = X(\Omega) \Big|_{\Omega = \frac{2\pi k}{N}} = X\left(\frac{2\pi k}{N}\right) = \sum_{n=0}^{N-1} x[n]e^{-jn\frac{2\pi k}{N}}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi k}{N}n}$$

Summary of DTFT and DFT definitions and their connection through sampled frequency values.

CHAPTER 5:

Fourier Analysis of Systems

Chapter 5 Map

Chapter 5 studies LTI systems in the Fourier domain. The key idea is simple: convolution in time becomes multiplication in frequency. Therefore a system is characterized by its frequency response. For continuous-time systems, the response is described by $H(\omega)$; for discrete-time systems, it is described by $H(\Omega)$.

Topic	Core question
CT system frequency response	How does an LTI system change the amplitude and phase of each frequency component?
Periodic input response	How are Fourier series coefficients modified by the system?
Aperiodic input response	How is the output recovered from $Y(\omega) = X(\omega)H(\omega)$?
Ideal filters	Which frequency components are passed and which are rejected?
DT system frequency response	What is the DT counterpart of $Y(\omega) = X(\omega)H(\omega)$?
Digital lowpass filtering	How does an ideal or practical DT lowpass filter process sampled data?

5.1 Fourier Analysis of Continuous-Time Systems

5.1.1 Zero-state response and frequency response

For a continuous-time LTI system with impulse response $h(t)$, the zero-state output is

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda) d\lambda.$$

If $h(t)$ is absolutely integrable,

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty,$$

then its Fourier transform exists:

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt.$$

This $H(\omega)$ is the **frequency response** or **system function** of the CT LTI system.

5.1 Fourier Analysis of Continuous-Time System

LTI system: **Zero State Response**

$$\begin{aligned} y(t) &= x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda) h(t - \lambda) d\lambda \\ &= \int_{-\infty}^{\infty} h(\lambda) x(t - \lambda) d\lambda = y_{zs}(t) \end{aligned}$$

where $h(t)$ is assumed to **be absolutely integrable**, that is

$$\int_{-\infty}^{\infty} |h(\lambda)| dt < \infty$$

Then the CTFT of $h(t)$ exists and is given by

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

Zero-state response of a CT LTI system and the definition of its frequency response.

5.1.2 Frequency-domain input-output relation

The convolution theorem gives

$$y(t) = x(t) * h(t) \iff Y(\omega) = X(\omega)H(\omega).$$

Thus the output spectrum is the input spectrum multiplied by the system frequency response:

$$|Y(\omega)| = |X(\omega)| |H(\omega)|, \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega).$$

Rule / Test

For an LTI system, $H(\omega)$ tells what happens to every frequency component:

- $|H(\omega)|$ changes the amplitude;
- $\angle H(\omega)$ changes the phase.

5.1 Fourier Analysis of Continuous-Time System

The **frequency-domain representation** of the given system can be given as

$$Y(\omega) = X(\omega)H(\omega) \quad \text{Where} \quad H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$$

$$|Y(\omega)| = |X(\omega)| |H(\omega)| \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

4

Frequency-domain representation of a CT LTI system.

5.1.3 Response to a sinusoidal input

For the input

$$x(t) = A \cos(\omega_0 t + \theta),$$

the output of a stable LTI system is

$$y(t) = A|H(\omega_0)| \cos(\omega_0 t + \theta + \angle H(\omega_0)).$$

The frequency remains ω_0 . Only the amplitude and phase are changed.

Note

A sinusoid is an eigenfunction of an LTI system. The system does not create a new sinusoidal frequency from a single sinusoidal input; it only scales and phase-shifts that sinusoid.

5.1.1 Response to a Sinusoidal Input

If $x(t) = A \cos(\omega_0 t + \theta)$, $-\infty < t < \infty$

$$\cos(\omega_0 t) \leftrightarrow \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)] \quad x(t - c) \leftrightarrow e^{-j\omega c} X(\omega)$$

$$\begin{aligned} \cos(\omega_0 t + \theta) &= \cos[\omega_0(t + \theta/\omega_0)] \leftrightarrow \pi[e^{j\omega(\theta/\omega_0)} \delta(\omega + \omega_0) + e^{j\omega(\theta/\omega_0)} \delta(\omega - \omega_0)] \\ &\leftrightarrow \pi[e^{-j\theta} \delta(\omega + \omega_0) + e^{j\theta} \delta(\omega - \omega_0)] \end{aligned}$$

$$Y(\omega) = H(\omega)X(\omega)$$

$$= A\pi |H(\omega)| e^{j\angle H(\omega)} [e^{-j\theta} \delta(\omega + \omega_0) + e^{j\theta} \delta(\omega - \omega_0)]$$

$$= A\pi [|H(\omega)| e^{-j[\theta - \angle H(\omega)]} \delta(\omega + \omega_0) + |H(\omega)| e^{j[\theta + \angle H(\omega)]} \delta(\omega - \omega_0)]$$

$$= A\pi [|H(-\omega_0)| e^{-j[\theta - \angle H(-\omega_0)]} \delta(\omega + \omega_0) + |H(\omega_0)| e^{j[\theta + \angle H(\omega_0)]} \delta(\omega - \omega_0)]$$

$$= A |H(\omega_0)| \pi [e^{-j[\theta + \angle H(\omega_0)]} \delta(\omega + \omega_0) + e^{j[\theta + \angle H(\omega_0)]} \delta(\omega - \omega_0)]$$

$$y(t) = A |H(\omega_0)| \cos[\omega_0 t + \theta + \angle H(\omega_0)]$$

Derivation of the CT sinusoidal response formula.

5.1.4 Differential-equation system example

For a system described by

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = x(t),$$

taking the Fourier transform gives

$$[(j\omega)^2 + 3j\omega + 2]Y(\omega) = X(\omega),$$

so

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{(j\omega)^2 + 3j\omega + 2}.$$

For $x(t) = X_m \cos \omega_0 t$, the output is

$$y(t) = X_m |H(\omega_0)| \cos[\omega_0 t + \angle H(\omega_0)].$$

5.1.5 RC lowpass example

For the RC circuit model

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t),$$

the frequency response is

$$H(\omega) = \frac{1/RC}{j\omega + 1/RC}.$$

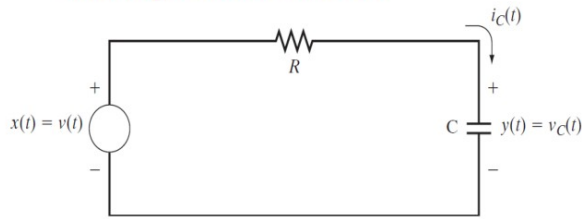
Therefore

$$|H(\omega)| = \frac{1/RC}{\sqrt{\omega^2 + (1/RC)^2}}, \quad \angle H(\omega) = -\tan^{-1}(\omega RC).$$

This is a first-order lowpass response: low frequencies pass with little attenuation, while high frequencies are attenuated.

5.1.1 Response to a Sinusoidal Input

Example: RC Circuit



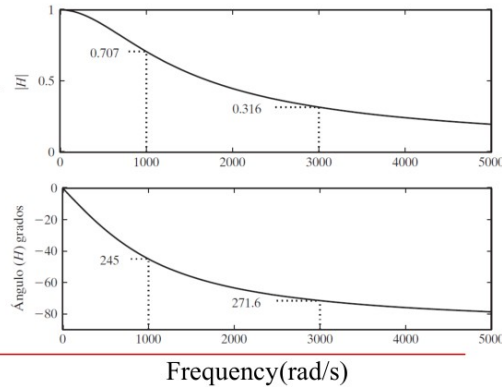
$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$

$$j\omega Y(\omega) + \frac{1}{RC} Y(\omega) = \frac{1}{RC} X(\omega)$$

$$Y(\omega) = \frac{1/RC}{j\omega + 1/RC} X(\omega)$$

$$H(\omega) = \frac{1/RC}{j\omega + 1/RC} \quad |H(\omega)| = \frac{1/RC}{[\omega^2 + (1/RC)^2]^{1/2}}$$

$$\angle H(\omega) = -\tan^{-1} \omega RC$$



RC circuit frequency response: amplitude decreases with frequency and phase becomes increasingly negative.

5.2 Response to Periodic and Nonperiodic Inputs

5.2.1 Response to periodic CT input: trigonometric form

If the periodic input is

$$x(t) = a_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k),$$

then the output is

$$y(t) = a_0 H(0) + \sum_{k=1}^{\infty} A_k |H(k\omega_0)| \cos(k\omega_0 t + \theta_k + \angle H(k\omega_0)).$$

Each harmonic is processed independently:

$$A_k^y = |H(k\omega_0)| A_k^x, \quad \theta_k^y = \theta_k^x + \angle H(k\omega_0).$$

5.2 Response to Periodic and Nonperiodic Inputs

$$Y(\omega) = X(\omega)H(\omega) \quad H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$$

Input: Periodic CT Signal

$$x(t) = a_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k), -\infty < t < \infty$$

Output:

$$y(t) = a_0 H(0) + \sum_{k=1}^{\infty} A_k |H(k\omega_0)| \cos(k\omega_0 t + \theta_k + \angle H(k\omega_0)), -\infty < t < \infty$$

$$A_k^y = |H(k\omega_0)| A_k^x$$

$$\theta_k^y = \angle H(k\omega_0) + \theta_k^x$$

Response of a CT LTI system to a periodic input represented by Fourier series.

5.2.2 Response to periodic CT input: complex form

If

$$x(t) = \sum_{k=-\infty}^{\infty} C_k^x e^{jk\omega_0 t},$$

then

$$y(t) = \sum_{k=-\infty}^{\infty} C_k^y e^{jk\omega_0 t}, \quad C_k^y = H(k\omega_0) C_k^x.$$

This form is often the cleanest for calculations because the system acts by multiplying each line-spectrum coefficient by the value of $H(\omega)$ at that line.

Example: rectangular pulse train through an RC system

For a rectangular pulse train, the nonzero Fourier coefficients occur at selected odd harmonics. If the system is an RC lowpass with

$$H(\omega) = \frac{1/RC}{j\omega + 1/RC},$$

then each harmonic coefficient is changed by

$$C_k^y = H(k\omega_0) C_k^x.$$

Higher harmonics have smaller $|H(k\omega_0)|$, so the output waveform becomes smoother than the input pulse train.

5.2 Response to Periodic and Nonperiodic Inputs

Example 5.4 Response to a Rectangular Pulse Train

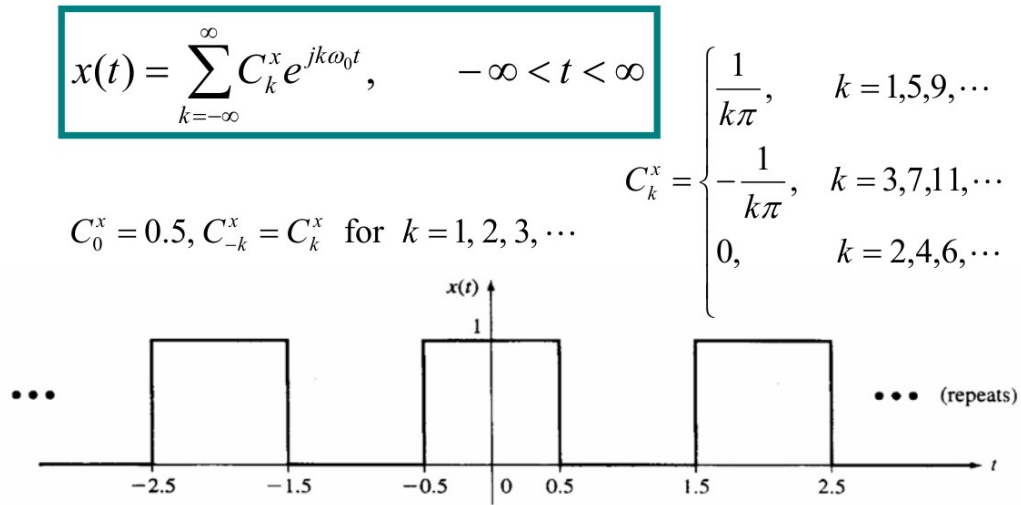


Figure 5.4 Periodic input signal in Example 5.4.

Fourier-series coefficients of a rectangular pulse train before system filtering.

5.2.3 Response to aperiodic input

For an aperiodic input with Fourier transform $X(\omega)$,

$$Y(\omega) = X(\omega)H(\omega).$$

The time-domain output is recovered by inverse CTFT:

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)H(\omega)e^{j\omega t} d\omega.$$

5.2.1 Response to Aperiodic Input

$$Y(\omega) = X(\omega)H(\omega)$$

$$\text{Where } H(\omega) = \int_{-\infty}^{\infty} h(\lambda)e^{-j\omega\lambda} d\lambda$$

$$|Y(\omega)| = |X(\omega)||H(\omega)|; \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

$$\Rightarrow y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega)X(\omega)e^{j\omega t} d\omega$$

Aperiodic-input response computed by multiplying the input spectrum by the frequency response.

5.2.4 Example: RC response to a pulse

For a rectangular pulse input, $X(\omega)$ has a sinc-shaped magnitude. Passing it through an RC lowpass gives

$$|Y(\omega)| = |X(\omega)||H(\omega)|, \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega).$$

Increasing $1/RC$ widens the effective passband and gives a faster time-domain response.

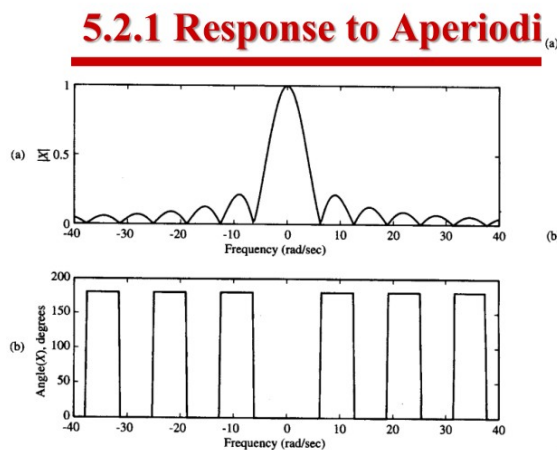


Figure 5.9 (a) Amplitude and (b) phase spectra of the input pulse.

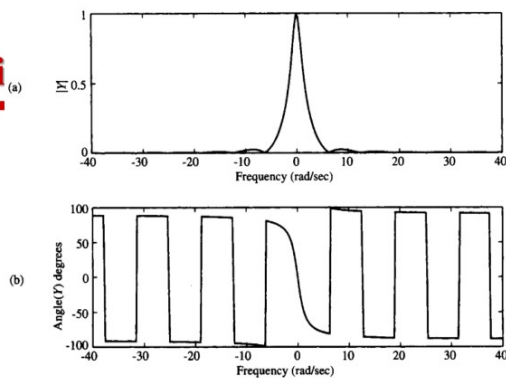


Figure 5.10 (a) Amplitude and (b) phase spectra of $y(t)$ when $1/RC = 1$.

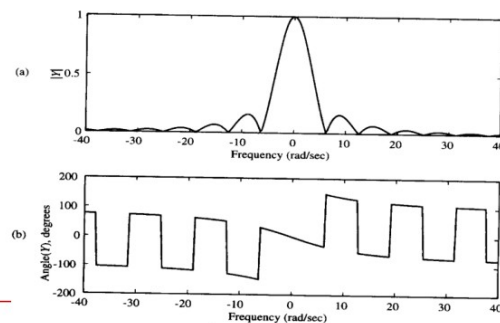


Figure 5.11 (a) Amplitude and (b) phase spectra of $y(t)$ when $1/RC = 10$.

Amplitude and phase spectra for the RC response to a pulse.

5.3 Analysis of Ideal Filters

5.3.1 Meaning of filtering

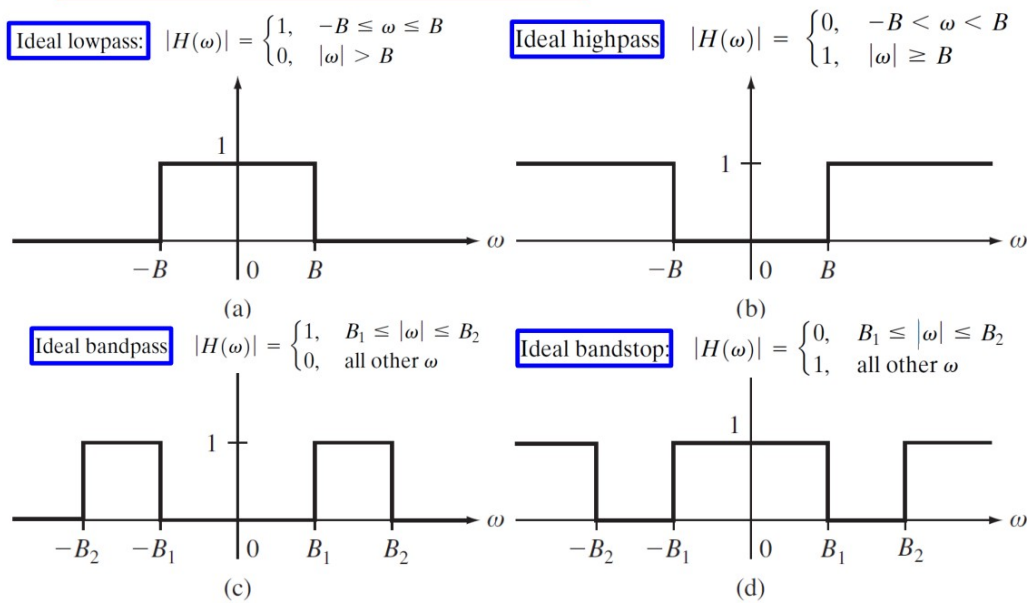
Filtering means rejecting sinusoidal components in some frequency ranges and passing sinusoidal components in other ranges. A system with this property is called a filter.

Definition

An ideal filter has a frequency response magnitude equal to either 1 or 0 over specified frequency intervals. The four basic ideal filters are:

- lowpass filter;
- highpass filter;
- bandpass filter;
- bandstop filter.

5.3 Analysis of Ideal Filters



Four basic ideal CT filters: lowpass, highpass, bandpass, and bandstop.

5.3.2 Ideal filter types

For an ideal lowpass filter with cutoff B ,

$$|H(\omega)| = \begin{cases} 1, & |\omega| \leq B, \\ 0, & |\omega| > B. \end{cases}$$

For an ideal highpass filter,

$$|H(\omega)| = \begin{cases} 0, & |\omega| < B, \\ 1, & |\omega| \geq B. \end{cases}$$

Bandpass and bandstop filters are defined similarly by retaining or rejecting a finite frequency band $B_1 \leq |\omega| \leq B_2$.

5.3.3 Modulation and demodulation through filtering

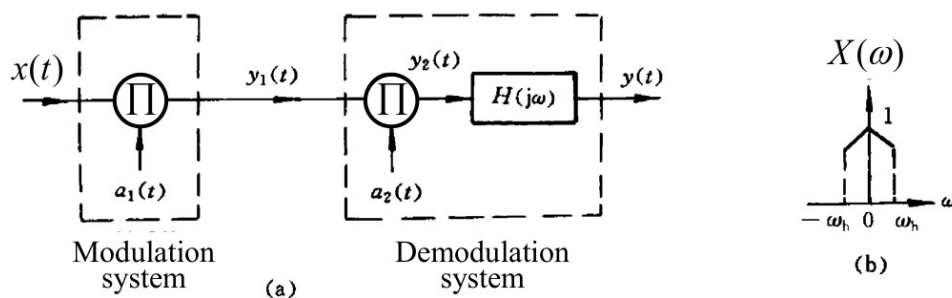
Multiplying a signal by $\cos \omega_0 t$ shifts its spectrum:

$$x(t) \cos \omega_0 t \iff \frac{1}{2}X(\omega + \omega_0) + \frac{1}{2}X(\omega - \omega_0).$$

Thus modulation moves a baseband spectrum to frequency neighborhoods around $\pm \omega_0$. Demodulation multiplies again by the same carrier and then uses a lowpass filter to recover the baseband part.

5.3 Analysis of Ideal Filters

Example: Modulation and Demodulation



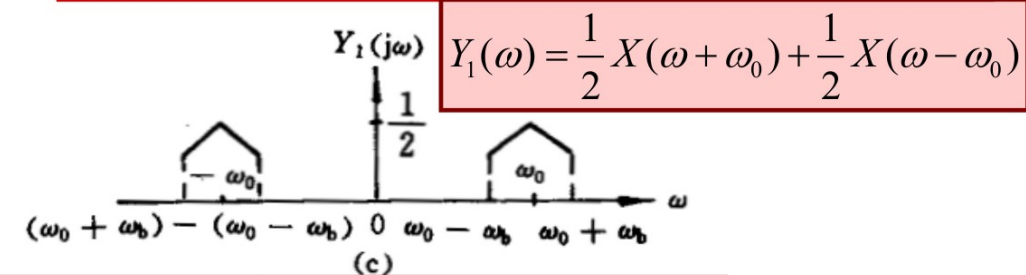
$$a_1(t) = a_2(t) = \cos \omega_0 t, \quad \omega_0 \gg \omega_b$$

1. $Y_1(\omega) = ?$ 2. $Y_2(\omega) = ?$ 3. If $y(t) = x(t)$, $H(\omega) = ?$

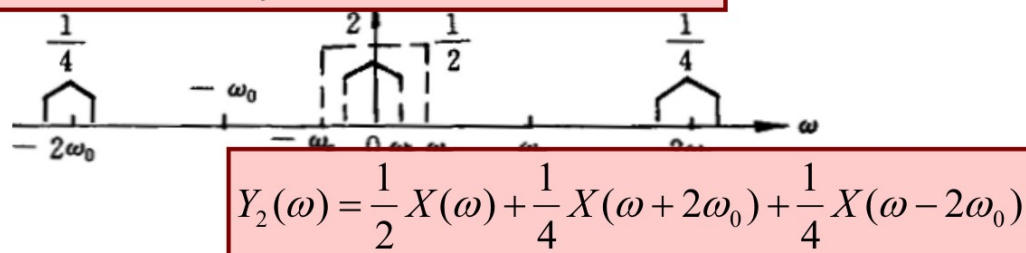
Modulation and demodulation system structure.

5.3 Analysis of Ideal Filters

Example



$$(3) H(\omega) = 2p_{2\omega_c}(\omega); \quad \omega_b \leq \omega_c \leq (2\omega_0 - \omega_b)$$



Spectrum movement in the modulation-demodulation example.

5.3.4 Example: ideal lowpass filtering

If

$$x(t) = 20 \cos(100t) \cos^2(10^4 t),$$

then

$$x(t) = 10 \cos(100t) + 5 \cos(20100t) + 5 \cos(19900t).$$

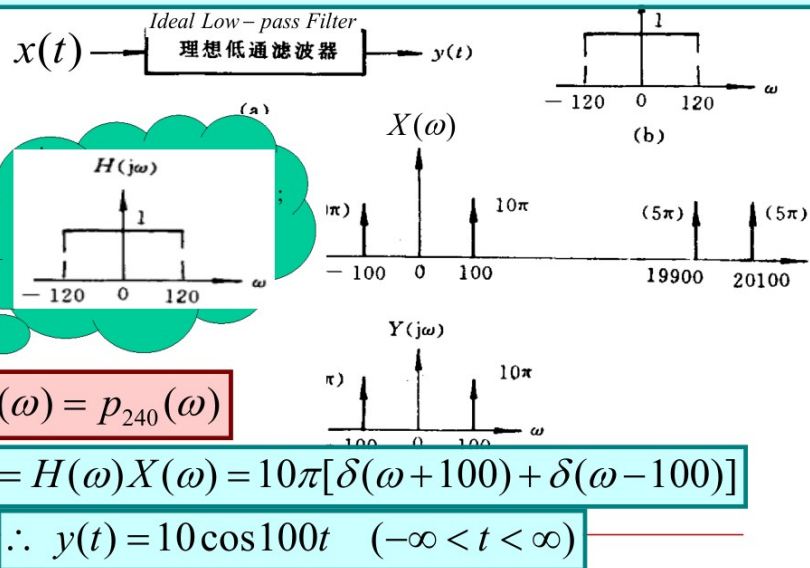
An ideal lowpass filter passing only $|\omega| \leq 120$ rejects the high-frequency terms and leaves

$$y(t) = 10 \cos(100t).$$

Solution

$$\therefore x(t) = 10 \cos 100t + 5[\cos 20100t + \cos 19900t]$$

$$\therefore X(\omega) = 10\pi[\delta(\omega + 100) + \delta(\omega - 100)] + 5\pi[\delta(\omega + 20100) + \delta(\omega - 20100) + \delta(\omega + 19900) + \delta(\omega - 19900)]$$



Ideal lowpass filtering keeps the 100 rad/s component and rejects the high-frequency components.

5.3.5 Phase distortion and linear phase

A filter avoids phase distortion if all passband frequency components experience the same time delay. This requires

$$\angle H(\omega) = -\omega t_d, \quad \omega \text{ in the passband.}$$

Equivalently,

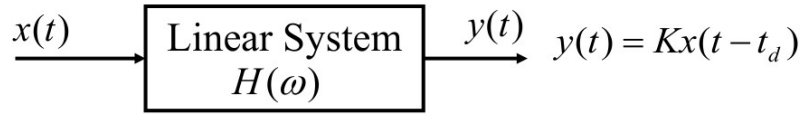
$$H(\omega) = K e^{-j\omega t_d}, \quad y(t) = K x(t - t_d).$$

Rule / Test

No phase distortion means the output is only a delayed and scaled version of the input. The required phase characteristic is linear in ω over the passband.

5.3 Analysis of Ideal Filters

The Condition to Avoid Phase Distortion in the Filtering or Transmission Process—Phase Function



$$Y(\omega) = KX(\omega)e^{-j\omega t_d} = H(\omega)X(\omega)$$

$$\Rightarrow H(\omega) = Ke^{-j\omega t_d} \quad \Rightarrow \begin{aligned} |H(\omega)| &= K \\ \angle H(\omega) &= -\omega t_d \end{aligned}$$

To avoid **phase distortion(相位畸变)** in the filtering process, a filter should have a **linear phase (线性相位)characteristic** over the passband of the filter

Linear phase condition for avoiding phase distortion.

5.4 Fourier Analysis of Discrete-Time Systems

5.4.1 DT LTI system frequency response

For a discrete-time LTI system with unit-pulse response $h[n]$, the zero-state response is

$$y[n] = x[n] * h[n] = \sum_{i=-\infty}^{\infty} x[i]h[n-i] = \sum_{i=-\infty}^{\infty} x[n-i]h[i].$$

If

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty,$$

then the DTFT of $h[n]$ exists:

$$H(\Omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\Omega n}.$$

The frequency-domain relation is

$$Y(\Omega) = H(\Omega)X(\Omega).$$

5.4 Fourier Analysis of DT Systems

Consider a **linear time-invariant discrete-time system** with **unit-pulse response** $h[n]$. The output response $y[n]$ resulting from the application of $x[n]$ with **initial zero energy** is :

$$y[n] = y_{zs}[n] = x[n] * h[n] = \sum_{i=-\infty}^{\infty} x[i]h[n-i] = \sum_{i=-\infty}^{\infty} x[n-i]h[i]$$

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad \Rightarrow \quad H(\Omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\Omega n}$$

frequency response function

$$\Rightarrow \quad Y(\Omega) = H(\Omega)X(\Omega)$$

Frequency response of a DT LTI system.

5.4.2 Magnitude and phase relation

From

$$Y(\Omega) = H(\Omega)X(\Omega),$$

we obtain

$$|Y(\Omega)| = |H(\Omega)| |X(\Omega)|, \quad \angle Y(\Omega) = \angle H(\Omega) + \angle X(\Omega).$$

This is the DT counterpart of the CT result. The only major difference is that all DTFT spectra are 2π -periodic in Ω .

5.4.3 Response to a sinusoidal input

For

$$x[n] = A \cos(\Omega_0 n + \theta),$$

the output is

$$y[n] = A |H(\Omega_0)| \cos(\Omega_0 n + \theta + \angle H(\Omega_0)).$$

Again, the frequency is unchanged; only amplitude and phase are modified.

5.4 Fourier Analysis of DT Systems

Response to a Sinusoidal Input

$$Y(\Omega) = \begin{cases} A\pi [H(-\Omega_0)e^{-j\theta}\delta(\Omega + \Omega_0) + H(\Omega_0)e^{j\theta}\delta(\Omega - \Omega_0)] & -\pi < \Omega < \pi \\ \text{periodic elsewhere} \end{cases}$$

Now $H(\Omega_0) = |H(\Omega_0)|e^{j\angle H(\Omega_0)}$
 $H(-\Omega_0) = \underbrace{|H(\Omega_0)|e^{-j\angle H(\Omega_0)}}_{\text{Used symmetry properties}}$

$$Y(\Omega) = \begin{cases} A\pi |H(\Omega_0)| [e^{-j(\theta + \angle H(\Omega_0))}\delta(\Omega + \Omega_0) + e^{+j(\theta + \angle H(\Omega_0))}\delta(\Omega - \Omega_0)] & -\pi < \Omega < \pi \\ \text{periodic elsewhere} \end{cases}$$

$$y[n] = A |H(\Omega_0)| \cos(\Omega_0 n + \theta + \angle H(\Omega_0))$$

Response of a DT LTI system to a sinusoidal sequence.

5.4.4 DFT-domain system analysis

If $x[n]$ and $h[n]$ are finite-length sequences and zero-padding is chosen so that the DFT length satisfies

$$L \geq N + Q - 1,$$

then linear convolution can be computed using the DFT:

$$Y[k] = X[k]H[k], \quad y[n] = \text{IDFT}\{Y[k]\}.$$

The zero-padding condition prevents circular convolution from aliasing the linear convolution result.

5.4 Fourier Analysis of DT Systems

System Analysis via the DFT

Assume: $x[n]=0, n < 0$ and $n \geq N$
 $h[n]=0, n < 0$ and $n \geq Q$

$$y[n] = y_{zs}[n] = x[n] * h[n] = \sum_{i=0}^Q x[i]h[n-i], \quad 0 \leq n < N+Q$$

For the length of this linear convolution is **$N+Q-1$** , Let us **pad** $x[n]$ and $h[n]$ with zeros, now the *DFT* domain representation can be generated by taking **$(N+Q)$ -point *DFT*** of both sides of the representation:

$$x[n] = 0, n = N, N+1, \dots, N+Q-1$$

$$h[n] = 0, n = Q+1, Q+2, \dots, Q+N-1$$

DFT-domain computation of finite-length DT system response.

5.5 Application to Lowpass Digital Filtering

5.5.1 Ideal DT lowpass filter

An ideal DT lowpass filter passes frequencies near $\Omega = 0$ and rejects frequencies near $\Omega = \pi$. Because DTFTs are periodic, the ideal DT lowpass frequency response is repeated every 2π :

$$H(\Omega) = \sum_{k=-\infty}^{\infty} p_{2B}(\Omega + 2\pi k).$$

For a sinusoidal input

$$x[n] = A \cos(\Omega_0 n),$$

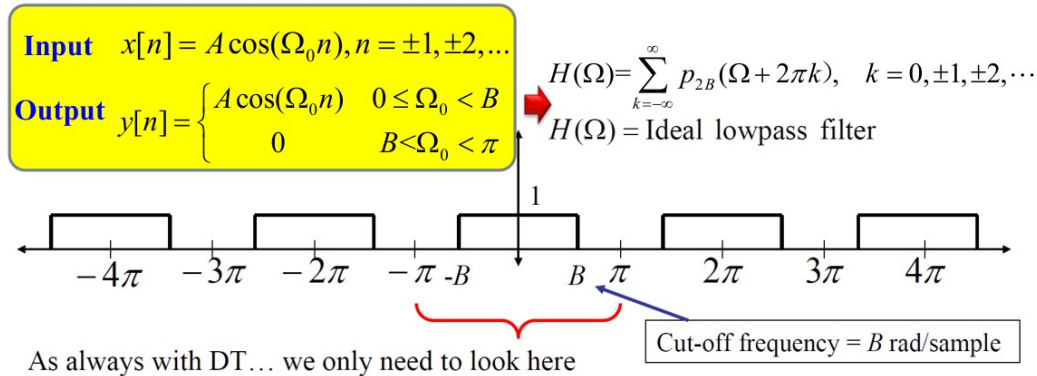
the ideal lowpass output is

$$y[n] = \begin{cases} A \cos(\Omega_0 n), & 0 \leq \Omega_0 < B, \\ 0, & B < \Omega_0 < \pi. \end{cases}$$

5.5 Application to Lowpass Digital Filtering

Analysis of Ideal D-T lowpass Filter (LPF)

Just as in the CT case... we can specify filters. We looked at the ideal lowpass filter for the CT case... here we look at it for the DT case.



Ideal DT lowpass response and its periodic repetition in frequency.

5.5.2 ADC - DT filter - DAC interpretation

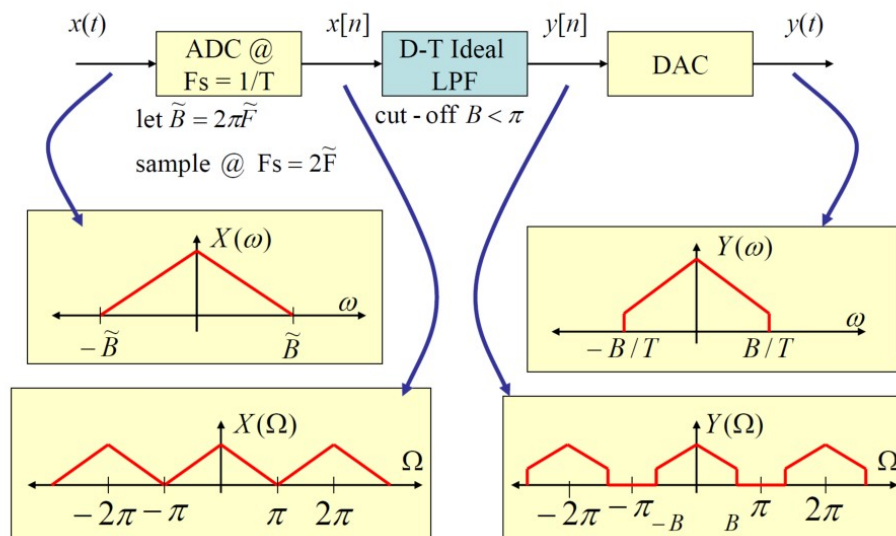
A practical digital filtering chain may be interpreted as

$$x(t) \longrightarrow x[n] \longrightarrow y[n] \longrightarrow y(t),$$

that is, analog-to-digital conversion, discrete-time filtering, and digital-to-analog reconstruction. Under proper sampling and reconstruction conditions, the entire chain can behave like an equivalent continuous-time lowpass system.

5.5 Application to Lowpass Digital Filtering

Analysis of Ideal D-T lowpass Filter (LPF)



Whole System (ADC - D-T filter - DAC) acts like an equivalent C-T system

ADC - DT lowpass filter - DAC chain and its frequency-domain interpretation.

5.5.3 Impulse response of the ideal DT lowpass filter

For the ideal lowpass response

$$H(\Omega) = \begin{cases} 1, & |\Omega| \leq B, \\ 0, & B < |\Omega| \leq \pi, \end{cases}$$

the impulse response is

$$h[n] = \frac{1}{2\pi} \int_{-B}^B e^{j\Omega n} d\Omega = \begin{cases} \frac{\sin(Bn)}{\pi n}, & n \neq 0, \\ \frac{B}{\pi}, & n = 0. \end{cases}$$

This impulse response is two-sided and infinite-length, so the ideal lowpass filter is noncausal and not directly realizable.

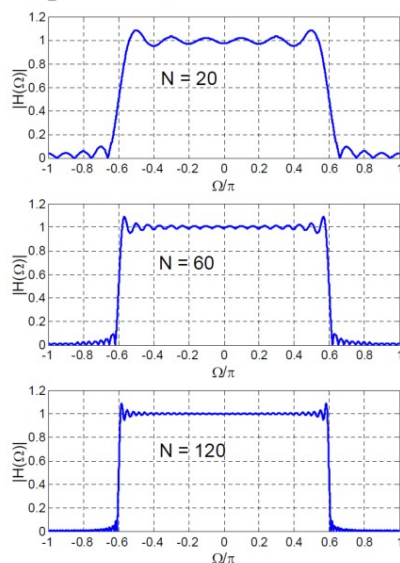
5.5.4 Causal approximation

In practice, a realizable lowpass filter approximates the ideal response by delaying and truncating the ideal impulse response. A longer truncated impulse response usually gives a closer approximation to the ideal magnitude response, but it also increases computation and delay.

5.5 Application to Lowpass Digital Filtering

Causal Lowpass Filter (But not Ideal)

In practice, the best we can do is try to **approximate** the ideal LPF



Frequency Response
of a filter truncated to
21 samples

Frequency Response
of a filter truncated to
61 samples

Frequency Response
of a filter truncated to
121 samples

Some general insight:
Longer lengths for
the truncated impulse
response Gives **better
approximation** to the
ideal filter response!!

Longer truncated impulse responses produce a better approximation to the ideal lowpass filter.

5.5.5 Moving-average filters

A length- M moving-average filter has the form

$$y[n] = \frac{1}{M} \sum_{m=0}^{M-1} x[n-m].$$

Its impulse response is

$$h[n] = \begin{cases} 1/M, & 0 \leq n \leq M-1, \\ 0, & \text{otherwise.} \end{cases}$$

The frequency response is

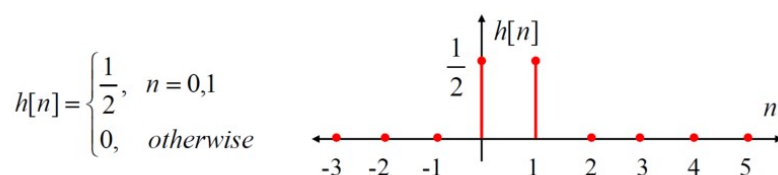
$$H(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} e^{-j\Omega m} = \frac{1}{M} e^{-j\Omega(M-1)/2} \frac{\sin(M\Omega/2)}{\sin(\Omega/2)}.$$

It attenuates high frequencies, but it is not an ideal stopband filter.

5.5 Application to Lowpass Digital Filtering

Moving Average Filters

Here is a very simple, low quality LPF:



To see how well this works as a lowpass filter we find its frequency response:

$$\begin{aligned} H(\Omega) &= \sum_{n=-\infty}^{\infty} h[n] e^{-j\Omega n} = \frac{1}{2} \times e^{-j\Omega 0} + \frac{1}{2} \times e^{-j\Omega 1} \\ &= \frac{1}{2} [1 + e^{-j\Omega}] \end{aligned}$$

By definition of the DTFT

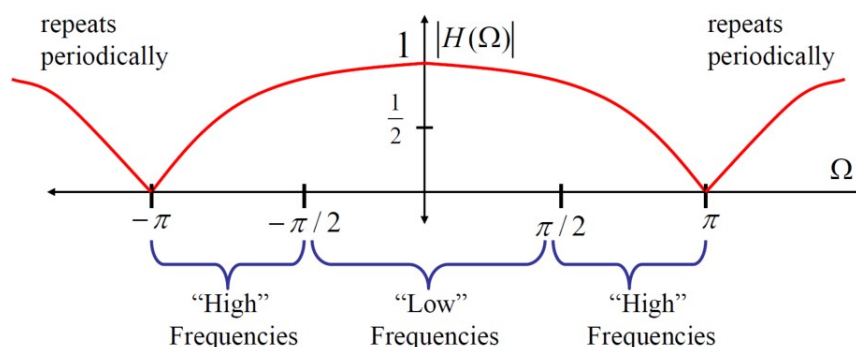
Only 2 non-zero terms in the sum

Moving-average filter structure.

5.5 Application to Lowpass Digital Filtering

Moving Average Filters

Here's a plot of this filter's freq. resp. magnitude:



This does attenuate high frequencies but doesn't really "stop" them!
It is a low pass filter but **not a very good one!**

Magnitude response of a moving-average filter: lowpass behavior with nonideal stopband attenuation.

5.6 Chapter 5 Review Table

Concept	Core formula or rule
CT LTI output	$Y(\omega) = X(\omega)H(\omega)$
CT frequency response	$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$
CT sinusoidal response	$A \cos(\omega_0 t + \theta) \mapsto A H(\omega_0) \cos(\omega_0 t + \theta + \angle H(\omega_0))$
Periodic-input coefficients	$C_k^y = H(k\omega_0)C_k^x$
Aperiodic-input output	$y(t) = \frac{1}{2\pi} \int X(\omega)H(\omega)e^{j\omega t} d\omega$
Ideal LPF	Pass $ \omega \leq B$, reject $ \omega > B$
No phase distortion	$\angle H(\omega) = -\omega t_d$ in the passband
DT LTI output	$Y(\Omega) = H(\Omega)X(\Omega)$
DT sinusoidal response	$A \cos(\Omega_0 n + \theta) \mapsto A H(\Omega_0) \cos(\Omega_0 n + \theta + \angle H(\Omega_0))$
DFT system analysis	Use zero-padding and compute $Y[k] = X[k]H[k]$
Ideal DT LPF impulse response	$h[n] = \sin(Bn)/(\pi n)$ for $n \neq 0$, $h[0] = B/\pi$
Moving average	Simple practical lowpass approximation, but not ideal.

CHAPTER 6:

Laplace Transform and Transfer Function Representation

Chapter 6 Map

Chapter 6 moves from the Fourier variable $j\omega$ to the complex variable

$$s = \sigma + j\omega.$$

The Laplace transform introduces a convergence factor, gives an algebraic method for solving differential-equation systems, and leads to the transfer-function representation

$$H(s) = \frac{Y(s)}{X(s)}.$$

This representation is derived under zero initial conditions.

Topic	What you should be able to do
Laplace transform	Compute $X(s)$, identify the region of convergence, and connect $X(s)$ with the Fourier transform.
Properties	Use shifting, scaling, multiplication by t^N , exponential modulation, differentiation, integration, convolution, and value theorems.
Inverse transform	Use partial fractions, residues, repeated poles, and complex-pole pairing to recover $x(t)$.
Differential equations	Convert input-output differential equations into algebraic equations in s .
Transfer function	Derive $H(s) = Y(s)/X(s)$ from differential equations, convolution, circuits, and block diagrams.
System behavior	Relate pole locations to the time-domain form and long-time behavior of a response.

6.1 Laplace Transform of a Signal

6.1.1 Why the Laplace transform is introduced

For some causal signals, the Fourier integral does not converge. A typical example is the unit-step signal $u(t)$:

$$\int_0^{\infty} e^{-j\omega t} dt$$

does not converge in the ordinary sense. The Laplace transform solves this by adding an exponential convergence factor $e^{-\sigma t}$:

$$e^{-\sigma t} e^{-j\omega t} = e^{-(\sigma + j\omega)t}.$$

With

$$s = \sigma + j\omega,$$

the transform becomes a function of a complex variable.

6.1 Laplace Transform of a Signal

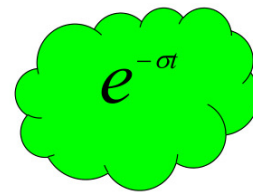
Given a continuous –time signal $x(t)$

The Fourier transform $X(\omega)$ of $x(t)$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

For the unit-step function $u(t)$

$$X(\omega) = \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-j\omega t} dt = \int_0^{\infty} e^{-\sigma t} e^{-j\omega t} dt$$



The integral does not exist.

We can circumvent this problem by introducing a **convergence factor** to Fourier transform

3

Motivation for adding the convergence factor $e^{-\sigma t}$.

6.1.2 Two-sided and one-sided Laplace transforms

Definition

The **two-sided**, or bilateral, Laplace transform is

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-st} dt.$$

The **one-sided** Laplace transform is

$$X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt.$$

The one-sided form is especially useful for causal systems with initial conditions.

Region of convergence

The **region of convergence**, abbreviated ROC, is the set of all complex numbers s for which the Laplace integral converges. The expression $X(s)$ alone is not complete without its ROC.

6.1.3 Unit-step example

For

$$x(t) = u(t),$$

we have

$$X(s) = \int_0^{\infty} e^{-st} dt = \frac{1}{s}, \quad \operatorname{Re}\{s\} > 0.$$

Thus

$$u(t) \longleftrightarrow \frac{1}{s}, \quad \text{ROC} : \operatorname{Re}\{s\} > 0.$$

6.1 Laplace Transform of a Signal

Note!!

$$X(s) = \frac{1}{s}$$

The **region of convergence** (收敛域) for the Laplace transform of the unit-step function.

$X(s)$ is a complex-valued function of the complex number s .

The function $X(s)$ given above is defined for only those complex numbers s for which the real part of s is strictly positive.

$$\operatorname{Re}s = \sigma > 0$$

5

Laplace transform and ROC of the unit-step signal.

6.1.4 Exponential example

For

$$x(t) = e^{-bt}u(t),$$

we obtain

$$X(s) = \int_0^{\infty} e^{-bt}e^{-st} dt = \frac{1}{s+b}, \quad \operatorname{Re}\{s\} > -b.$$

So

$$e^{-bt}u(t) \longleftrightarrow \frac{1}{s+b}, \quad \text{ROC} : \operatorname{Re}\{s\} > -b.$$

6.1.5 Relationship with the Fourier transform

Note

The Laplace transform is a generalization of the Fourier transform. Formally,

$$X(\omega) = X(s)|_{s=j\omega},$$

but this substitution is valid only when the ROC of $X(s)$ includes the $j\omega$ axis, that is, $\text{Re}\{s\} = 0$.

6.1 Laplace Transform of a Signal

1. Relationship between the Fourier and Laplace Transform

The **two-sided Laplace transform** of a signal $x(t)$ can be viewed as a **generalization of the definition of the Fourier transform** of $x(t)$; that is the two-sided Laplace transform is the Fourier transform with the addition of an exponential factor.

Fourier Transform:

$$\left. \begin{array}{l} \text{Laplace Transform:} \\ X(s) = \int_0^{\infty} x(t)e^{-(\sigma + j\omega)t} dt = \int_0^{\infty} x(t)e^{-st} dt \end{array} \right\} X(\omega) = X(s) \big|_{s=j\omega}$$

It is valid if and only if the region of convergence for $X(s)$ includes $\text{Re } s=0$. ($j\omega$ axis)

Relationship between the Fourier transform and the Laplace transform.

6.2 Properties of the Laplace Transform

Let

$$x(t) \longleftrightarrow X(s), \quad v(t) \longleftrightarrow V(s).$$

The following properties are frequently used to compute transforms without direct integration.

6.2.1 Basic algebraic properties

Property	Laplace-domain rule
Linearity	$ax(t) + bv(t) \longleftrightarrow aX(s) + bV(s)$
Right shift in time	$x(t - c)u(t - c) \longleftrightarrow e^{-cs}X(s), c > 0$
Time scaling	$x(at) \longleftrightarrow \frac{1}{a}X\left(\frac{s}{a}\right), a > 0$
Multiplication by t^N	$t^N x(t) \longleftrightarrow (-1)^N \frac{d^N}{ds^N} X(s)$
Multiplication by exponential	$e^{at}x(t) \longleftrightarrow X(s - a)$
Convolution	$x(t) * v(t) \longleftrightarrow X(s)V(s)$

6.2 Properties of The Laplace Transform

Most of the properties correspond directly to the properties of the Fourier transform.

Let: $x(t) \leftrightarrow X(s)$ $v(t) \leftrightarrow V(s)$

1. Linearity

$$ax(t) + bv(t) \leftrightarrow aX(s) + bV(s)$$

2. Right shift in time

$$x(t-c)u(t-c) \leftrightarrow e^{-cs}X(s) \quad c - \text{positive real number}$$

3. Time scaling

$$x(at) \leftrightarrow \frac{1}{a}X\left(\frac{s}{a}\right) \quad c - \text{positive real number}$$

4. Multiplication by a power of t

$$(t)^N x(t) \leftrightarrow (-1)^N \frac{d^N}{ds^N} X(s)$$

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Basic Laplace-transform properties.

6.2.2 Multiplication by sinusoidal signals

Using Euler's formula,

$$x(t) \cos \omega_0 t \longleftrightarrow \frac{1}{2} [X(s - j\omega_0) + X(s + j\omega_0)],$$

$$x(t) \sin \omega_0 t \longleftrightarrow \frac{j}{2} [X(s + j\omega_0) - X(s - j\omega_0)].$$

This is the s -domain analogue of frequency shifting in Fourier analysis.

6.2 Properties of the Laplace Transform

Example 6.10 Laplace Transform of a Cosine

$$v(t) = (\cos \omega t)u(t)$$

$$u(t) \leftrightarrow \frac{1}{s}$$



$$\begin{aligned} V(s) &= \frac{1}{2} \left(\frac{1}{s + j\omega} + \frac{1}{s - j\omega} \right) \\ &= \frac{1}{2} \frac{s - j\omega + s + j\omega}{s^2 + \omega^2} \\ &= \frac{s}{s^2 + \omega^2} \end{aligned}$$

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Example: deriving the Laplace transform of $\cos \omega t u(t)$.

6.2.3 Differentiation, integration, and convolution

For the one-sided Laplace transform,

$$\dot{x}(t) \longleftrightarrow sX(s) - x(0^-).$$

If $x(t) = 0$ for $t < 0$ and the initial value is zero, this reduces to

$$\dot{x}(t) \longleftrightarrow sX(s).$$

For the second derivative,

$$\frac{d^2x(t)}{dt^2} \longleftrightarrow s^2X(s) - sx(0^-) - \dot{x}(0^-).$$

The integral property is

$$\int_0^t x(\lambda) d\lambda \longleftrightarrow \frac{1}{s}X(s).$$

6.2 Properties of the Laplace Transform

7. Differentiation in the time domain

$$\dot{x}(t) \leftrightarrow sX(s) - x(0)$$

It should be pointed out that if $x(t)$ is **discontinuous** at $t=0$ or if $x(t)$ contains an impulse or derivative of an impulse located at $t=0$, it is necessary to take the initial time in $t=0^-$.

$$\dot{x}(t) \leftrightarrow sX(s) - x(0^-)$$

If $x(t)=0$ for $t<0$

$$\dot{x}(t) \leftrightarrow sX(s)$$

6.2 Properties of the Laplace Transform

8. Integration

$$\int_0^t x(\lambda) d\lambda \leftrightarrow \frac{1}{s} X(s)$$

The Laplace transform of the integral of $x(t)$ is equal to $X(s)$ divided by s .

9. Convolution

Convolution in the time domain corresponds to a product in the Laplace transform domain.

$$x(t) * v(t) \leftrightarrow X(s)V(s)$$

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Integration and convolution properties.

6.2.4 Initial-value and final-value theorems

Rule / Test

Initial-value theorem:

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s),$$

provided the required limit exists.

Final-value theorem:

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s),$$

provided the ROC of $sX(s)$ includes the origin and the signal has a finite final value.

6.2.5 Convolution example

If

$$x(t) = u(t) - u(t-1),$$

then

$$X(s) = \frac{1 - e^{-s}}{s}.$$

Because convolution in time corresponds to multiplication in the Laplace domain,

$$\mathcal{L}\{x(t) * x(t)\} = X^2(s) = \left(\frac{1 - e^{-s}}{s} \right)^2.$$

The inverse transform gives a triangular waveform:

$$x(t) * x(t) = tu(t) - 2(t-1)u(t-1) + (t-2)u(t-2).$$

6.2 Properties of the Laplace Transform

Example 6.15 Convolution

Let $x(t) = u(t) - u(t-1)$, to determine the convolution $x(t) * x(t)$.

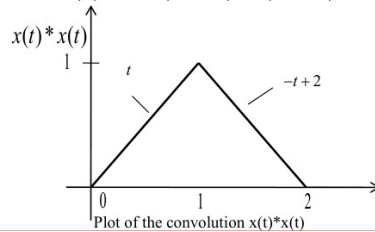
$$X(s) = (1 - e^{-s}) / s.$$

The convolution $x(t) * x(t)$ is equal to $X^2(s)$:

$$X^2(s) = \left(\frac{1 - e^{-s}}{s}\right)^2 = \frac{1 - 2e^{-s} + e^{-2s}}{s^2}$$

Inverse
Laplace
transform

$$x(t) * x(t) = tu(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$$



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Example 6.15: convolution computed by multiplying Laplace transforms.

6.3 Computation of the Inverse Laplace Transform

6.3.1 Rational Laplace transforms

Most system functions in this course are rational functions:

$$X(s) = \frac{B(s)}{A(s)} = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_1 s + b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0}.$$

The roots of $A(s) = 0$ are the **poles** of $X(s)$; the roots of $B(s) = 0$ are the **zeros**.

6.3 Computation of the Inverse Laplace Transform

1. Rational Laplace transform

Rational function

$$X(s) = \frac{B(s)}{A(s)} = \frac{b_M s^M + b_{M-1} s^{M-1} \cdots b_1 s + b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots a_1 s + a_0}$$

$X(s) = B(s)/A(s)$ is a rational function of s .

The order of the
rational function.

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Rational Laplace transform and order of a rational function.

6.3.2 Distinct poles and residues

If the poles $p_1, p_2, \dots, p_N$ are distinct, then

$$X(s) = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_N}{s - p_N}.$$

The coefficients c_i are residues:

$$c_i = [(s - p_i)X(s)]_{s=p_i}.$$

The inverse transform is

$$x(t) = \sum_{i=1}^N c_i e^{p_i t} u(t).$$

Hence the pole locations determine the time-domain modes.

6.3 Computation of the Inverse Laplace Transform

2. Distinct Poles

$$X(s) = \frac{B(s)}{A(s)} = \frac{b_M s^M + b_{M-1} s^{M-1} \dots b_1 s + b_0}{a_N (s - p_1)(s - p_2) \dots (s - p_N)}$$

The poles of $X(s)$: $p_1, \dots, p_N$ are distinct real or complex number, Then

$$X(s) = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_N}{s - p_N}$$

where

$$c_i = [(s - p_i)X(s)]_{s=p_i}$$

Constants c_i are called the **residues**.

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Residue formula for distinct poles.

Example 6.17: Distinct-pole case

For

$$X(s) = \frac{s + 2}{s^3 + 4s^2 + 3s} = \frac{s + 2}{s(s + 1)(s + 3)},$$

partial fractions give

$$X(s) = \frac{2/3}{s} - \frac{1/2}{s + 1} - \frac{1/6}{s + 3}.$$

Therefore

$$x(t) = \left(\frac{2}{3} - \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t} \right) u(t).$$

6.3 Computation of the Inverse Laplace Transform

Example 6.17 Distinct Pole Case

$$X(s) = \frac{s+2}{s^3 + 4s^2 + 3s} = \frac{c_1}{s-0} + \frac{c_2}{s-(-1)} + \frac{c_3}{s-(-3)} = \frac{c_1}{s} + \frac{c_2}{s+1} + \frac{c_3}{s+3}$$

$$A(s) = s^3 + 4s^2 + 3s = s(s+1)(s+3)$$

$$c_1 = [sX(s)]_{s=0} = \left. \frac{s+2}{(s+1)(s+3)} \right|_{s=0} = \frac{2}{3}$$

$$c_2 = [(s+1)X(s)]_{s=-1} = \left. \frac{s+2}{s(s+3)} \right|_{s=-1} = \frac{1}{-2} \quad x(t) = \frac{2}{3} - \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t}, t \geq 0$$

$$c_3 = [(s+3)X(s)]_{s=-3} = \left. \frac{s+2}{s(s+1)} \right|_{s=-3} = \frac{-1}{6}$$

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Worked distinct-pole inverse Laplace example.

6.3.3 Complex poles

For a complex-conjugate pole pair

$$p_{1,2} = \sigma \pm j\omega,$$

the corresponding real time-domain term can be written as

$$Ce^{\sigma t} \cos(\omega t + \theta)u(t).$$

This form is usually clearer than keeping two conjugate exponential terms separately.

6.3 Computation of the Inverse Laplace Transform

3. Distinct Poles with Two or More Poles Complex

$X(s)$ has two complex poles,

$$p_1 = \sigma + j\omega; p_2 = \sigma - j\omega$$

$$X(s) = \frac{c_1}{s-p_1} + \frac{\bar{c}_1}{s-\bar{p}_1} + \frac{c_3}{s-p_3} + \cdots \frac{c_N}{s-p_N}$$

$$x(t) = c_1 e^{p_1 t} + \bar{c}_1 e^{\bar{p}_1 t} + c_3 e^{p_3 t} + \cdots c_N e^{p_N t}$$

$$c_1 e^{p_1 t} + \bar{c}_1 e^{\bar{p}_1 t} = 2 |c_1| e^{\sigma t} \cos(\omega t + \angle c_1)$$

$$x(t) = 2 |c_1| e^{\sigma t} \cos(\omega t + \angle c_1) + c_3 e^{p_3 t} + \cdots c_N e^{p_N t}$$

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Complex-conjugate poles combine into a real damped sinusoid.

6.3.4 Repeated poles

If a pole p is repeated r times, the partial-fraction expansion contains

$$\frac{c_1}{s-p} + \frac{c_2}{(s-p)^2} + \cdots + \frac{c_r}{(s-p)^r}.$$

In the time domain, repeated poles generate polynomial factors multiplied by exponentials:

$$\frac{1}{(s-p)^m} \longleftrightarrow \frac{t^{m-1}}{(m-1)!} e^{pt} u(t).$$

6.3 Computation of the Inverse Laplace Transform

4. Repeated Poles

Suppose that pole p_1 of $X(s)$ is repeated r times and the other $N-r$ poles are distinct.

$$X(s) = \frac{c_1}{s-p_1} + \frac{c_2}{(s-p_1)^2} + \frac{c_r}{(s-p_1)^r} + \cdots + \frac{c_{r+1}}{(s-p_{r+1})} + \cdots + \frac{c_N}{s-p_N}$$

$$c_r = [(s-p_1)^r X(s)]_{s=p_1}$$

$$c_{r-i} = \frac{1}{i!} \frac{d^i}{ds^i} [(s-p_1)^r X(s)]_{s=p_1}, i = 1, 2, \dots, r-1$$

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Repeated-pole partial-fraction expansion.

6.3.5 Pole locations and signal forms

Pole pattern	Time-domain form
Non-repeated real pole p	$ce^{pt}u(t)$
Repeated real pole p	$(c_1 + c_2t + \cdots + c_rt^{r-1})e^{pt}u(t)$
Complex pair $\sigma \pm j\omega$	$Ce^{\sigma t} \cos(\omega t + \theta)u(t)$
Repeated complex pair	Polynomial times $e^{\sigma t} \cos(\omega t + \theta)$
All poles in left half-plane	Natural modes decay as $t \rightarrow \infty$
Pole at $s = 0$	May produce a nonzero final value.

6.3 Computation of the Inverse Laplace Transform

6. Poles Locations and the Form of a Signal

1. If $X(s)$ has a **non-repeated pole** p which is real, then $x(t)$ contains a term of the form ce^{pt} for some constant c .
2. If the **pole** p is **repeated twice**, $x(t)$ contains the term for some constant c_1 and c_2 . $c_1e^{pt} + c_2te^{pt}$
3. If $X(s)$ has a **non-repeated pair** $\sigma \pm j\omega$ of complex poles, then $x(t)$ contains the form $ce^{\sigma t} \cos(\omega t + \theta)$
4. If the **complex pair** $\sigma \pm j\omega$ is **repeated twice**, $x(t)$ contains the term $c_1e^{\sigma t} \cos(\omega t + \theta_1) + c_2te^{\sigma t} \cos(\omega t + \theta_2)$

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Pole locations determine the qualitative time-domain signal form.

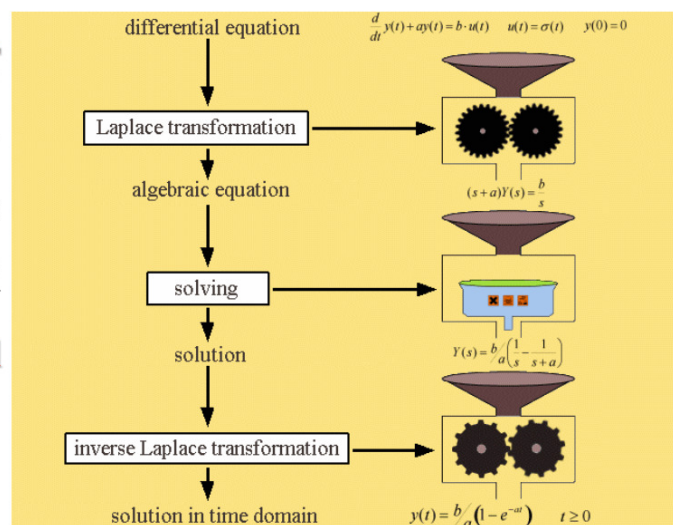
6.4 Transform of the Input/Output Differential Equation

6.4.1 From differential equation to algebraic equation

For a continuous-time LTI system, the input-output relation can be described by a differential equation. The Laplace transform converts derivatives into multiplication by s plus initial-condition terms. This changes the problem from solving a differential equation to solving an algebraic equation in s .

6.4 Transform of the Input/Output Differential Equation

For a **LTI CT system**, the input-output relationship can be described by a **differential equation**.



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Laplace transformation converts a differential-equation problem into an algebraic problem.

6.4.2 First-order case

For

$$\frac{dy(t)}{dt} + ay(t) = bx(t),$$

the one-sided Laplace transform gives

$$sY(s) - y(0^-) + aY(s) = bX(s).$$

Thus

$$Y(s) = \frac{y(0^-)}{s+a} + \frac{b}{s+a}X(s).$$

The first term is caused by initial energy; the second term is caused by the input.

6.4.3 Second-order case

For

$$\frac{d^2y(t)}{dt^2} + a_1\frac{dy(t)}{dt} + a_0y(t) = b_1\frac{dx(t)}{dt} + b_0x(t),$$

we get

$$Y(s) = \frac{y(0^-)s + \dot{y}(0^-) + a_1y(0^-)}{s^2 + a_1s + a_0} + \frac{b_1s + b_0}{s^2 + a_1s + a_0}X(s).$$

If the initial energy is zero, the transfer-function form is

$$Y(s) = H(s)X(s), \quad H(s) = \frac{b_1s + b_0}{s^2 + a_1s + a_0}.$$

6.4 Transform of the Input/Output Differential Equation

2. Second-order case

If there is **no initial energy** in the system at time $t=0$, it reduces to the transfer function representations:

$$Y(s) = \frac{y(0^-)s + \dot{y}(0^-) + a_1y(0^-)}{s^2 + a_1s + a_0} + \frac{b_1s + b_0}{s^2 + a_1s + a_0}X(s) \quad Y(s) = \frac{b_1s + b_0}{s^2 + a_1s + a_0}X(s)$$

The transfer function $H(s)$ is a second-order rational function of s given by

$$H(s) = \frac{b_1s + b_0}{s^2 + a_1s + a_0} \quad \frac{d^2y(t)}{dt^2} + a_1\frac{dy(t)}{dt} + a_0y(t) = b_1\frac{dx(t)}{dt} + b_0x(t)$$

6.4.4 Nth-order case

For the general differential equation

$$\frac{d^N y(t)}{dt^N} + \sum_{i=0}^{N-1} a_i \frac{d^i y(t)}{dt^i} = \sum_{i=0}^M b_i \frac{d^i x(t)}{dt^i},$$

the Laplace-domain solution has the form

$$Y(s) = \frac{C(s)}{A(s)} + \frac{B(s)}{A(s)} X(s).$$

Here $C(s)/A(s)$ is determined by initial conditions, while $B(s)X(s)/A(s)$ is the zero-state part due to the input. Under zero initial conditions,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{B(s)}{A(s)}.$$

6.4 Transform of the Input/Output Differential Equation

3. Nth-order case

$$\mathbf{Y}(s) = \frac{\mathbf{C}(s)}{\mathbf{A}(s)} + \frac{\mathbf{B}(s)}{\mathbf{A}(s)} \mathbf{X}(s)$$

If initial condition $y(0^-), y^{(1)}(0^-), \dots, y^{(N-1)}(0^-)$ are zero

$$Y(s) = \frac{B(s)}{A(s)} X(s) = \frac{b_M s^M + \dots b_1 s + b_0}{s^N + a_{N-1} s + \dots a_1 s + a_0} X(s)$$

$$H(s) = \frac{b_M s^M + \dots b_1 s + b_0}{s^N + a_{N-1} s + \dots a_1 s + a_0}$$

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Nth-order case: zero initial conditions reduce the representation to $Y(s) = H(s)X(s)$.

6.4.5 Output response computation

The standard workflow is:

$$X(s), H(s) \implies Y(s) = X(s)H(s) \implies y(t) = \mathcal{L}^{-1}\{Y(s)\}.$$

Example 6.30: Output response

Given

$$H(s) = \frac{s^2 + 2s + 16}{s^3 + 4s^2 + 8s}, \quad x(t) = e^{-2t}u(t),$$

we have

$$X(s) = \frac{1}{s + 2}.$$

Then

$$Y(s) = H(s)X(s),$$

and inverse transformation gives

$$y(t) = e^{-2t} \cos 2t - \frac{1}{2} e^{-2t} \sin 2t + 1 - 2e^{-2t}, \quad t \geq 0.$$

6.4 Transform of the Input/Output Differential Equation

4. Computation of output response

Example 6.30 Computation of Output Response

$$H(s) = \frac{s^2 + 2s + 16}{s^3 + 4s^2 + 8s} \quad x(t) = e^{-2t} u(t) \quad X(s) = \frac{1}{s + 2}$$

$$Y(s) = H(s)X(s) = \frac{s^2 + 2s + 16}{[(s + 2)^2 + 4]s(s + 2)}$$

$$y(t) = e^{-2t} \cos 2t - \frac{1}{2} e^{-2t} \sin 2t + 1 - 2e^{-2t}, t \geq 0$$



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Example 6.30: computing output response through $Y(s) = H(s)X(s)$.

6.5 Transform of the Input/Output Convolution Integral

For a causal LTI system,

$$y(t) = h(t) * x(t) = \int_0^t h(\lambda)x(t - \lambda) d\lambda, \quad t \geq 0.$$

Taking the Laplace transform gives

$$Y(s) = H(s)X(s),$$

where

$$h(t) \longleftrightarrow H(s).$$

Therefore

$$H(s) = \frac{Y(s)}{X(s)}.$$

Note

The transfer function $H(s)$ is determined by the system structure and component values, not by a particular input signal. Different inputs $X(s)$ pass through the same $H(s)$.

6.5 Transform of the Input/Output Convolution Integral

$$y(t) = h(t) * x(t) = \int_0^t h(\lambda)x(t-\lambda)d\lambda, t \geq 0$$

$$Y(s) = H(s)X(s)$$

System impulse
response

$$h(t) \leftrightarrow H(s)$$

$$H(s) = \frac{Y(s)}{X(s)}$$

Transfer
function

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Transfer function obtained from the input-output convolution integral.

6.6 Direct Construction of the Transfer Function

6.6.1 Circuit elements in the s domain

For zero initial energy, the common element relations are

$$\text{resistor: } V(s) = RI(s),$$

$$\text{capacitor: } V(s) = \frac{1}{Cs}I(s),$$

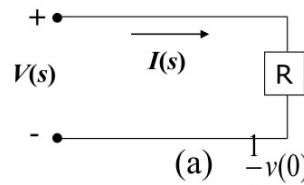
$$\text{inductor: } V(s) = LsI(s).$$

With nonzero initial energy, voltage-source or current-source terms associated with initial capacitor voltage or inductor current must be included.

6.6 Direction Construction of the Transfer Function

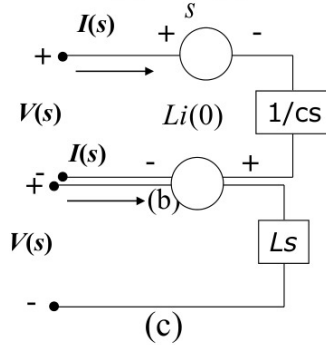
1. RLC circuit

R: $v(t) = Ri(t) \leftrightarrow V(s) = RI(s)$



C: $\frac{dv(t)}{dt} = \frac{1}{C}i(t) \leftrightarrow sV(s) - v(0) = \frac{1}{C}I(s)$

$$V(s) = \frac{1}{Cs}I(s) + \frac{1}{s}v(0)$$



L: $v(t) = L\frac{di(t)}{dt} \leftrightarrow V(s) = LsI(s) - Li(0)$



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s-domain representations of resistor, capacitor, and inductor.

6.6.2 Series and parallel impedances

For series impedances,

$$Z(s) = Z_1(s) + Z_2(s),$$

and voltage division gives

$$\frac{V_2(s)}{V(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}.$$

For parallel impedances, current division gives

$$\frac{I_2(s)}{I(s)} = \frac{Z_1(s)}{Z_1(s) + Z_2(s)}.$$

Series RLC transfer functions

For a series RLC circuit, taking the capacitor voltage as output gives

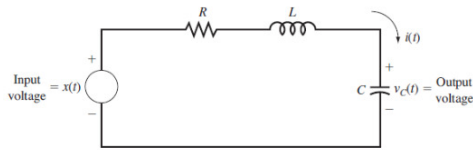
$$H_C(s) = \frac{1/(LC)}{s^2 + (R/L)s + 1/(LC)}.$$

Taking the resistor voltage as output gives

$$H_R(s) = \frac{(R/L)s}{s^2 + (R/L)s + 1/(LC)}.$$

6.6 Direction Construction of the Transfer Function

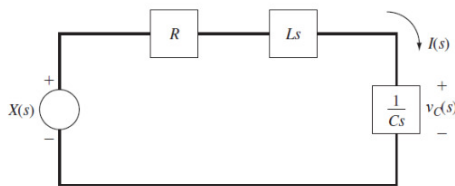
Example 6.37 Series RC circuit



$$V_c(s) = \frac{1/Cs}{Ls + R + (1/Cs)} X(s)$$

$$= \frac{1/L}{s^2 + (R/L)s + (1/LC)} X(s)$$

$$H(s) = \frac{1/LC}{s^2 + (R/L)s + (1/LC)}$$



$$V_R(s) = \frac{R}{Ls + R + (1/Cs)} X(s)$$

$$= \frac{(R/L)s}{s^2 + (R/L)s + (1/LC)} X(s)$$

$$H(s) = \frac{(R/L)s}{s^2 + (R/L)s + (1/LC)}$$

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Example 6.37: series RLC circuit transfer functions.

6.6.3 Integrator interconnections

An integrator has transfer function

$$\frac{1}{s}.$$

To compute the transfer function of an interconnection of integrators, adders, subtractors, and scalar multipliers:

1. Redraw the interconnection in the s domain.
2. Write the Laplace-domain equation for each integrator output.
3. Write the output transform $Y(s)$.
4. Eliminate internal variables and solve for $H(s) = Y(s)/X(s)$.

6.6 Direction Construction of the Transfer Function

2. Interconnections of Integrators

The integrator has the transfer function of $1/s$.

Consider the interconnection of integrators, adders, subtracters and scalar multipliers. To compute the transfer function:

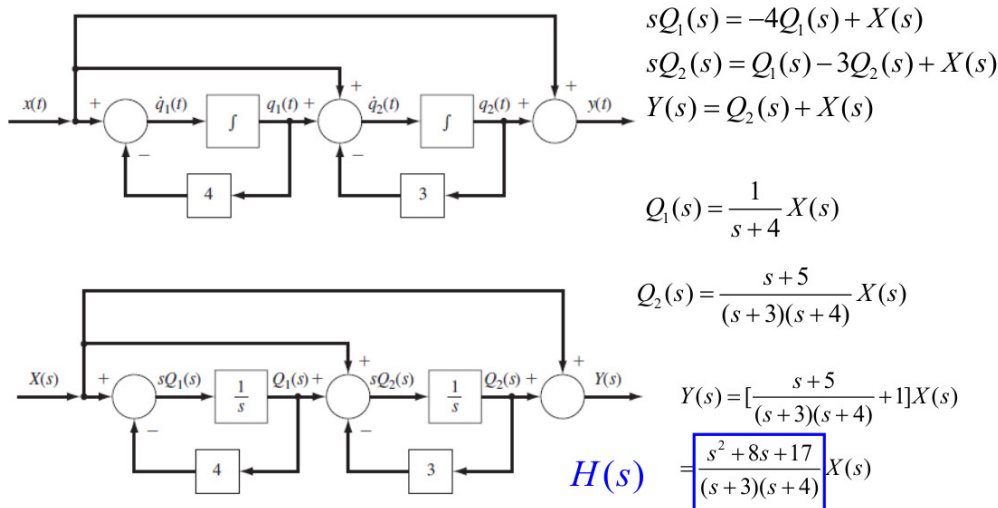
1. Redraw the interconnection in s domain.
2. Write the Laplace transform of each integrator in the interconnection.
3. Write the output transform.
4. Combine the equations to combine algebraically to derive the transfer function relationship.

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Procedure for constructing transfer functions from integrator interconnections.

6.6 Direction Construction of the Transfer Function

Example 6.39 Integrator Interconnection



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Example 6.39: algebraic derivation of $H(s)$ from an integrator interconnection.

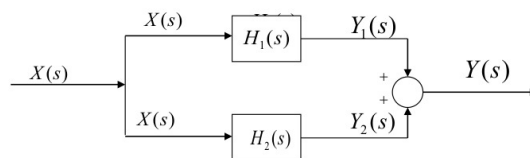
6.6.4 Block diagram reduction

Interconnection	Equivalent transfer function
Parallel	$H(s) = H_1(s) + H_2(s)$
Series	$H(s) = H_1(s)H_2(s)$
Negative feedback	$H(s) = \frac{H_1(s)}{1 + H_1(s)H_2(s)}$
Positive feedback	$H(s) = \frac{H_1(s)}{1 - H_1(s)H_2(s)}$

6.6 Direction Construction of the Transfer Function

3. Transfer Function of the Block Diagram

(1) Parallel Interconnection



$$Y(s) = [H_1(s) + H_2(s)]X(s) \quad (6.127)$$

$$H(s) = H_1(s) + H_2(s) \quad (6.128)$$

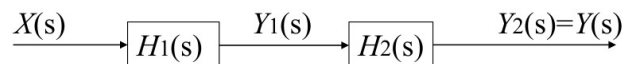
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Parallel interconnection.

6.6 Direction Construction of the Transfer Function

3. Transfer Function of the Block Diagram

(2) Series Connection



$$H(s) = H_1(s)H_2(s) \quad (6.131)$$

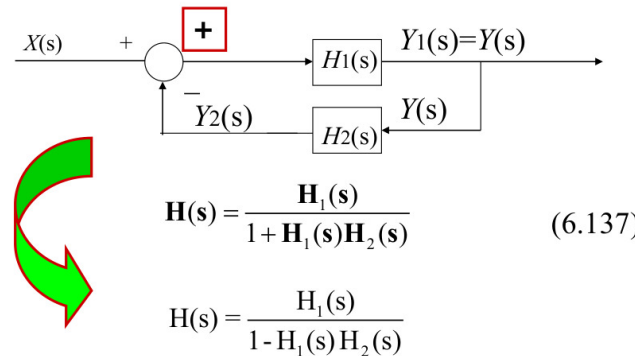
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Series interconnection.

6.6 Direction Construction of the Transfer Function

3. Transfer Function of the Block Diagram

(3) Feedback Connection



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Feedback interconnection.

6.7 Chapter 6 Review Table

Concept	Core formula or rule
Laplace transform	$X(s) = \int x(t)e^{-st} dt, s = \sigma + j\omega$
ROC	Set of s values for which the transform converges
Unit step	$u(t) \longleftrightarrow 1/s, \text{Re}\{s\} > 0$
Exponential	$e^{-bt}u(t) \longleftrightarrow 1/(s + b)$
Fourier relation	$X(\omega) = X(s) _{s=j\omega}$ only if ROC includes the $j\omega$ axis
Differentiation	$\dot{x}(t) \longleftrightarrow sX(s) - x(0^-)$
Integration	$\int_0^t x(\lambda)d\lambda \longleftrightarrow X(s)/s$
Convolution	$x(t) * v(t) \longleftrightarrow X(s)V(s)$
Distinct pole	$c_i = [(s - p_i)X(s)]_{s=p_i}$
Repeated pole	Generates $t^k e^{pt}$ terms
Complex pole pair	Generates damped sinusoids $e^{\sigma t} \cos(\omega t + \theta)$
Transfer function	$H(s) = Y(s)/X(s)$ under zero initial conditions
Output response	$Y(s) = H(s)X(s)$, then inverse Laplace transform
Feedback block	$H = H_1/(1 + H_1H_2)$ for negative feedback

CHAPTER 7:

The z -Transform and Discrete-Time System

Chapter 7 Map

Chapter 7 is the discrete-time counterpart of Chapter 6. The Laplace transform represents continuous-time signals and systems in the complex variable s ; the z -transform represents discrete-time sequences and systems in the complex variable z . For LTI discrete-time systems, the transfer-function form is

$$Y(z) = H(z)X(z), \quad H(z) = \frac{Y(z)}{X(z)}$$

under zero initial conditions.

Topic	Core task
z-transform	Convert $x[n]$ into $X(z) = \sum x[n]z^{-n}$ and state the ROC.
DTFT relation	Use $X(\Omega) = X(z) _{z=e^{j\Omega}}$ when the unit circle lies in the ROC.
Properties	Apply shifting, multiplication by n , summation, convolution, and value theorems.
Inverse transform	Recover $x[n]$ by series expansion, long division, partial fractions, or residues.
Transfer function	Convert difference equations or convolution sums into $H(z)$.
System analysis	Judge stability and sinusoidal response from poles and $H(e^{j\Omega})$.

7.1 z -Transform of Discrete-Time Signals

7.1.1 Motivation and definition

The z -transform is introduced by adding a convergence factor to the DTFT. Since

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n},$$

we insert a real radial factor ρ^{-n} and set

$$z = \rho e^{j\Omega}.$$

This gives the two-sided z -transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

For causal sequences, the one-sided form is usually

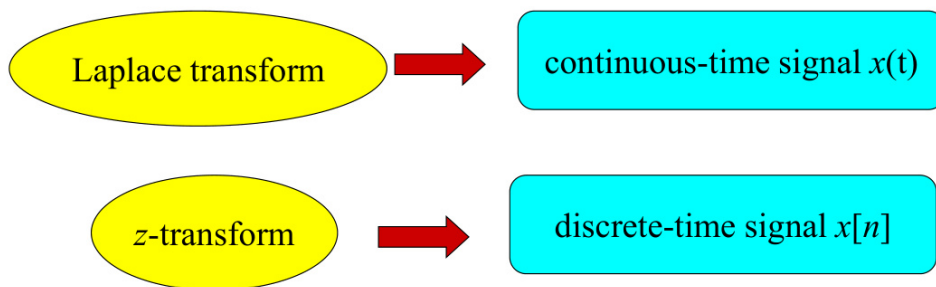
$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}.$$

Definition

The **region of convergence** (ROC) is the set of all complex numbers z for which the defining summation of $X(z)$ converges.

7.1 z -Transform of Discrete-time Signal

This chapter deals with the z -transform, which is the discrete-time counterpart of the Laplace transform.



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The z -transform is the discrete-time counterpart of the Laplace transform.

7.1 z -Transform of Discrete-time Signal

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] \rho^{-n} e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n] (\rho e^{j\Omega})^{-n} \quad z = \rho e^{j\Omega}$$



$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

z -transform of $x[n]$

two sided z -transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

one sided z -transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

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From the DTFT to the z -transform by using $z = \rho e^{j\Omega}$.

7.1.2 Basic examples

Sequence	z -transform	ROC
$\delta[n]$	1	all z
$\delta[n - q]$	z^{-q}	$ z > 0$ for $q > 0$
$u[n]$	$\frac{z}{z-1} = \frac{1}{1-z^{-1}}$	$ z > 1$
$a^n u[n]$	$\frac{z}{z-a} = \frac{1}{1-az^{-1}}$	$ z > a $

Example 7.3: Unit-step sequence

For

$$u[n] = \begin{cases} 1, & n = 0, 1, 2, \dots, \\ 0, & n = -1, -2, \dots, \end{cases}$$

we have

$$U(z) = \sum_{n=0}^{\infty} z^{-n} = 1 + z^{-1} + z^{-2} + \dots$$

Multiplying by $z - 1$ gives

$$U(z) = \frac{z}{z-1} = \frac{1}{1-z^{-1}}, \quad |z| > 1.$$

7.1.3 Relationship with the DTFT

Rule / Test

The DTFT is the z -transform evaluated on the unit circle:

$$X(\Omega) = X(z) \Big|_{z=e^{j\Omega}}.$$

This is valid only when the ROC of $X(z)$ includes the unit circle $|z| = 1$.

7.1 z -Transform of Discrete-time Signal

Relationship between the DTFT and the z -Transform

The z -transform $X(z)$ of a discrete-time signal $x[n]$ can be viewed as **a generalization** of the discrete-time Fourier transform (DTFT)

when the region of convergence of $X(z)$ includes all complex numbers z such that $|z|=1$,

$$X(\Omega) = X(z) \Big|_{z=e^{j\Omega}}$$

The z transform pair

$$x[n] \leftrightarrow X(z)$$

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The DTFT is obtained from the z -transform on the unit circle when the ROC includes $|z| = 1$.

7.1.4 ROC patterns

Sequence type	Typical ROC	Interpretation
Finite-length	Entire z -plane except possibly 0 or ∞	No infinite tail except finite powers of z
Right-sided	Exterior of outermost pole	Causal-type sequence
Left-sided	Interior of innermost pole	Anticausal-type sequence
Two-sided	Annulus between poles	Has both left and right tails

7.2 Properties of the z -Transform

Let

$$x[n] \longleftrightarrow X(z), \quad v[n] \longleftrightarrow V(z).$$

The most used properties are summarized below.

Property	Formula
Linearity	$ax[n] + bv[n] \longleftrightarrow aX(z) + bV(z)$
Right shift	$x[n - q] \longleftrightarrow z^{-q}X(z)$ if the missing initial samples are zero
Left shift	$x[n + q] \longleftrightarrow z^qX(z) - z^qx[0] - z^{q-1}x[1] - \dots - zx[q - 1]$
Multiplication by n	$nx[n] \longleftrightarrow -z \frac{dX(z)}{dz}$
Multiplication by a^n	$a^n x[n] \longleftrightarrow X(z/a)$
Multiplication by $\cos \Omega n$	$x[n] \cos \Omega n \longleftrightarrow \frac{1}{2}[X(e^{j\Omega}z) + X(e^{-j\Omega}z)]$
Summation	$\sum_{i=0}^n x[i] \longleftrightarrow \frac{z}{z-1}X(z)$
Convolution	$x[n] * v[n] \longleftrightarrow X(z)V(z)$
Initial value	$x[0] = \lim_{z \rightarrow \infty} X(z)$
Final value	If all poles of $(z-1)X(z)$ have magnitudes less than one, then $\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z-1)X(z)$

7.2 Properties of the z-Transform

Properties of the Laplace Transform and z Transform

Property	Laplace Transform Pair/Property	z Transform Pair/Property
Linearity	$ax(t) + bv(t) \leftrightarrow aX(s) + bV(s)$	$ax[n] + bv[n] \leftrightarrow aX(z) + bV(z)$
Right Shift in Time	$x(t-c)u(t-c) \leftrightarrow e^{-cs}X(s), c > 0$	$x[n-q]u[n-q] \leftrightarrow z^{-q}X(z)$ $x[n-q] \leftrightarrow z^{-q}X(z) + x[-q] + z^{-1}x[-q+1] + \dots + z^{-q+1}x[-1]$
Left Shift in Time		$x[n+q] \leftrightarrow z^qX(z) - x[0]z^q - x[1]z^{q-1} - \dots - x[q-1]z$
Time Scaling	$x(at) \leftrightarrow (1/a)X(s/a), a > 0$	
Mult. by a power of t/n	$t^N x(t) \leftrightarrow (-1)^N \frac{d^N}{ds^N} X(s), N = 1, 2, \dots$	$nx[n] \leftrightarrow -z \frac{d}{dz} X(z), n^2 x[n] \leftrightarrow z \frac{d}{dz} X(z) + z^2 \frac{d^2}{dz^2} X(z)$
Mult. by an exponential	$e^{at}x(t) \leftrightarrow X(s-a), a \text{ real or complex}$	$a^n x[n] \leftrightarrow X(z/a)$
Mult. by $\sin \omega t / \sin \Omega n$	$x(t) \sin \omega t \leftrightarrow \frac{j}{2}[X(s+j\omega) - X(s-j\omega)]$	$x[n] \sin \Omega n \leftrightarrow \frac{j}{2}[X(e^{j\Omega}z) - X(e^{-j\Omega}z)]$
Mult. by $\cos \omega t / \cos \Omega n$	$x(t) \cos \omega t \leftrightarrow \frac{1}{2}[X(s+j\omega) + X(s-j\omega)]$	$x[n] \cos \Omega n \leftrightarrow \frac{1}{2}[X(e^{j\Omega}z) + X(e^{-j\Omega}z)]$
Diff. in the time domain	$\dot{x}(t) \leftrightarrow sX(s) - x(0)$	
Second derivative	$\ddot{x}(t) \leftrightarrow s^2 X(s) - sx(0) - \dot{x}(0)$	
Nth derivative	$x^{(N)}(t) \leftrightarrow s^N X(s) - s^{N-1}x(0) - s^{N-2}\dot{x}(0) - \dots - sx^{(N-2)}(0) - x^{(N-1)}(0)$	
Integration/Summation	$\int_0^t x(\lambda) d\lambda \leftrightarrow \frac{1}{s} X(s)$	$\sum_{i=0}^n x[i] \leftrightarrow \frac{z}{z-1} X(z)$
Convolution	$x(t) * v(t) \leftrightarrow X(s)V(s)$	$x[n] * v[n] \leftrightarrow X(z)V(z)$
Initial-value theorem	$x(0) = \lim_{s \rightarrow \infty} sX(s)$ $\dot{x}(0) = \lim_{s \rightarrow \infty} [s^2 X(s) - sx(0)]$ $x^{(N)}(0) \leftrightarrow \lim_{s \rightarrow \infty} [s^{N+1} X(s) - s^N x(0) - s^{N-1} \dot{x}(0) - \dots - sx^{(N-1)}(0)]$	$x[0] = \lim_{z \rightarrow \infty} X(z)$ $x[1] = \lim_{z \rightarrow \infty} [zX(z) - zx[0]]$ $x[q] \leftrightarrow \lim_{z \rightarrow \infty} [z^q X(z) - z^q x[0] - z^{q-1}x[1] - \dots - zx[q-1]]$
Final-value theorem	If $\lim_{t \rightarrow \infty} x(t)$ exists, then $\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s)$	If $X(z)$ is rational and the poles of $(z-1)X(z)$ have magnitudes < 1 , then $\lim_{n \rightarrow \infty} x[n] = [(z-1)X(z)]_{z=1}$

Comparison of Laplace-transform and z-transform properties.

7.2 Properties of the z-Transform

Common Laplace Transform and z Transform Pairs

Common Laplace Transform Pairs	Common z Transform Pairs
$u(t) \leftrightarrow \frac{1}{s}$	$u[n] \leftrightarrow \frac{z}{z-1}$
$u(t) - u(t-c) \leftrightarrow \frac{1-e^{-cs}}{s}, c > 0$	$u[n] - u[n-q] \leftrightarrow \frac{z^q - 1}{z^{q-1}(z-1)}, q = 1, 2, \dots$
$t^N u(t) \leftrightarrow \frac{N!}{s^{N+1}}, N = 1, 2, 3, \dots$	$nu[n] \leftrightarrow \frac{z}{(z-1)^2}, n^2 u[n] \leftrightarrow \frac{z(z+1)}{(z-1)^3},$
$\delta(t) \leftrightarrow 1$	$\delta[n] \leftrightarrow 1$
$\delta(t-c) \leftrightarrow e^{-cs}, c > 0$	$\delta[n-q] \leftrightarrow z^{-q}, q = 1, 2, \dots$
$e^{-bt} u(t) \leftrightarrow \frac{1}{s+b}, b \text{ real or complex}$	$a^n u[n] \leftrightarrow \frac{z}{z-a}, a \text{ real or complex}$
$t^N e^{-bt} u(t) \leftrightarrow \frac{N!}{(s+b)^{N+1}}, N = 1, 2, 3, \dots$	$na^n u[n] \leftrightarrow \frac{az}{(z-a)^2}, n^2 a^n u[n] \leftrightarrow \frac{az(z+a)}{(z-a)^3}$
$(\cos \omega t) u(t) \leftrightarrow \frac{s}{s^2 + \omega^2}, (\cos^2 \omega t) u(t) \leftrightarrow \frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)}$	$(\cos \Omega n) u[n] \leftrightarrow \frac{z^2 - (\cos \Omega)z}{z^2 - (2 \cos \Omega)z + 1}$
$(\sin \omega t) u(t) \leftrightarrow \frac{\omega}{s^2 + \omega^2}, (\sin^2 \omega t) u(t) \leftrightarrow \frac{2\omega^2}{s(s^2 + 4\omega^2)}$	$(\sin \Omega n) u[n] \leftrightarrow \frac{(\sin \Omega)z}{z^2 - (2 \cos \Omega)z + 1}$
$(e^{-bt} \cos \omega t) u(t) \leftrightarrow \frac{s+b}{(s+b)^2 + \omega^2}, (e^{-bt} \sin \omega t) u(t) \leftrightarrow \frac{\omega}{(s+b)^2 + \omega^2}$	$a^n (\cos \Omega n) u[n] \leftrightarrow \frac{z^2 - (a \cos \Omega)z}{z^2 - (2a \cos \Omega)z + a^2}, a^n (\sin \Omega n) u[n] \leftrightarrow \frac{(a \sin \Omega)z}{z^2 - (2a \cos \Omega)z + a^2}$
$(t \cos \omega t) u(t) \leftrightarrow \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}, (t \sin \omega t) u(t) \leftrightarrow \frac{2\omega s}{(s^2 + \omega^2)^2}$	
$(te^{-bt} \cos \omega t) u(t) \leftrightarrow \frac{(s+b)^2 - \omega^2}{[(s+b)^2 + \omega^2]^2}, (te^{-bt} \sin \omega t) u(t) \leftrightarrow \frac{2\omega(s+b)}{[(s+b)^2 + \omega^2]^2}$	

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Common Laplace-transform and z-transform pairs.

Example 7.6: Finite pulse sequence

For

$$p[n] = \begin{cases} 1, & n = 0, 1, 2, \dots, q-1, \\ 0, & \text{otherwise,} \end{cases}$$

write it as

$$p[n] = u[n] - u[n-q].$$

Using the shift property,

$$P(z) = \frac{z}{z-1} - z^{-q} \frac{z}{z-1} = \frac{z(1-z^{-q})}{z-1} = \frac{z^q - 1}{z^{q-1}(z-1)}.$$

7.2 Properties of the z-Transform

Example 7.6 z transform of a pulse

$$p[n] = \begin{cases} 1, & n = 0, 1, 2, \dots, q-1 \\ 0, & \text{all other } n \end{cases}$$

$$p[n] = u[n] - u[n - q]$$

$$\begin{aligned} & \begin{array}{c} \downarrow \qquad \qquad \downarrow \\ \text{z transform of } p[n] \end{array} \quad \frac{z}{z-1} \quad z^{-q} \frac{z}{z-1} = \frac{z^{-q+1}}{z-1} \\ & \frac{z}{z-1} - \frac{z^{-q+1}}{z-1} = \frac{z(1 - z^{-q})}{z-1} = \frac{z^q - 1}{z^{q-1}(z-1)} \end{aligned}$$

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Example 7.6: z-transform of a finite pulse sequence.

Example 7.14: Limiting value

For

$$X(z) = \frac{3z^2 - 2z + 4}{z^3 - 2z^2 + 1.5z - 0.5},$$

there is a pole-zero cancellation in $(z - 1)X(z)$. Thus

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z) = \frac{3z^2 - 2z + 4}{z^2 - z + 0.5} \Big|_{z=1} = 10.$$

7.3 Computation of the Inverse z-Transform

The inverse z-transform is formally

$$x[n] = \frac{1}{j2\pi} \oint X(z) z^{n-1} dz.$$

In practice, rational transforms are usually inverted without evaluating this contour integral.

Method	Use
Power series expansion	Expand $X(z)$ into powers of z^{-1} and read coefficients as $x[n]$.
Long division	Divide numerator by denominator to obtain a power series.
Partial fractions	Decompose rational $X(z)$ into basic transform pairs.
Residue method	Use contour integration or pole residues directly.

7.3 Computation of the Inverse z-Transform

1. Inverse z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$x[n] = \frac{1}{j2\pi} \int X(z)z^{n-1} dz \quad (7.59)$$

When the transform $X(z)$ is a rational function of z , the inverse z -transform can be computed without having to evaluate the integral in (7.59).

1. Power Series Expansion(幂级数展开)

2. Long Division(长除法)

3. Partial Fraction(部分分式展开)

4. Loop integral—Residue Method(留数法)

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Four standard methods for computing the inverse z -transform.

7.3.1 Power series expansion

Because

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n},$$

the coefficient of z^{-n} is the sample value $x[n]$.

Power series example

If

$$X(z) = e^{-a/z},$$

then

$$X(z) = 1 - az^{-1} + \frac{a^2}{2!}z^{-2} - \frac{a^3}{3!}z^{-3} + \cdots$$

Therefore

$$x[0] = 1, \quad x[n] = \frac{(-a)^n}{n!}, \quad n \geq 1.$$

7.3 Computation of the Inverse z-Transform

1. Power Series Expansion

Example:

$$X(z) = e^{-a/z}, \quad x[n] = ?$$

Solution:

$$\because e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots$$

$$X(z) = 1 + \left(-\frac{a}{z}\right) + \frac{1}{2!}\left(-\frac{a}{z}\right)^2 + \frac{1}{3!}\left(-\frac{a}{z}\right)^3 + \dots$$

$$\therefore x[n] = \{1, -a, \frac{(-a)^2}{2!}, \frac{(-a)^3}{3!}, \dots, \frac{(-a)^n}{n!}, \dots\} = \begin{cases} \frac{(-a)^n}{n!} & n \geq 1 \\ 1 & n = 0 \end{cases}$$

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Inverse z-transform by power-series expansion.

7.3.2 Partial fractions and pole forms

If $X(z)$ is rational, factor the denominator and use partial fractions. The pole locations determine the time-domain form.

Pole pattern	Time-domain term
Non-repeated real pole p	cp^n
Repeated real pole p of order 2	$c_1p^n + c_2np^n$
Non-repeated complex pair $\sigma \pm j\Omega$	$c\sigma^n \cos(\Omega n + \theta)$
Repeated complex pair	$c_1\sigma^n \cos(\Omega n + \theta_1) + c_2n\sigma^n \cos(\Omega n + \theta_2)$

7.3 Computation of the Inverse z-Transform

4. Poles Locations and the Form of a Signal

1. If $X(z)$ has a non-repeated pole p which is real, then $x[n]$ contains a term of the form $c(p)^n$ for some constant c .
2. If the pole p is repeated twice, $x[n]$ contains the term for some constant c_1 and c_2 . $c_1(p)^n$ and $c_2 n(p)^n$
3. If $X(z)$ has a non-repeated pair $\sigma \pm j\Omega$ of complex poles, then $x[n]$ contains the form $c\sigma^n \cos(\Omega n + \theta)$
4. If the complex pair $\sigma + j\Omega$ is repeated twice, $x[n]$ contains the term $c_1 \sigma^n \cos(\Omega n + \theta_1) + c_2 n \sigma^n \cos(\Omega n + \theta_2)$



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Pole locations and the corresponding time-domain signal forms.

7.4 Transfer Function Representation

7.4.1 First-order case

For the difference equation

$$y[n] + ay[n-1] = bx[n],$$

taking the z -transform gives an initial-condition term and an input-driven term. If the initial condition is zero,

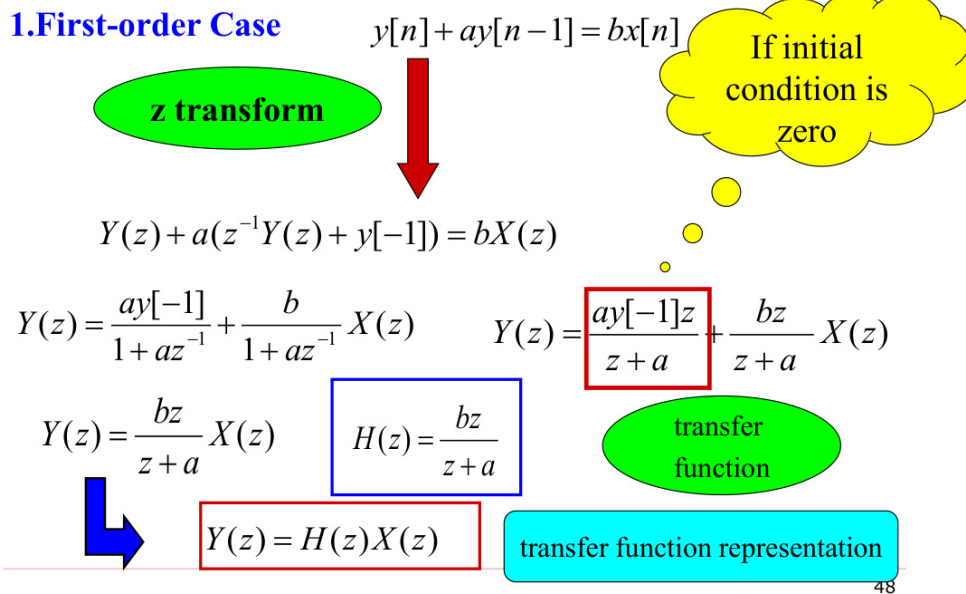
$$Y(z) = \frac{bz}{z+a} X(z),$$

so

$$H(z) = \frac{Y(z)}{X(z)} = \frac{bz}{z+a}.$$

7.4 Transfer Function Representation

1. First-order Case



First-order difference equation and its transfer-function representation.

7.4.2 Nth-order case

For a general difference equation

$$y[n] + \sum_{i=1}^N a_i y[n-i] = \sum_{i=1}^M b_i x[n-i],$$

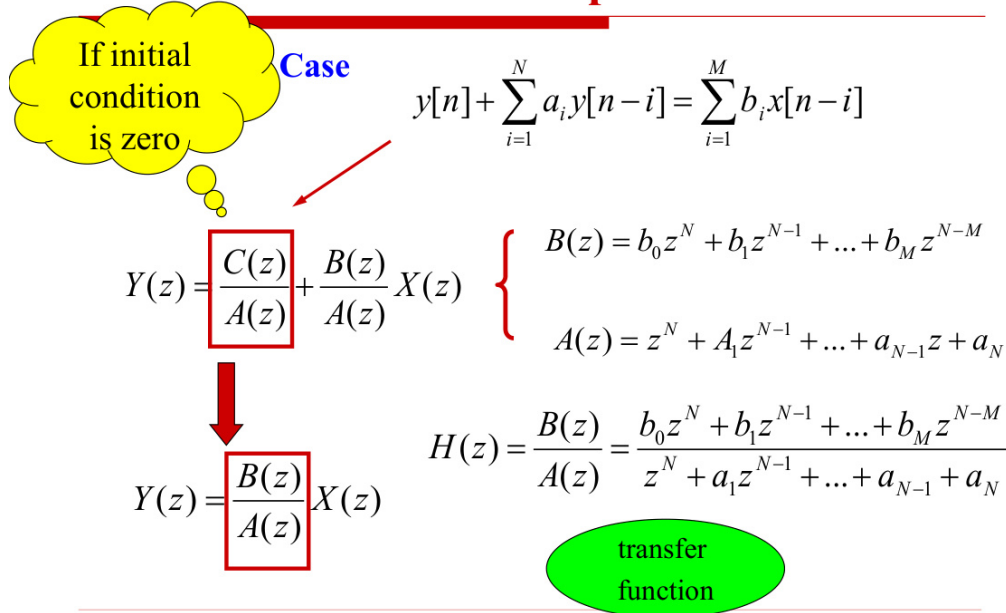
the z -domain form is

$$Y(z) = \frac{C(z)}{A(z)} + \frac{B(z)}{A(z)} X(z).$$

Under zero initial conditions,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{B(z)}{A(z)}.$$

7.4 Transfer Function Representation



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Nth-order difference equation: zero initial conditions lead to $H(z) = B(z)/A(z)$.

7.4.3 Convolution sum and transfer function

For an LTI discrete-time system,

$$y[n] = h[n] * x[n] = \sum_{i=-\infty}^{\infty} h[i] x[n-i].$$

Taking the z -transform gives

$$Y(z) = H(z)X(z), \quad H(z) = \frac{Y(z)}{X(z)}.$$

Here $h[n]$ is the unit-pulse response and $H(z)$ is the transfer function.

Example 7.19: Step response of a first-order system

For

$$y[n] + ay[n-1] = bx[n], \quad x[n] = u[n],$$

we use

$$X(z) = \frac{z}{z-1}, \quad H(z) = \frac{bz}{z+a}.$$

If the initial condition is zero, then

$$Y(z) = H(z)X(z),$$

and the response is

$$y[n] = \frac{b}{a+1} [-(-a)^{n+1} + 1], \quad n = 0, 1, 2, \dots$$

7.4 Transfer Function Representation

Example 7.19 Step Response

$$y[n] + ay[n-1] = bx[n]$$

$$x[n] = u[n] \quad \longrightarrow \quad X(z) = \frac{z}{z-1} \quad Y(z) = \frac{ay[-1]z}{z+a} + \frac{bz}{z+a} X(z)$$

$$Y(z) = H(z)X(z) \quad Y(z) = \frac{ay[-1]z}{z+a} + \frac{bz}{z+a} \frac{z}{z-1}$$

$$= \frac{ay[-1]z}{z+a} + \frac{ab/(a+1)}{z+a} + \frac{b/(a+1)}{z-1}$$

$$y[n] = -ay[-1](-a)^n + \frac{b}{a+1}[-(-a)^{n+1} + 1], n = 0, 1, 2, \dots$$

$$y[n] = \frac{b}{a+1}[-(-a)^{n+1} + 1], n = 0, 1, 2, \dots$$

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Example 7.19: step response of a first-order discrete-time system.

7.4.4 Transfer functions of interconnections

The unit-delay block satisfies

$$y[n] = x[n-1] \quad \Longleftrightarrow \quad Y(z) = z^{-1}X(z).$$

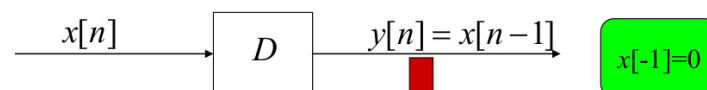
Hence a unit delay has transfer function $z^{-1} = 1/z$.

7.4 Transfer Function Representation

5. Transfer Function of Interconnections unit-delay elements

$$y[n] = x[n-1]$$

If initial condition is zero



z transform

$$Y(z) = z^{-1}X(z)$$

The transfer function for any such interconnection can be computed by working in the **z-domain** with the unit delays represented by their transfer function $1/z$.

Unit-delay element represented by z^{-1} in the z domain.

Basic reductions are analogous to continuous-time block diagrams:

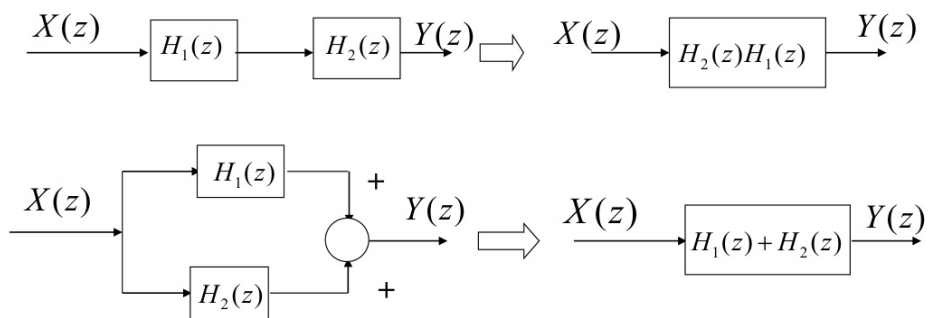
$$H_{\text{series}}(z) = H_1(z)H_2(z), \quad H_{\text{parallel}}(z) = H_1(z) + H_2(z),$$

$$H_{\text{negative feedback}}(z) = \frac{H_1(z)}{1 + H_1(z)H_2(z)}.$$

7.4 Transform Function Representation

5. Transfer Function of Interconnections

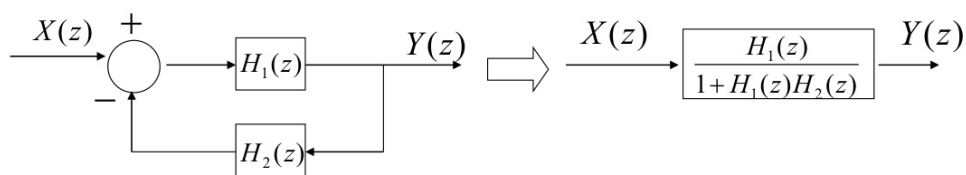
Transfer function of basic interconnections



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Series and parallel interconnections in the z domain.

7.4 Transform Function Representation



Block-diagram reduction

By using the equivalences above and the equivalence for moving pick-off points and adders, it is possible to reduce a given block diagram in order to compute the transfer function of the system.

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Negative-feedback reduction in the z domain.

7.5 System Analysis Using the Transfer Function Representation

7.5.1 Stability of discrete-time systems

For a rational $H(z)$, stability is determined by pole locations.

Rule / Test

A causal rational discrete-time LTI system is BIBO stable if all poles of $H(z)$ lie strictly inside the unit circle:

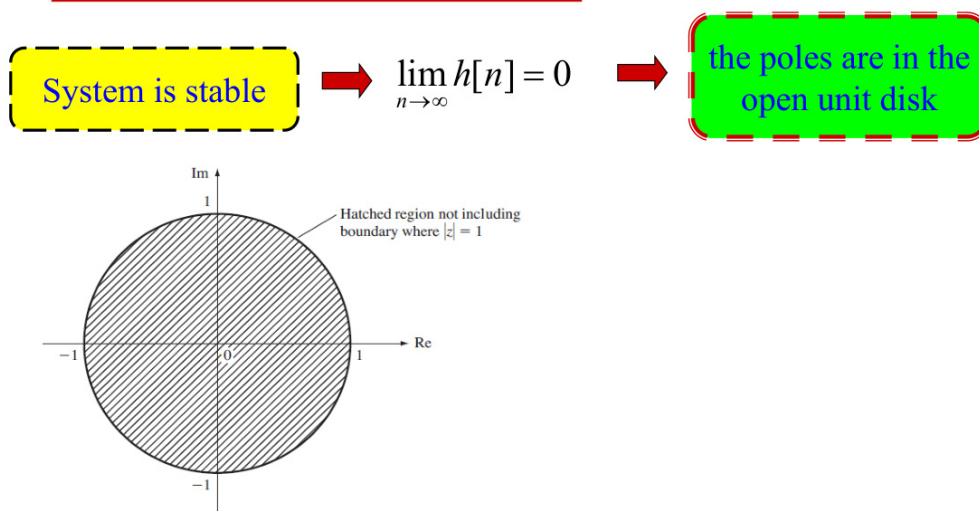
$$|p_i| < 1, \quad i = 1, 2, \dots, N.$$

Equivalently, the impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Pole location	System behavior
All poles inside $ z < 1$	Stable; impulse response decays.
Non-repeated poles on $ z = 1$, repeated poles inside	Marginally stable; bounded but not absolutely summable.
Any pole outside $ z = 1$, or repeated pole on $ z = 1$	Unstable.

7.5 Stability of Discrete-Time System



A stable causal rational system has poles in the open unit disk.

7.5 Stability of Discrete-Time System

① Stable

all system poles lie in the open unit disk of the complex plane.

② Marginally Stable

$$|h[n]| < c \quad \text{for all } n \quad (7.73)$$

where c is a **finite positive constant**.

$$\begin{aligned} |p_i| &\leq 1 && \text{for all non-repeated poles of } H(z), \text{ and} \\ |p_i| &< 1 && \text{for all repeated poles.} \end{aligned}$$

All poles lie in the open disk ,
except that **non-repeated poles can be located on the unit circle**.

Stable, marginally stable, and unstable pole-location cases.

Stability example

For

$$y[n+2] + 0.2y[n+1] - 0.24y[n] = x[n+2] + x[n+1],$$

the transfer function is

$$H(z) = \frac{z^2 + z}{z^2 + 0.2z - 0.24} = \frac{z(z+1)}{(z-0.4)(z+0.6)}.$$

The poles are $p_1 = 0.4$ and $p_2 = -0.6$. Since

$$|p_1| < 1, \quad |p_2| < 1,$$

the causal system is stable.

7.5 Stability of Discrete-Time System

Example : The stability of a DT system

$$y[n+2] + 0.2y[n+1] - 0.24y[n] = x[n+2] + x[n+1]$$

$$H(z) = ? \quad h[n] = ? \quad \text{Is the system stable?}$$

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{z^2 + z}{z^2 + 0.2z - 0.24} = \frac{z(z+1)}{(z-0.4)(z+0.6)} \\ &= \frac{1.4z}{z-0.4} - \frac{0.4z}{z+0.6} \end{aligned}$$

$$\therefore h[n] = \{1.4(0.4)^n - 0.4(-0.6)^n\}u[n]$$

$$\because |p_1| = 0.4 < 1, \quad |p_2| = 0.6 < 1, \quad \therefore \text{The system is stable.}$$

Example: pole locations at 0.4 and -0.6 imply stability.

7.5.2 Second-order stability intuition

For

$$H(z) = \frac{1}{1 - (2r \cos \theta)z^{-1} + r^2 z^{-2}},$$

the poles are

$$z_1 = re^{j\theta}, \quad z_2 = re^{-j\theta}.$$

For a causal system, the ROC is outside the outermost pole. Thus:

- if $r < 1$, the poles lie inside the unit circle and the system is stable;
- if $r > 1$, the poles lie outside the unit circle and the system is unstable.

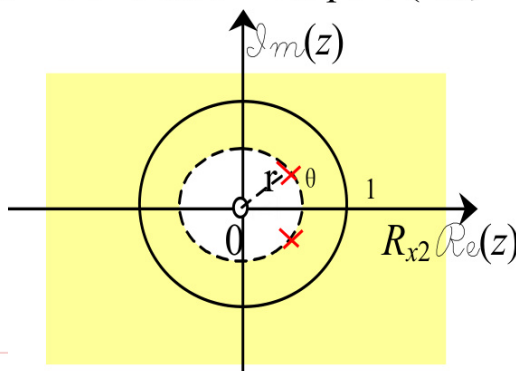
7.5 Stability of Discrete-Time System

Example : The stability of a second-order system

$$H(z) = \frac{1}{1 - (2r \cos \theta)z^{-1} + r^2 z^{-2}} \quad \text{Two poles located at } z_1 = re^{j\theta} \text{ and } z_2 = re^{-j\theta}.$$

Assume that it is causality, the ROC is outermost pole (i.e., $|z| > |r|$).

For $r < 1$, the poles are inside the unit circle, **the ROC include the circle**, and therefore the system is stable.



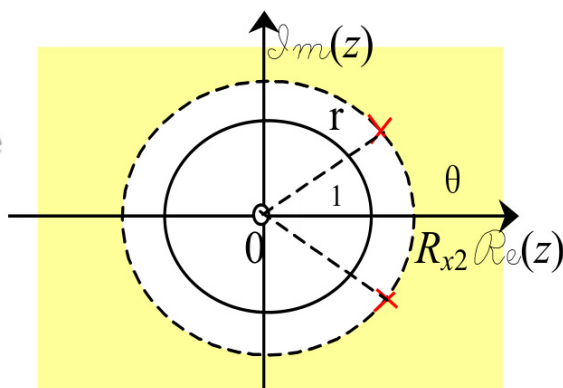
Second-order case with $r < 1$: the ROC includes the unit circle.

7.5 Stability of Discrete-Time System

Example : The stability of a second-order system

$$H(z) = \frac{1}{1 - (2r \cos \theta)z^{-1} + r^2 z^{-2}} \quad \text{Two poles located at } z_1 = re^{j\theta} \text{ and } z_2 = re^{-j\theta}.$$

For $r > 1$, the poles are outside the unit circle, **the ROC does not include the unit circle**, and therefore the system is unstable.



Second-order case with $r > 1$: the ROC does not include the unit circle.

7.5.3 Response to a sinusoidal input

For a stable discrete-time LTI system, if

$$x[n] = C \cos(\Omega_0 n),$$

then the steady-state response is determined by the frequency response

$$H(e^{j\Omega}).$$

The output sinusoid keeps the same frequency, with amplitude scaled by $|H(e^{j\Omega_0})|$ and phase shifted by $\angle H(e^{j\Omega_0})$:

$$y[n] = C|H(e^{j\Omega_0})| \cos(\Omega_0 n + \angle H(e^{j\Omega_0})).$$

7.5 Stability of Discrete-Time System

2. The Response to A Sinusoidal Input

$$H(z) = \frac{B(z)}{A(z)} = \frac{b_M z^M + b_{M-1} z^{M-1} + \dots + b_1 z + b_0}{a_N z^N + a_{N-1} z^{N-1} + \dots + a_1 z + a_0}$$

$x[n] = C \cos(\Omega_0 n), \quad n = 0, 1, 2, \dots$
z transform
 $X(z) = \frac{C[z^2 - (\cos \Omega_0)z]}{z^2 - (2 \cos \Omega_0)z + 1}$

$Y(z) = H(z)X(z) = \frac{CB(z)[z^2 - (\cos \Omega_0)z]}{A(z)[z^2 - (2 \cos \Omega_0)z + 1]}$

$z^2 - (2 \cos \Omega_0)z + 1 = (z - e^{j\Omega_0})(z - e^{-j\Omega_0})$

$Y(z) = \frac{CB(z)[z^2 - (\cos \Omega_0)z]}{A(z)(z - e^{j\Omega_0})(z - e^{-j\Omega_0})}$

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The z -domain derivation of the sinusoidal response of a discrete-time LTI system.

7.6 Chapter 7 Review Table

Concept	Core formula or rule
Two-sided z -transform	$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$
One-sided z -transform	$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$
ROC	Set of z values for which the transform converges
DTFT relation	$X(\Omega) = X(z) _{z=e^{j\Omega}}$, if $ z = 1$ is in the ROC
Unit step	$u[n] \longleftrightarrow z/(z-1)$
Right-sided exponential	$a^n u[n] \longleftrightarrow z/(z-a)$
Right shift	$x[n-q] \longleftrightarrow z^{-q}X(z)$ if initial missing samples are zero
Multiplication by n	$nx[n] \longleftrightarrow -z \frac{dX(z)}{dz}$
Summation	$\sum_{i=0}^n x[i] \longleftrightarrow \frac{z}{z-1}X(z)$
Convolution	$x[n] * v[n] \longleftrightarrow X(z)V(z)$
Final value theorem	$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z-1)X(z)$ under the pole condition
Transfer function	$H(z) = Y(z)/X(z)$ under zero initial conditions
Unit delay	z^{-1}
Stability	causal rational system stable $\iff$ all poles of $H(z)$ are inside $ z < 1$
Sinusoidal response	amplitude $ H(e^{j\Omega_0}) $, phase $\angle H(e^{j\Omega_0})$